



IEEE Standard for Ethernet

Amendment 7: Physical Layer Specifications and Management Parameters for Multi-Gigabit Glass Optical Fiber Automotive Ethernet

IEEE Computer Society

Developed by the
LAN/MAN Standards Committee

IEEE Std 802.3cz™-2023

(Amendment to IEEE Std 802.3™-2022
as amended by IEEE Std 802.3dd™-2022,
IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022,
IEEE Std 802.3ck™-2022, IEEE Std 802.3de™-2022,
and IEEE Std 802.3cx™-2023)

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Approved 30 March 2023
IEEE SA Standards Board

Abstract: This amendment to IEEE Std 802.3-2022 adds Physical Layer specifications and management parameters for 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, and 50 Gb/s operation on glass optical fiber in an automotive environment.

Keywords: 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, 50GBASE-AU, amendment, automotive Ethernet, Ethernet, IEEE 802.3™, IEEE 802.3cz™, Medium Dependent Interface, Physical Coding Sublayer, Physical Layer, Physical Medium Attachment

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Introduction

This introduction is not part of IEEE Std 802.3cz-2023, IEEE Standard for Ethernet—Amendment 7: Physical Layer Specifications and Management Parameters for Multi-Gigabit Glass Optical Fiber Automotive Ethernet.

IEEE Std 802.3™ was first published in 1985. Since the initial publication, many projects have added functionality or provided maintenance updates to the specifications and text included in the standard. Each IEEE 802.3 project/amendment is identified with a suffix (e.g., IEEE Std 802.3ba™-2010).

The half duplex Media Access Control (MAC) protocol specified in IEEE Std 802.3-1985 is Carrier Sense Multiple Access with Collision Detection (CSMA/CD). This MAC protocol was key to the experimental Ethernet developed at Xerox Palo Alto Research Center, which had a 2.94 Mb/s data rate. Ethernet at 10 Mb/s was jointly released as a public specification by Digital Equipment Corporation (DEC), Intel and Xerox in 1980. “Local Area Networks: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications” was approved as an IEEE standard by the IEEE Standards Board in 1983 and subsequently published in 1985 as IEEE Std 802.3-1985. Since 1985, new media options, new speeds of operation, and new capabilities have been added to IEEE Std 802.3. A full duplex MAC protocol and the ability to use an EtherType to specify the MAC client protocol were added in 1997. The title was changed to Standard for Ethernet with the 2012 Revision.

Some of the major additions to IEEE Std 802.3 are identified in the marketplace with their project number. This is most common for projects adding higher speeds of operation or new protocols. For example, IEEE Std 802.3u™ added 100 Mb/s operation (also called Fast Ethernet), IEEE Std 802.3z™ added 1000 Mb/s operation (also called Gigabit Ethernet), IEEE Std 802.3ae™ added 10 Gb/s operation (also called 10 Gigabit Ethernet), IEEE Std 802.3ah™ specified access network Ethernet (also called Ethernet in the First Mile) and IEEE Std 802.3ba added 40 Gb/s operation (also called 40 Gigabit Ethernet) and 100 Gb/s operation (also called 100 Gigabit Ethernet). These major additions are all now included in and are superseded by IEEE Std 802.3-2022 and are not maintained as separate documents.

At the date of IEEE Std 802.3cz-2023 publication, IEEE Std 802.3 was composed of the following documents:

IEEE Std 802.3-2022

Section One—Includes Clause 1 through Clause 20 and Annex A through Annex K and Annex 4A. Section One includes the specifications for 10 Mb/s operation and the MAC, frame formats and service interfaces used for all speeds of operation.

Section Two—Includes Clause 21 through Clause 33 and Annex 22A through Annex 33A. Section Two includes management attributes for multiple protocols and speed of operation as well as specifications for providing power over twisted pair cabling for multiple operational speeds. It also includes general information on 100 Mb/s operation as well as most of the 100 Mb/s Physical Layer specifications.

Section Three—Includes Clause 34 through Clause 43 and Annex 36A through Annex 43C. Section Three includes general information on 1000 Mb/s operation as well as most of the 1000 Mb/s Physical Layer specifications.

Section Four—Includes Clause 44 through Clause 55 and Annex 44A through Annex 55B. Section Four includes general information on 10 Gb/s operation as well as most of the 10 Gb/s Physical Layer specifications.

Section Five—Includes Clause 56 through Clause 77 and Annex 57A through Annex 76A. Clause 56 through Clause 67 and Clause 75 through Clause 77, as well as associated annexes, specify subscriber access and other Physical Layers and sublayers for operation from 512 kb/s to 10 Gb/s, and defines services and protocol elements that enable the exchange of IEEE Std 802.3 format frames between stations in a subscriber access network. Clause 68 specifies a 10 Gb/s Physical Layer specification. Clause 69 through Clause 74 and associated annexes specify Ethernet operation over electrical backplanes at speeds of 1000 Mb/s and 10 Gb/s.

Section Six—Includes Clause 78 through Clause 95 and Annex 83A through Annex 93C. Clause 78 specifies Energy-Efficient Ethernet. Clause 79 specifies IEEE 802.3 Organizationally Specific Link Layer Discovery Protocol (LLDP) type, length, and value (TLV) information elements. Clause 80 through Clause 95 and associated annexes include general information on 40 Gb/s and 100 Gb/s operation as well as 40 Gb/s and 100 Gb/s Physical Layer specifications. Clause 90 specifies Ethernet support for time synchronization protocols.

Section Seven—Includes Clause 96 through Clause 115 and Annex 97A through Annex 115A. Clause 96 through Clause 98, Clause 104, and associated annexes, specify Physical Layers and optional features for 100 Mb/s and 1000 Mb/s operation over a single twisted pair. Clause 100 through Clause 103, as well as associated annexes, specify Physical Layers for the operation of the EPON protocol over coaxial distribution networks. Clause 105 through Clause 114 and associated annexes include general information on 25 Gb/s operation as well as 25 Gb/s Physical Layer specifications. Clause 99 specifies a MAC merge sublayer for the interspersing of express traffic. Clause 115 and its associated annex specify a Physical Layer for 1000 Mb/s operation over plastic optical fiber.

Section Eight—Includes Clause 116 through Clause 140 and Annex 119A through Annex 136D. Clause 116 through Clause 124 and associated annexes include general information on 200 Gb/s and 400 Gb/s operation as well as 200 Gb/s and 400 Gb/s Physical Layer specifications. Clause 125 includes general information on 2.5 Gb/s and 5 Gb/s operation. Clause 126 through Clause 130 and associated annexes include 2.5 Gb/s and 5 Gb/s Physical Layer specifications. Clause 131 provides general information on 50 Gb/s operation. Clause 132 through Clause 140 and associated annexes include 50 Gb/s Physical Layer specifications and additional 100 Gb/s, 200 Gb/s, and 400 Gb/s Physical Layer specifications.

Section Nine—Includes Clause 141 through Clause 160 and Annex 142A through Annex 154A. Clause 141 through Clause 144 and associated annexes specify symmetric and asymmetric operation of Ethernet passive optical networks over multiple 25 Gb/s channels. Clause 145 and associated annexes specify increased power delivery using all four pairs in the structured wiring plant. Clause 146 through Clause 149 and associated annexes specify Physical Layers for 10 Mb/s, 2.5 Gb/s, 5 Gb/s, and 10 Gb/s operation over a single balanced pair of conductors. Clause 150 and Clause 151 include additional 400 Gb/s Physical Layer specifications. Clause 153 and Clause 154 specify 100 Gb/s operation over DWDM channels. Clause 157 through Clause 160 include 10 Gb/s, 25 Gb/s, and 50 Gb/s bidirectional Physical Layer specifications.

IEEE Std 802.3dd™-2022

Amendment 1—This amendment includes editorial and technical corrections, refinements, and clarifications to Clause 104, Power over Data Lines of Single Pair Ethernet, and related portions of the standard.

IEEE Std 802.3cs™-2022

Amendment 2—This amendment to IEEE Std 802.3-2022 defines Super-PON optical subscriber access networks, in the family of Ethernet passive optical networks (EPONs). Super-PON has a reach of up to 50 km and up to 1024 ONUs over a point-to-multipoint passive optical distribution network (ODN)

through wavelength division multiplexing (WDM). A Super-PON ODN contains a passive wavelength router that determines the channels used by the ODN. This standard specifies the Super-PON Reconciliation Sublayer (RS), Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer at a MAC data rate of 10 Gb/s in the downstream direction and of 10 Gb/s or 2.5 Gb/s in the upstream direction.

IEEE Std 802.3db™-2022

Amendment 3—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 167. This amendment adds Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s over one, two, and four pairs of multimode fiber based on 100 Gb/s optical signaling.

IEEE Std 802.3ck™-2022

Amendment 4—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 161 through Clause 163, Annex 120F, Annex 120G, Annex 162A through Annex 162D, Annex 163A, and Annex 163B. This amendment includes Physical Layer specifications and management parameters for 100 Gb/s, 200 Gb/s, and 400 Gb/s electrical interfaces based on 100 Gb/s signaling.

IEEE Std 802.3de™-2022

Amendment 5—This amendment includes changes to IEEE Std 802.3-2022 to add 10 Mb/s Single-Pair Ethernet point-to-point PHYs to the PHYs supporting the MAC Merge function and the Time Synchronization Service Interface (TSSI).

IEEE Std 802.3cx™-2023

Amendment 6—This amendment to IEEE Std 802.3-2022 modifies Clause 30, Clause 45, and Clause 90, and adds Annex 90A to enhance support for time synchronization protocols by providing options for sub-nanosecond reporting of the transmit and receive path data delays, selection of the data delay measurement point, and dynamic reporting of path data delay variation.

IEEE Std 802.3cz™-2023

Amendment 7—This amendment includes changes to IEEE Std 802.3-2022 and adds Clause 166. This amendment adds 2.5 Gb/s, 5 Gb/s, 10 Gb/s, 25 Gb/s, and 50 Gb/s Physical Layer specifications and management parameters for optical automotive Ethernet using graded-index glass optical fiber.

Two companion documents exist, IEEE Std 802.3.1 and IEEE Std 802.3.2. IEEE Std 802.3.1 describes Ethernet management information base (MIB) modules for use with the Simple Network Management Protocol (SNMP). IEEE Std 802.3.2 describes YANG data models for Ethernet. IEEE Std 802.3.1 and IEEE Std 802.3.2 are updated to add management capability for enhancements to IEEE Std 802.3 after approval of those enhancements.

IEEE Std 802.3 will continue to evolve. New Ethernet capabilities are anticipated to be added within the next few years as amendments to this standard.

Contents

1. Introduction.....	20
1.4 Definitions	20
30. Management.....	21
30.3 Layer management for DTEs.....	21
30.3.2 PHY device managed object class	21
30.3.2.1 PHY device attributes	21
30.3.2.1.2 aPhyType	21
30.3.2.1.3 aPhyTypeList	21
30.5 Layer management for medium attachment units (MAUs)	22
30.5.1 MAU managed object class.....	22
30.5.1.1 MAU attributes	22
30.5.1.1.2 aMAUType	22
44. Introduction to 10 Gb/s baseband networks	23
44.1 Overview.....	23
44.1.1 Scope	23
44.1.3 Relationship of 10 Gigabit Ethernet to the ISO OSI reference model	24
44.1.4 Summary of 10 Gigabit Ethernet sublayers	25
44.1.4.1 Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII).....	25
44.1.4.4 Physical Layer signaling systems	25
44.3 Delay constraints.....	26
44.4 Protocol implementation conformance statement (PICS) proforma.....	26
45. Management Data Input/Output (MDIO) Interface.....	27
45.2 MDIO Interface registers	27
45.2.1 PMA/PMD registers.....	27
45.2.1.6 PMA/PMD control 2 register (Register 1.7)	27
45.2.1.6.3 PMA/PMD type selection (1.7.6:0)	28
45.2.1.60a BASE-AU PMA/PMD extended ability register (Register 1.72).....	28
45.2.1.158a BASE-AU PMA/PMD control register (Register 1.901).....	28
45.2.1.158a.1 Type selection (1.901.3:0)	29
45.2.3 PCS registers	29
45.2.3.88 BASE-U OAM transmit registers (Registers 3.2330 through 3.2338).....	30
45.2.3.88.1 TXO_REQ (3.2330.15).....	31
45.2.3.88.2 TXO_PHYT (3.2330.14)	31
45.2.3.88.3 TXO_MERT (3.2330.13).....	31
45.2.3.88.4 TXO_MSGT (3.2330.12).....	31
45.2.3.88.5 TXO_DATAx (Bits 3.2330.11:0 and registers 3.2331 through 3.2338)	31
45.2.3.89 BASE-U OAM receive registers (Registers 3.2339 through 3.2347)	31
45.2.3.89.1 RXO_VAL (3.2339.15)	32
45.2.3.89.2 RXO_MSGT (3.2339.12)	32
45.2.3.89.3 RXO_DATAx (Bits 3.2339.11:0 and registers 3.2340 through 3.2347)	32
45.2.3.90 BASE-U PCS control register (Register 3.2348)	33
45.2.3.90.1 Operation mode (3.2348.15:13).....	33
45.2.3.90.2 Loopback mode (3.2348.12:10).....	33

45.2.3.90.3	BASE-U OAM advertisement enable (3.2348.1)	33
45.2.3.90.4	EEE advertisement enable (3.2348.0)	34
45.2.3.91	BASE-U PCS status 1 register (Register 3.2349)	34
45.2.3.91.1	Local receiver status (3.2349.15)	35
45.2.3.91.2	Remote receiver status (3.2349.14)	35
45.2.3.91.3	Link status (3.2349.13)	35
45.2.3.91.4	Local PHD reception status (3.2349.12)	35
45.2.3.91.5	Remote PHD reception status (3.2349.11)	35
45.2.3.91.6	PHD lock status (3.2349.10)	35
45.2.3.91.7	Tx Assert LPI received (3.2349.8)	35
45.2.3.91.8	Rx Assert LPI generated (3.2349.7)	36
45.2.3.91.9	Tx LPI indication (3.2349.6)	36
45.2.3.91.10	Rx LPI indication (3.2349.5)	36
45.2.3.91.11	Remote BASE-U OAM advertisement (3.2349.3)	36
45.2.3.91.12	Remote EEE advertisement (3.2349.2)	36
45.2.3.91.13	Local BASE-U OAM ability (3.2349.1)	36
45.2.3.91.14	Local EEE ability (3.2349.0)	36
45.2.3.92	BASE-U PCS status 2 register (Register 3.2350)	36
45.2.3.92.1	Local link margin (3.2350.7:0)	37
45.2.3.93	BASE-U PCS status 3 register (Register 3.2351)	37
45.2.3.93.1	Remote link margin (3.2351.7:0)	37
45.2.3.94	BASE-U PCS status 4 register (Register 3.2352)	37
45.2.3.95	BASE-U PCS status 5 register (Register 3.2353)	38
45.5	Protocol implementation conformance statement (PICS) proforma for Clause 45, Management Data Input/Output (MDIO) Interface	39
45.5.3	PICS proforma tables for the Management Data Input Output (MDIO) Interface	39
45.5.3.2	PMA/PMD MMD options	39
45.5.3.3	PMA/PMD management functions	39
45.5.3.6	PCS options	40
45.5.3.7	PCS management functions	40
78.	Energy-Efficient Ethernet (EEE)	43
78.1	Overview	43
78.1.4	PHY types optionally supporting EEE	43
78.5	Communication link access latency	44
105.	Introduction to 25 Gb/s networks	45
105.1	Overview	45
105.1.1	Scope	45
105.1.2	Relationship of 25 Gigabit Ethernet to the ISO OSI reference model	45
105.1.3	Nomenclature	46
105.2	Physical Layer signaling systems	47
105.3	Summary of 25 Gigabit Ethernet sublayers	47
105.3.2	Physical Coding Sublayer (PCS)	47
105.3.4	Physical Medium Attachment (PMA) sublayer	47
105.3.5	Physical Medium Dependent (PMD) sublayer	48
105.5	Delay constraints	48
105.7	Protocol implementation conformance statement (PICS) proforma	49
125.	Introduction to 2.5 Gb/s and 5 Gb/s networks	50
125.1	Overview	50

125.1.2	Relationship of 2.5 Gigabit and 5 Gigabit Ethernet to the ISO OSI reference model.....	50
125.1.3	Nomenclature	50
125.1.4	Physical Layer signaling systems.....	52
125.2	Summary of 2.5 Gigabit and 5 Gigabit Ethernet sublayers	53
125.2.2	Physical coding sublayer (PCS).....	53
125.2.3	Physical Medium Attachment sublayer (PMA)	53
125.2.3a	Physical Medium Dependent sublayer (PMD).....	53
125.3	Delay Constraints.....	53
131.	Introduction to 50 Gb/s networks	54
131.1	Overview.....	54
131.1.2	Relationship of 50 Gigabit Ethernet to the ISO OSI reference model.....	54
131.1.3	Nomenclature	54
131.1.4	Physical Layer signaling systems.....	55
131.2	Summary of 50 Gigabit Ethernet sublayers	56
131.2.2	Physical Coding Sublayer (PCS).....	56
131.2.4	Physical Medium Attachment (PMA) sublayer	56
131.2.5	Physical Medium Dependent (PMD) sublayer.....	56
131.4	Delay constraints.....	56
166.	Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU	57
166.1	Overview.....	57
166.1.1	Conventions.....	57
166.1.2	Relationship to other Ethernet PHY types	58
166.1.3	Relationship of BASE-AU PHYs to other standards	58
166.1.4	Operation of BASE-AU PHYs.....	59
166.2	Physical Coding Sublayer (PCS)	62
166.2.1	PCS functions.....	62
166.2.2	PCS transmit function	63
166.2.2.1	Physical header data path	63
166.2.2.1.1	Physical header data (PHD) structure	63
166.2.2.1.2	Physical header encoding.....	65
166.2.2.1.3	Physical header CRC16	66
166.2.2.1.4	Physical header splitting function	66
166.2.2.1.5	Physical header three repetition code (TRC)	66
166.2.2.2	Payload data path.....	67
166.2.2.3	PCS transmit ordering	67
166.2.2.4	RS-FEC encoder	67
166.2.2.5	Scrambler.....	69
166.2.2.6	PCS transmit bit order	70
166.2.2.7	PCS 64B/65B encoding.....	74
166.2.2.7.1	Notation conventions	74
166.2.2.7.2	Transmit process	74
166.2.2.7.3	65-bit block structure	74
166.2.2.7.4	Control codes	76
166.2.2.7.5	Idle (/I)	78
166.2.2.7.6	LPI (/LI)	78
166.2.2.7.7	Start (/S)	78
166.2.2.7.8	Terminate (/T).....	78

166.2.2.7.9	Ordered set (/O/)	78
166.2.2.7.10	Error (/E/)	79
166.2.2.8	PCS 64B/65B transmit state diagram parameters	79
166.2.2.8.1	Constants	79
166.2.2.8.2	Variables	79
166.2.2.8.3	Functions	80
166.2.2.9	PCS 64B/65B transmit state diagram	81
166.2.3	PCS receive function	81
166.2.3.1	Descrambler	83
166.2.3.2	RS-FEC decoder	83
166.2.3.3	PCS receiver ordering	83
166.2.3.4	PHD decoding	83
166.2.3.5	Invalid 65-bit blocks	84
166.2.3.6	PCS receive bit order	85
166.2.3.7	PCS 64B/65B receive state diagram parameters	87
166.2.3.7.1	Constants	87
166.2.3.7.2	Variables	87
166.2.3.7.3	Functions	87
166.2.3.7.4	Counters	89
166.2.3.8	PCS 64B/65B receive state diagram	89
166.3	Physical Medium Attachment (PMA) sublayer	91
166.3.1	PMA transmit function	91
166.3.2	PMA receive function	92
166.3.3	Interface to the PMD	92
166.3.4	PHY control	92
166.3.4.1	PHY control state variables	92
166.3.4.2	PHY Tx control state diagram	94
166.3.4.3	PHY Rx control state diagram	95
166.3.4.4	Link monitor state diagram	97
166.3.4.5	PHD monitor state diagrams	98
166.3.4.6	PHY quality monitor	100
166.3.4.6.1	PHY quality criterion	100
166.3.4.6.2	PHY quality assessment	100
166.3.4.6.3	PHY quality monitor state variables	101
166.3.4.6.4	PHY quality monitor state diagram	101
166.4	Energy-Efficient Ethernet (EEE)	103
166.4.1	EEE capability enable	103
166.4.2	Fast wake mode LPI operation	103
166.4.3	LPI transmit operation mode	103
166.4.3.1	LPI refresh codewords	103
166.4.3.2	LPI wake codewords	103
166.4.3.3	Transition of PCS transmit function from normal to LPI operation mode	104
166.4.3.4	Transition of PCS transmit function from LPI to normal operation mode	105
166.4.4	LPI receive operation mode	106
166.5	Test Modes	107
166.5.1	BER test mode	107
166.5.2	SSQWP test pattern	108
166.5.3	FSQWP test pattern	108
166.5.4	SSPR-NRZ test pattern	108
166.5.5	SSPR-PAM4 test pattern	109
166.5.6	Pattern for stressed receiver sensitivity measurement	110
166.6	Physical Medium Dependent (PMD) sublayer	111
166.6.1	PMD service interface	111
166.6.1.1	PMD_COMSIGNAL.request	111

166.6.1.1.1	Semantics of the primitive	111
166.6.1.1.2	When generated.....	111
166.6.1.1.3	Effect of receipt.....	111
166.6.1.2	PMD_COMSIGNAL.indication.....	111
166.6.1.2.1	Semantics of the primitive	111
166.6.1.2.2	When generated.....	112
166.6.1.2.3	Effect of receipt.....	112
166.6.1.3	PMD_RXDETECT.indication.....	112
166.6.1.3.1	Semantics of the service primitive	112
166.6.1.3.2	When generated.....	112
166.6.1.3.3	Effect of receipt.....	112
166.6.2	PMD functional specifications	112
166.6.2.1	PMD block diagram.....	112
166.6.2.2	PMD transmit function.	113
166.6.2.3	PMD receive function.....	113
166.6.2.4	PMD signal detect function	113
166.6.3	PMD to MDI optical specifications	114
166.6.3.1	Operating range	114
166.6.3.2	BASE-AU transmitter optical characteristics.....	114
166.6.3.3	BASE-AU receiver optical characteristics	116
166.6.3.4	BASE-AU illustrative link power budget.....	117
166.6.4	Definition of optical parameters and measurement methods	118
166.6.4.1	Test patterns for optical parameters.....	118
166.6.4.2	Center wavelength and RMS spectral width	119
166.6.4.3	Average optical power.....	119
166.6.4.4	Outer Optical Modulation Amplitude (OMA_{outer}).....	119
166.6.4.4.1	OMA_{outer} test setup.....	119
166.6.4.4.2	OMA_{outer} measurement	120
166.6.4.5	Extinction ratio	120
166.6.4.6	Relative intensity noise ($RIN_{12}OMA$).....	120
166.6.4.7	Uncorrelated random jitter.....	121
166.6.4.8	Transmitter distortion figure of merit (TDFOM)	121
166.6.4.8.1	TDFOM conformance test setup.....	121
166.6.4.8.2	TDFOM measurement method	122
166.6.4.8.3	TDFOM reference equalizer	123
166.6.4.8.4	BER calculation	125
166.6.4.8.5	OMA calculation at equalization output	126
166.6.4.8.6	TDFOM calculation	126
166.6.4.9	OMA_{outer} calculation.....	127
166.6.4.10	Extinction ratio calculation.....	127
166.6.4.11	Average optical power calculation	127
166.6.4.12	Receiver sensitivity.....	127
166.6.4.13	Stressed receiver sensitivity.....	128
166.6.4.13.1	Stressed receiver conformance test block diagram.....	128
166.6.4.13.2	Stressed receiver conformance test signal characteristics and calibration.	130
166.6.4.13.3	Stressed receiver conformance test signal verification	131
166.6.4.13.4	Sinusoidal jitter for receiver conformance test	131
166.6.5	Fiber optic cabling model.....	132
166.6.6	Characteristics of the fiber optic cabling (channel)	132
166.6.6.1	Optical fiber cable	132
166.6.6.2	Optical fiber connection	133
166.6.6.2.1	Optical fiber connection insertion loss.....	133
166.6.6.2.2	Maximum discrete reflectance	133
166.6.7	Medium Dependent Interface (MDI)	134

166.7	BASE-U operations, administration, and maintenance (BASE-U OAM) channel	134
166.8	Loopback modes	134
166.9	Management interface.....	135
166.10	Environmental specifications.....	136
166.10.1	Temperature classes	136
166.10.2	General safety.....	136
166.10.3	Environmental safety.....	136
166.10.4	Electromagnetic compatibility	137
166.10.5	Optical safety.....	137
166.11	Delay constraints.....	137
166.12	Protocol implementation conformance statement (PICS) proforma for Clause 166, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50BASE-AU	139
166.12.1	Introduction	139
166.12.2	Identification	139
166.12.2.1	Implementation identification.....	139
166.12.2.2	Protocol summary	139
166.12.3	Major capabilities/options	140
166.12.4	General	140
166.12.5	Physical Coding Sublayer (PCS).....	141
166.12.6	Physical Medium Attachment (PMA).....	145
166.12.7	Energy-Efficient Ethernet (EEE)	146
166.12.8	Test modes.....	147
166.12.9	Physical Medium Dependent (PMD)	148
166.12.10	Optical measurement methods	148
166.12.11	Characteristics of the fiber optic cabling (channel)	149
166.12.12	BASE-U operations, administration, and maintenance (BASE-U OAM) channel.....	150
166.12.13	Loopback modes	150
166.12.14	Management Interface.....	151
166.12.15	Environmental specifications	151
166.12.16	Delay constraints	151
Annex 166A	(informative) BASE-U scrambler sequence.....	152
166A.1	Introduction.....	152
166A.2	2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U scrambler sequence	152
166A.3	50GBASE-U scrambler sequence.....	153
Annex 166B	(informative) RS-FEC(544,522) codeword example	155
166B.1	Introduction.....	155
166B.2	Output of the RS-FEC(544,522) encoder	155

IEEE Standard for Ethernet

Amendment 7: Physical Layer Specifications and Management Parameters for Multi-Gigabit Glass Optical Fiber Automotive Ethernet

(This amendment is based on IEEE Std 802.3™-2022 as amended by IEEE Std 802.3dd™-2022, IEEE Std 802.3cs™-2022, IEEE Std 802.3db™-2022, IEEE Std 802.3ck™-2022, IEEE Std 802.3de™-2022, and IEEE Std 802.3cx™-2023.)

NOTE—The editing instructions contained in this amendment define how to merge the material contained therein into the existing base standard and its amendments to form the comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: change, delete, insert, and replace. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed by using ~~striking through~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. Deletions and insertions may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.

Cross references that refer to clauses, tables, equations, or figures not covered by this amendment are highlighted in green.⁶

⁶ Notes in text, tables, and figures are given for information only and do not contain requirements needed to implement the standard.

1. Introduction

1.4 Definitions

Insert a new definition for “10GBASE-AU” after 1.4.62 “10BROAD36” as follows:

1.4.62a 10GBASE-AU: IEEE 802.3 Physical Layer specification for 10 Gb/s full duplex Ethernet operation over multimode glass optical fiber in an automotive environment. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “2.5GBASE-AU” after 1.4.95 “1G-EPON” as follows:

1.4.95a 2.5GBASE-AU: IEEE 802.3 Physical Layer specification for 2.5 Gb/s full duplex Ethernet operation over multimode glass optical fiber in an automotive environment. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “25GBASE-AU” after 1.4.116 “25GBASE” as follows:

1.4.116a 25GBASE-AU: IEEE 802.3 Physical Layer specification for 25 Gb/s full duplex Ethernet operation over multimode glass optical fiber in an automotive environment. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “50GBASE-AU” after 1.4.165 “50/50G-EPON” as follows:

1.4.165a 50GBASE-AU: IEEE 802.3 Physical Layer specification for 50 Gb/s full duplex Ethernet operation over multimode glass optical fiber in an automotive environment. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “5GBASE-AU” after 1.4.178 “50G-EPON” as follows:

1.4.178a 5GBASE-AU: IEEE 802.3 Physical Layer specification for 5 Gb/s full duplex Ethernet operation over multimode glass optical fiber in an automotive environment. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “BASE-AU” after 1.4.204 “Base Page” as follows:

1.4.204a BASE-AU: The set of PHYs that use a BASE-U PCS and PMA with PMD specifications for operation over multimode glass optical fiber in an automotive environment, including 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU. (See IEEE Std 802.3, Clause 166.)

Insert a new definition for “BASE-U” after 1.4.206 “BASE-T1” as follows:

1.4.206a BASE-U: IEEE 802.3 PCS and PMA sublayer specifications used by a family of Physical Layer devices. (See IEEE Std 802.3, Clause 166.)

Change 1.4.464 as follows:

1.4.464 physical header data (PHD): ~~Side An~~ information block embedded inside a Transmit Block that is used to exchange control information and for negotiation of PCS and PMA parameters between two link partners. (See IEEE Std 802.3, Clause 115 and Clause 166.)

30. Management

30.3 Layer management for DTEs

30.3.2 PHY device managed object class

30.3.2.1 PHY device attributes

30.3.2.1.2 aPhyType

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “1000BASE-X” as follows:

2.5GBASE-AU Clause 166 2.5 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “2.5GBASE-X” as follows:

5GBASE-AU Clause 166 5 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “10/2.5GBASE-SP” (inserted by IEEE Std 802.3cs-2022) as follows:

10GBASE-AU Clause 166 10 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “25/10GBASE-PQ” as follows:

25GBASE-AU Clause 166 25 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.2 after the entry for “50/25GBASE-PQ” as follows:

50GBASE-AU Clause 166 50 Gb/s PAM4

30.3.2.1.3 aPhyTypeList

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “1000BASE-X” as follows:

2.5GBASE-AU Clause 166 2.5 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “2.5GBASE-X” as follows:

5GBASE-AU Clause 166 5 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “10/2.5GBASE-SP” (inserted by IEEE Std 802.3cs-2022) as follows:

10GBASE-AU Clause 166 10 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “25/10GBASE-PQ” as follows:

25GBASE-AU Clause 166 25 Gb/s NRZ

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.3.2.1.3 after the entry for “50/25GBASE-PQ” as follows:

50GBASE-AU Clause 166 50 Gb/s PAM4

30.5 Layer management for medium attachment units (MAUs)

30.5.1 MAU managed object class

30.5.1.1 MAU attributes

30.5.1.1.2 aMAUType

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “1000BASE-XHD” as follows:

2.5GBASE-AU Glass optical fiber PHY as specified in Clause 166

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “2.5GBASE-X” as follows:

5GBASE-AU Glass optical fiber PHY as specified in Clause 166

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “10/2.5GBASE-SPI-Uxy” (inserted by IEEE Std 802.3cs-2022) as follows:

10GBASE-AU Glass optical fiber PHY as specified in Clause 166

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “25/10GBASE-PQX-U3” as follows:

25GBASE-AU Glass optical fiber PHY as specified in Clause 166

Insert the following new entry in the APPROPRIATE SYNTAX section of 30.5.1.1.2 after the entry for “50/25GBASE-PQX-U3” as follows:

50GBASE-AU Glass optical fiber PHY as specified in Clause 166

44. Introduction to 10 Gb/s baseband networks

44.1 Overview

44.1.1 Scope

Change the first paragraph of 44.1.1 as follows:

10 Gigabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 10 Gb/s, coupled with any IEEE 802.3 10GBASE Physical Layer implementation, ~~connected through a 10-Gigabit Media Independent Interface (XGMII) to Physical Layer devices (PHYs) such as 10GBASE-SR, 10GBASE-LX4, 10GBASE-CX4, 10GBASE-LRM, 10GBASE-LR, 10GBASE-ER, 10GBASE-SW, 10GBASE-LW, 10GBASE-EW, 10GBASE-T, and 10GBASE-T1.~~

44.1.3 Relationship of 10 Gigabit Ethernet to the ISO OSI reference model

Replace Figure 44–1 with the following figure, which adds 10GBASE-AU:

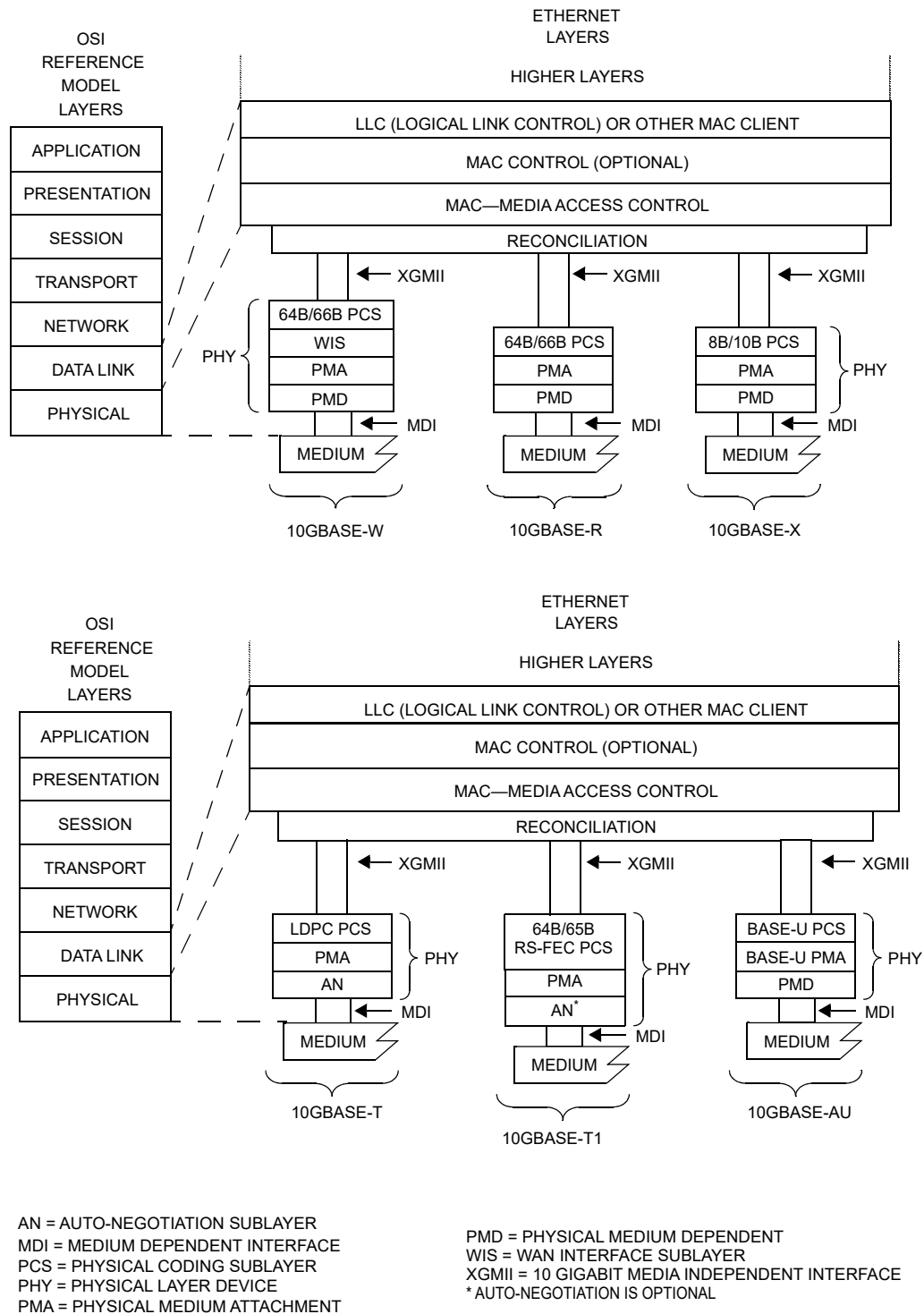


Figure 44–1—Architectural positioning of 10 Gigabit Ethernet

Change item d) of 44.1.3 as follows:

- d) The MDI as specified in [Clause 53](#) for 10GBASE-LX4, in [Clause 54](#) for 10GBASE-CX4, in [Clause 55](#) for 10GBASE-T, in [Clause 68](#) for 10GBASE-LRM, in [Clause 149](#) for 10GBASE-T1, in [Clause 166](#) for 10GBASE-AU, and in [Clause 52](#) for other PMD types.

44.1.4 Summary of 10 Gigabit Ethernet sublayers

44.1.4.1 Reconciliation Sublayer (RS) and 10 Gigabit Media Independent Interface (XGMII)

Change the second paragraph of 44.1.4.1 as follows:

While the XGMII is an optional interface, it is used extensively in this standard as a basis for functional specification and provides a common service interface for [Clause 47](#), [Clause 48](#), [Clause 49](#), [Clause 55](#), [and Clause 149](#), and [Clause 166](#).

44.1.4.4 Physical Layer signaling systems

Change Table 44–1 as follows:

Table 44–1—Nomenclature and clause correlation

Nomenclature	Clause ^a												
	48	49	50	51	52			53	54	55	68	149	166
	8B/10B PCS & PMA	64B/66B PCS	WIS	Serial PMA	850 nm Serial PMD	1310 nm Serial PMD	1550 nm Serial PMD	1310 nm WDM PMD	4-Lane electrical PMD	LDPC PCS & 4-pair PMA	1310 nm Serial MMF PMD	RS-FEC PCS & 1-pair PMA	10GBASE-AU PCS/PMA/PMD
10GBASE-CX4	M								M				
10GBASE-T1												M	
10GBASE-T										M			
10GBASE-AU													M
10GBASE-LRM		M		M							M		
10GBASE-SR		M		M	M								
10GBASE-SW		M	M	M	M								
10GBASE-LX4	M							M					
10GBASE-LR		M		M		M							
10GBASE-LW		M	M	M		M							
10GBASE-ER		M		M			M						
10GBASE-EW		M	M	M			M						

^a M = Mandatory

Insert a new paragraph after the seventh paragraph of 44.1.4.4, which begins “The term 10GBASE-T1, ...”, as follows:

The term 10GBASE-AU, specified in Clause 166, refers to a specific Physical Layer implementation based upon 64B/65B data coding method placed in RS-FEC codewords that are transmitted using NRZ modulation on glass optical fiber in an automotive environment.

Change former paragraph 8 (now paragraph 9 due to the above insertion) in 44.1.4.4 as follows:

Physical Layer device specifications are contained in [Clause 52](#), [Clause 53](#), [Clause 54](#), [Clause 55](#), [Clause 68](#), ~~and [Clause 149](#)~~, and [Clause 166](#).

44.3 Delay constraints

Insert a new row at the end of Table 44–2 as follows (unchanged rows not shown):

Table 44–2—Round-trip delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Notes ^a
...			
10GBASE-AU	11 264	22	Does not include delay of medium. See 166.11.

^a Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

44.4 Protocol implementation conformance statement (PICS) proforma

Change the first paragraph of 44.4 as follows:

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 45 through [Clause 55](#), [Clause 68](#), ~~and [Clause 149](#)~~, and [Clause 166](#), demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

Change the rows for register addresses 1.72 to 1.79 and 1.901 to 1.999 in Table 45–3 as follows (unchanged rows not shown):

Table 45–3—PMA/PMD registers

Register address	Register name	Subclause
...		
<u>1.72</u>	<u>BASE-AU PMA/PMD extended ability</u>	<u>45.2.1.60a</u>
1.72 1.73 through 1.79	Reserved	
...		
<u>1.901</u>	<u>BASE-AU PMA/PMD control</u>	<u>45.2.1.158a</u>
1.901 1.902 through 1.999	Reserved	
...		

45.2.1.6 PMA/PMD control 2 register (Register 1.7)

Change the description for bits 1.7.6:0 in Table 45–7 as follows (unchanged table rows and bit description lines not shown):

Table 45–7—PMA/PMD control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.7.6:0	PMA/PMD type selection	6 5 4 3 2 1 0 ... 0 1 1 1 1 1 x = reserved 0 1 1 1 1 1 1 = reserved <u>0 1 1 1 1 1 0 = BASE-AU PMA/PMD^d</u> ...	R/W
...			

^aR/W = Read/Write, RO = Read only

^bIf BASE-T1 is selected, bits 1.2100.3:0 are used to differentiate which BASE-T1 PMA/PMD is selected.

^cIf BASE-H PMA/PMD is selected, register 1.900 is used to differentiate which BASE-H PMA/PMD is selected.

^dIf BASE-AU PMA/PMD is selected, register 1.901 is used to differentiate which BASE-AU PMA/PMD is selected.

45.2.1.6.3 PMA/PMD type selection (1.7.6:0)

Change the first paragraph of 45.2.1.6.3 as follows:

The PMA/PMD type of the PMA/PMD shall be selected using bits 6 to 0. The PMA/PMD type abilities of the PMA/PMD are advertised in bits 9 and 7 through 0 of the PMA/PMD status 2 register; the PMA/PMD extended ability register; the 40G/100G PMA/PMD extended ability register; the 50G PMA/PMD extended ability register; the 200G PMA/PMD extended ability register; ~~and the 400G PMA/PMD extended ability register;~~ and the BASE-AU PMA/PMD extended ability register. A PMA/PMD shall ignore writes to the PMA/PMD type selection bits that select PMA/PMD types it has not advertised. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY.

Insert new subclause 45.2.1.60a, including Table 45–58a, after 45.2.1.60 as follows:

45.2.1.60a BASE-AU PMA/PMD extended ability register (Register 1.72)

The assignment of bits in the BASE-AU PMA/PMD extended ability register is shown in Table 45–58a.

Table 45–58a—BASE-AU PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.72.15:5	Reserved	Value always 0	RO
1.72.4	50GBASE-AU ability	1 = PMA/PMD is able to perform 50GBASE-AU 0 = PMA/PMD is not able to perform 50GBASE-AU	RO
1.72.3	25GBASE-AU ability	1 = PMA/PMD is able to perform 25GBASE-AU 0 = PMA/PMD is not able to perform 25GBASE-AU	RO
1.72.2	10GBASE-AU ability	1 = PMA/PMD is able to perform 10GBASE-AU 0 = PMA/PMD is not able to perform 10GBASE-AU	RO
1.72.1	5GBASE-AU ability	1 = PMA/PMD is able to perform 5GBASE-AU 0 = PMA/PMD is not able to perform 5GBASE-AU	RO
1.72.0	2.5GBASE-AU ability	1 = PMA/PMD is able to perform 2.5GBASE-AU 0 = PMA/PMD is not able to perform 2.5GBASE-AU	RO

^aRO = Read only

All of the bits in the BASE-AU PMA/PMD extended ability register are read only; a write to the BASE-AU PMA/PMD extended ability register shall have no effect.

Insert new subclause 45.2.1.158a, including Table 45–127a, after 45.2.1.158 as follows:

45.2.1.158a BASE-AU PMA/PMD control register (Register 1.901)

The assignment of bits in the BASE-AU PMA/PMD control register is shown in Table 45–127a.

Table 45–127a—BASE-AU PMA/PMD control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.901.15:4	Reserved	Value always 0	RO
1.901.3:0	Type selection	3 2 1 0 1 x x x = Reserved 0 1 1 x = Reserved 0 1 0 1 = Reserved 0 1 0 0 = 50GBASE-AU 0 0 1 1 = 25GBASE-AU 0 0 1 0 = 10GBASE-AU 0 0 0 1 = 5GBASE-AU 0 0 0 0 = 2.5GBASE-AU	R/W

^aRO = Read only, R/W = Read/Write

45.2.1.158a.1 Type selection (1.901.3:0)

Bits 1.901.3:0 shall select only one of the modes of operation specified in Table 45–127a and advertised by register 1.72.

When these bits are set to binary 0000, the mode of operation is 2.5GBASE-AU. When these bits are set to binary 0001, the mode of operation is 5GBASE-AU. When these bits are set to binary 0010, the mode of operation is 10GBASE-AU. When these bits are set to binary 0011, the mode of operation is 25GBASE-AU. When these bits are set to binary 0100, the mode of operation is 50GBASE-AU.

Writes of reserved values or modes of operation not advertised by register 1.72, to bits 1.901.3:0 shall have no effect.

45.2.3 PCS registers

Change the rows for register addresses 3.2325 to 3.32767 in Table 45–233 as follows (unchanged rows not shown):

Table 45–233—PCS registers

Register address	Register name	Subclause
...		
3.2325 through 3.32767	Reserved	
<u>3.2330</u>	<u>BASE-U OAM transmit control</u>	<u>45.2.3.88</u>
<u>3.2331 through 3.2338</u>	<u>BASE-U OAM transmit message</u>	<u>45.2.3.88</u>
<u>3.2339</u>	<u>BASE-U OAM receive control</u>	<u>45.2.3.89</u>
<u>3.2340 through 3.2347</u>	<u>BASE-U OAM receive message</u>	<u>45.2.3.89</u>
<u>3.2348</u>	<u>BASE-U PCS control</u>	<u>45.2.3.90</u>
<u>3.2349</u>	<u>BASE-U PCS status 1</u>	<u>45.2.3.91</u>
<u>3.2350</u>	<u>BASE-U PCS status 2</u>	<u>45.2.3.92</u>
<u>3.2351</u>	<u>BASE-U PCS status 3</u>	<u>45.2.3.93</u>

Table 45–233—PCS registers (*continued*)

Register address	Register name	Subclause
<u>3.2352</u>	<u>BASE-U PCS status 4</u>	<u>45.2.3.94</u>
<u>3.2353</u>	<u>BASE-U PCS status 5</u>	<u>45.2.3.95</u>
<u>3.2354 through 3.32767</u>	<u>Reserved</u>	
...		

Insert 45.2.3.88 through 45.2.3.95 after 45.2.3.87 as follows:

45.2.3.88 BASE-U OAM transmit registers (Registers 3.2330 through 3.2338)

Registers 3.2330 through 3.2338 are used to transmit messages over the BASE-U OAM channel established between two link partners (see 166.7). The BASE-U OAM message is part of the physical header data (PHD) defined for BASE-U (see Table 166–2).

The transmit registers are used to compose a BASE-U OAM message and to control its transmission to the link partner. Register 3.2330 contains control, status, identification, and the beginning of a BASE-U OAM message. Eight additional registers (3.2331 through 3.2338) hold the remainder of the BASE-U OAM message.

The assignment of bits in the BASE-U OAM transmit registers is shown in Table 45–313a.

Table 45–313a—BASE-U OAM transmit register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2330.15	TXO_REQ	1 = Transmission of a BASE-U OAM message is pending; write as 1 to request transmission 0 = BASE-U transmit registers are available for a new BASE-U OAM message	R/W, SC
3.2330.14	TXO_PHYT	The identifier of the last BASE-U OAM message received by the remote PHY	RO
3.2330.13	TXO_MERT	The identifier of the last BASE-U OAM message read by the STA attached to the remote PHY	RO
3.2330.12	TXO_MSGT	BASE-U OAM message identifier; its value changes with each new transmitted message	RO
3.2330.11:0	TXO_DATA0	Transmit BASE-U OAM message 12-bit data word	R/W
3.2331.15:0	TXO_DATA1	Transmit BASE-U OAM message 16-bit data word 1	R/W
3.2332.15:0	TXO_DATA2	Transmit BASE-U OAM message 16-bit data word 2	R/W
3.2333.15:0	TXO_DATA3	Transmit BASE-U OAM message 16-bit data word 3	R/W
3.2334.15:0	TXO_DATA4	Transmit BASE-U OAM message 16-bit data word 4	R/W
3.2335.15:0	TXO_DATA5	Transmit BASE-U OAM message 16-bit data word 5	R/W
3.2336.15:0	TXO_DATA6	Transmit BASE-U OAM message 16-bit data word 6	R/W

Table 45–313a—BASE-U OAM transmit register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
3.2337.15:0	TXO_DATA7	Transmit BASE-U OAM message 16-bit data word 7	R/W
3.2338.15:0	TXO_DATA8	Transmit BASE-U OAM message 16-bit data word 8	R/W

^aR/W = Read/Write, RO = Read only, SC = Self-clearing

45.2.3.88.1 TXO_REQ (3.2330.15)

Bit 3.2330.15 set to one requests the BASE-U based PHY to transmit the BASE-U OAM message written to registers 3.2330 through 3.2338. Bit 3.2330.15 is set to zero by the BASE-U based PHY to indicate that the BASE-U OAM message has been accepted for transmission, and that the BASE-U OAM transmit message registers are free to accept a new BASE-U OAM message.

Bit 3.2330.15, together with bits 3.2330.14 (TXO_PHYT), 3.2330.13 (TXO_MERT), and 3.2330.12 (TXO_MSGT), indicates the status of the BASE-U OAM transmission channel (see 166.7).

45.2.3.88.2 TXO_PHYT (3.2330.14)

Bit 3.2330.14 reflects the value of the bit TXO_MSGT in the last BASE-U OAM message received by the remote BASE-U based PHY.

45.2.3.88.3 TXO_MERT (3.2330.13)

Bit 3.2330.13 reflects the value of the bit TXO_MSGT in the last BASE-U OAM message read by the STA attached to the remote BASE-U based PHY.

45.2.3.88.4 TXO_MSGT (3.2330.12)

Bit 3.2330.12 is used for BASE-U OAM message identification. Bit 3.2330.12 is changed by the BASE-U based PHY when it accepts a new BASE-U OAM message for transmission, acting as a one bit sequence number. Simultaneously with the bit 3.2330.12 change, bit 3.2330.15 is set to zero.

45.2.3.88.5 TXO_DATAx (Bits 3.2330.11:0 and registers 3.2331 through 3.2338)

Bits 3.2330.11:0 (TXO_DATA0) contain the first 12 bits of a BASE-U OAM message, and registers 3.2331 through 3.2338 (TXO_DATA1 through TXO_DATA8) contain the remaining 128 bits of that BASE-U OAM message.

TXO_DATA0 and TXO_DATA1 shall be a 28-bit BASE-U OAM protocol identifier composed of a 24-bit Organizationally Unique Identifier (OUI) or Company ID (CID) followed by a 4-bit protocol number; where TXO_DATA0[11:0] = OUI[23:12], TXO_DATA1[15:4] = OUI[11:0], and TXO_DATA1[3:0] = protocol number. The protocol number and the content of TXO_DATA2 through TXO_DATA8 shall be vendor specific to the assignee of the OUI or CID.

45.2.3.89 BASE-U OAM receive registers (Registers 3.2339 through 3.2347)

Registers 3.2339 through 3.2347 store messages received over the BASE-U OAM channel established between two link partners (see 166.7). Register 3.2339 contains status information, identification, and the first 12 bits of a BASE-U OAM message. Registers 3.2340 through 3.2347 contain the following 128 bits of the received BASE-U OAM message.

The assignment of bits in the BASE-U OAM receive registers is shown in Table 45–313b.

Table 45–313b—BASE-U OAM receive register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2339.15	RXO_VAL	1 = A new BASE-U OAM message has arrived 0 = No new BASE-U OAM message arrived since either the last message was read or PMA reset	RO
3.2339.14:13	Reserved	Value always 0	RO
3.2339.12	RXO_MSGT	Identifier of the received BASE-U OAM message	RO
3.2339.11:0	RXO_DATA0	Receive BASE-U OAM message first 12 bits	RO
3.2340.15:0	RXO_DATA1	Receive BASE-U OAM message 16-bit data word 1	RO
3.2341.15:0	RXO_DATA2	Receive BASE-U OAM message 16-bit data word 2	RO
3.2342.15:0	RXO_DATA3	Receive BASE-U OAM message 16-bit data word 3	RO
3.2343.15:0	RXO_DATA4	Receive BASE-U OAM message 16-bit data word 4	RO
3.2344.15:0	RXO_DATA5	Receive BASE-U OAM message 16-bit data word 5	RO
3.2345.15:0	RXO_DATA6	Receive BASE-U OAM message 16-bit data word 6	RO
3.2346.15:0	RXO_DATA7	Receive BASE-U OAM message 16-bit data word 7	RO
3.2347.15:0	RXO_DATA8	Receive BASE-U OAM message 16-bit data word 8	RO

^aRO = Read only

45.2.3.89.1 RXO_VAL (3.2339.15)

The BASE-U based PHY sets bit 3.2339.15 to one to indicate the reception of a new BASE-U OAM message. The BASE-U based PHY sets bit 3.2339.15 to zero when the last register (3.2347) containing the BASE-U OAM message is read (see [Figure 115–43](#)). The BASE-U based PHY does not update the receive message registers until the message is read by the STA as indicated by a read of register 3.2347.

45.2.3.89.2 RXO_MSGT (3.2339.12)

Bit 3.2339.12 is used for message identification. Bit 3.2339.12 changes with every new received BASE-U OAM message, acting as a one bit sequence number.

45.2.3.89.3 RXO_DATAx (Bits 3.2339.11:0 and registers 3.2340 through 3.2347)

Bits 3.2339.11:0 (RXO_DATA0) contain the first 12 bits of a received BASE-U OAM message, and registers 3.2340 through 3.2347 (RXO_DATA1 through RXO_DATA8) contain the remaining 128 data bits of that received BASE-U OAM message.

45.2.3.90 BASE-U PCS control register (Register 3.2348)

The assignment of bits in the BASE-U PCS control register is shown in Table 45–313c. The default value for each bit of the BASE-U PCS control register is chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

Table 45–313c—BASE-U PCS control register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2348.15:13	Operation mode	15 14 13 0 0 0 = Normal operation 0 0 1 = BER test mode 0 1 0 = SSQWP test pattern 0 1 1 = FSQWP test pattern 1 0 0 = SSPR-NRZ test pattern 1 0 1 = SSPR-PAM4 test pattern 1 1 x = Reserved	R/W
3.2348.12:10	Loopback mode	12 11 10 0 0 0 = No loopback 0 0 1 = xMII level loopback 0 1 0 = PMD interface level loopback 0 1 1 = Line loopback 1 x x = Reserved	R/W
3.2348.9:2	Reserved	Value always 0	RO
3.2348.1	BASE-U OAM advertisement enable	1 = Enable advertisement of BASE-U OAM ability 0 = Disable advertisement of BASE-U OAM ability	R/W
3.2348.0	EEE advertisement enable	1 = Enable advertisement of EEE ability 0 = Disable advertisement of EEE ability	R/W

^aR/W = Read/Write, RO = Read only

45.2.3.90.1 Operation mode (3.2348.15:13)

Bits 3.2348.15:13 shall select one of the BASE-U PCS test mode patterns defined in 166.5 and mapped per Table 45–313c. Bits 3.2348.15:13 shall have a default value of binary 000, selecting normal BASE-U PCS operation. Changes in the operation mode value shall only take effect after a PMA reset (see 166.3.4.1).

45.2.3.90.2 Loopback mode (3.2348.12:10)

Bits 3.2348.12:10 shall select one of the loopback modes defined in 166.8 and mapped per Table 45–313c. Bits 3.2348.12:10 shall have a default value of binary 000 (no loopback). Loopback modes are only available when a BASE-U PCS is in the normal operation mode (binary 000 is selected in 3.2348.15:13). Changes in the loopback mode value shall only take effect after a PMA reset (see 166.3.4.1).

45.2.3.90.3 BASE-U OAM advertisement enable (3.2348.1)

Bit 3.2348.1 controls establishing the BASE-U OAM channel with the link partner (see 166.7). If the BASE-U PCS does not have BASE-U OAM ability (bit 3.2349.1 = 0, see 45.2.3.91.13) setting bit 3.2348.1 shall have no effect. Setting bit 3.2348.1 to zero shall prevent establishment of a BASE-U OAM channel with the link partner. Changes in the BASE-U OAM advertisement enable value shall only take effect after a PMA reset (see 166.3.4.1). Bit 3.2348.1 has no specified default value.

45.2.3.90.4 EEE advertisement enable (3.2348.0)

Setting bit 3.2348.0 to one shall enable the advertisement of local PHY EEE ability (see 166.4). Setting bit 3.2348.0 to zero shall prevent establishment of EEE operation with the link partner. If the BASE-U PHY does not have EEE ability (bit 3.2349.0 = 0, see 45.2.3.91.14) setting bit 3.2348.0 has no effect. Changes in the EEE advertisement enable value shall only take effect after a PMA reset (see 166.3.4.1). Bit 3.2348.0 has no specified default value.

45.2.3.91 BASE-U PCS status 1 register (Register 3.2349)

The assignment of bits in the BASE-U PCS status 1 register is shown in Table 45–313d.

Table 45–313d— BASE-U PCS status 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2349.15	Local receiver status	1 = The local receive operation is reliable 0 = The local receive operation is unreliable	RO
3.2349.14	Remote receiver status	1 = The receive operation of the remote PHY is reliable 0 = The receive operation of the remote PHY is unreliable	RO
3.2349.13	Link status	1 = A bidirectional reliable link is established 0 = A bidirectional reliable link is not established	RO, LL
3.2349.12	Local PHD reception status	1 = PHD reception is reliable for the local PHY 0 = PHD reception is unreliable for the local PHY	RO
3.2349.11	Remote PHD reception status	1 = PHD reception is reliable for the remote PHY 0 = PHD reception is unreliable for the remote PHY	RO
3.2349.10	PHD lock status	1 = PHD transmission and reception are reliable 0 = PHD transmission or reception are unreliable	RO, LL
3.2349.9	Reserved	Value always 0	RO
3.2349.8	Tx Assert LPI received	1 = Tx PCS has received LPI 0 = LPI not received by Tx PCS	RO, LH
3.2349.7	Rx Assert LPI received	1 = Rx PCS has received LPI 0 = LPI not received by Rx PCS	RO, LH
3.2349.6	Tx LPI indication	1 = Tx PCS is currently transmitting LPI 0 = Tx PCS is not currently transmitting LPI	RO
3.2349.5	Rx LPI indication	1 = Rx PCS is currently transmitting LPI 0 = Rx PCS is not currently transmitting LPI	RO
3.2349.4	Reserved	Value always 0	RO
3.2349.3	Remote BASE-U OAM advertisement	1 = The remote PHY has BASE-U OAM ability and it is advertised 0 = The remote PHY does not have BASE-U OAM ability or it is not advertised	RO
3.2349.2	Remote EEE advertisement	1 = The remote PHY has EEE ability and it is advertised 0 = The remote PHY does not have EEE ability or it is not advertised	RO

Table 45–313d— BASE-U PCS status 1 register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
3.2349.1	Local BASE-U OAM ability	1 = The PHY has BASE-U OAM ability 0 = The PHY does not have BASE-U OAM ability	RO
3.2349.0	Local EEE ability	1 = The PHY has EEE ability 0 = The PHY does not have EEE ability	RO

^aRO = Read only, LH = Latching high, LL = Latching low

45.2.3.91.1 Local receiver status (3.2349.15)

Bit 3.2349.15 reflects the value of the state variable `loc_rcvr_status` as determined by the PHY quality monitor state diagram (see 166.3.4.6.4).

45.2.3.91.2 Remote receiver status (3.2349.14)

Bit 3.2349.14 reflects the value of the state variable `rem_rcvr_status` as determined by the link monitor state diagram (see 166.3.4.4).

45.2.3.91.3 Link status (3.2349.13)

Bit 3.2349.13 reflects the value of the state variable `link_status` as determined by the link monitor state diagram (see 166.3.4.4). Bit 3.2349.13 shall have latching low behavior. Bit 1.1.2, bit 3.1.2, and bit 3.2349.13 are identical for BASE-U; a read to any of these three bits shall release the latch for all the bits.

45.2.3.91.4 Local PHD reception status (3.2349.12)

Bit 3.2349.12 reflects the value of the state variable `loc_rcvr_hdr_lock` as determined by the local PHD reception monitor state diagram (see 166.3.4.5).

45.2.3.91.5 Remote PHD reception status (3.2349.11)

Bit 3.2349.11 reflects the value of the state variable `rem_rcvr_hdr_lock` as determined by the remote PHD reception monitor state diagram (see 166.3.4.5).

45.2.3.91.6 PHD lock status (3.2349.10)

Bit 3.2349.10 reflects the value of the state variable `rcvr_hdr_lock` as determined by the PHD monitor state diagram (see 166.3.4.5). Bit 3.2349.10 shall have latching low behavior.

45.2.3.91.7 Tx Assert LPI received (3.2349.8)

When read as a one, bit 3.2349.8 indicates that the transmit BASE-U PCS has received Assert LPI signaling from the transmit RS one or more times since the register was last read. When read as a zero, bit 3.2349.8 indicates that the BASE-U PCS transmitter has not received Assert LPI. Bit 3.2349.8 shall have latching high behavior. Bit 3.2349.8 is a copy of bit 3.1.11; a read of either bit shall release the latch for both of the bits.

45.2.3.91.8 Rx Assert LPI generated (3.2349.7)

When read as a one, bit 3.2349.7 indicates that the receive BASE-U PCS has generated Assert LPI to the receive RS one or more times since the register was last read. When read as a zero, bit 3.2349.7 indicates that the BASE-U PCS receiver has not generated Assert LPI to the receive RS. Bit 3.2349.7 shall have latching high behavior. Bit 3.2349.7 is a copy of bit 3.1.10; a read of either bit shall release the latch for all the bits.

45.2.3.91.9 Tx LPI indication (3.2349.6)

Bit 3.2349.6 indicates the current LPI mode status of the local BASE-U based PHY transmit path. When read as a one, the PHY transmitter is currently in LPI mode (transmitting refresh). When read as a zero, the PHY transmitter is not in LPI mode (see 166.4). Bit 3.2349.6 is a copy of bit 3.1.9.

45.2.3.91.10 Rx LPI indication (3.2349.5)

Bit 3.2349.5 indicates the current LPI mode status of the local BASE-U based PHY receive path. When read as a one, the PHY is receiving LPI mode signals (refresh). When read as a zero, the PHY is not receiving LPI mode signals (see 166.4). Bit 3.2349.5 is a copy of bit 3.1.8.

45.2.3.91.11 Remote BASE-U OAM advertisement (3.2349.3)

Bit 3.2349.3 indicates the BASE-U OAM ability of the remote PHY received in the PHD field PHD.CAP.OAM (see Table 166–2). When read as a one, bit 3.2349.3 indicates that the remote PHY has the BASE-U OAM ability and that BASE-U OAM advertisement is enabled. When read as a zero, bit 3.2349.3 indicates either that the remote PHY does not have BASE-U OAM ability or that BASE-U OAM advertisement is disabled.

45.2.3.91.12 Remote EEE advertisement (3.2349.2)

Bit 3.2349.2 indicates the EEE ability of the remote PHY received in the PHD field PHD.CAP.LPI (see Table 166–2). When read as a one, bit 3.2349.2 indicates that the remote PHY has the EEE ability and that EEE advertisement is enabled. When read as a zero, bit 3.2349.2 indicates either that the remote PHY does not have the EEE ability or that EEE advertisement is disabled.

45.2.3.91.13 Local BASE-U OAM ability (3.2349.1)

Bit 3.2349.1 indicates the BASE-U OAM ability of the BASE-U PHY (see 166.7). When read as a one, bit 3.2349.1 indicates that the BASE-U PHY supports the BASE-U OAM protocol. When read as a zero, bit 3.2349.1 indicates the BASE-U PHY does not support the BASE-U OAM protocol.

A PHY indicating in bit 3.2349.1 support of BASE-U OAM shall provide the functions of registers 3.2330 through 3.2347 as specified in 166.7.

45.2.3.91.14 Local EEE ability (3.2349.0)

Bit 3.2349.0 indicates the EEE ability of the BASE-U PHY. When read as a one, bit 3.2349.0 indicates that the BASE-U PHY implements EEE, and is able to enter LPI mode (see 166.4). When read as a zero, bit 3.2349.0 indicates the BASE-U PHY does not implement EEE operation.

45.2.3.92 BASE-U PCS status 2 register (Register 3.2350)

The assignment of bits in the BASE-U PCS status 2 register is shown in Table 45–313e.

Table 45–313e— BASE-U PCS status 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2350.15:8	Reserved	Value always 0	RO
3.2350.7:0	Local link margin	Reports the local link margin	RO

^aRO = Read only

45.2.3.92.1 Local link margin (3.2350.7:0)

Bits 3.2350.7:0 indicate the link margin of the receiver. Link margin is defined in 166.3.4.6.2 and is fixed-point formatted (8, 3). See 166.1.1 for fixed-point format definition.

45.2.3.93 BASE-U PCS status 3 register (Register 3.2351)

The assignment of bits in the BASE-U PCS status 3 register is shown in Table 45–313f.

Table 45–313f— BASE-U PCS status 3 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2351.15:8	Reserved	Value always 0	RO
3.2351.7:0	Remote link margin	Link margin reported by the link partner	RO

^aRO = Read only

45.2.3.93.1 Remote link margin (3.2351.7:0)

Bits 3.2351.7:0 report the link margin of the remote PHY receiver as it is received in the PHD field PHD.RX.LINKMARGIN (see Table 166–2). Link margin is defined in 166.3.4.6.2 and is fixed-point formatted (8, 3). See 166.1.1 for fixed-point format definition.

45.2.3.94 BASE-U PCS status 4 register (Register 3.2352)

The assignment of bits in the BASE-U PCS status 4 register is shown in Table 45–313g.

Table 45–313g— BASE-U PCS status 4 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2352.15:0	BER test mode counter	A 16-bit counter used when operating in BER test mode	RO, NR

^aRO = Read only, NR = Non Roll-over

When the BASE-U based PHY receiver is operating in BER test mode (see 166.5.1), bits 3.2352.15:0 are a 16-bit counter that counts the number of erroneous bits received at the input of the 64B/65B PCS decoder.

These bits shall be reset to all zeros when the counter is read. The counter shall be held at all ones in the case of overflow.

45.2.3.95 BASE-U PCS status 5 register (Register 3.2353)

The assignment of bits in the BASE-U PCS status 5 register is shown in Table 45–313h.

Table 45–313h— BASE-U PCS status 5 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.2353.15:0	RS-FEC codeword error counter	A 16-bit counter used when operating in normal and BER test mode	RO, NR

^aRO = Read only, NR = Non Roll-over

When the BASE-U based PHY receiver is operating in normal or BER test mode (see 166.5.1), bits 3.2353.15:0 are a 16-bit counter that counts the number of erroneous RS-FEC codewords at the input of the 64B/65B PCS decoder (see 166.2.3.2).

These bits shall be reset to all zeros when the counter is read. The counter shall be held at all ones in the case of overflow.

45.5 Protocol implementation conformance statement (PICS) proforma for Clause 45, Management Data Input/Output (MDIO) Interface⁷

45.5.3 PICS proforma tables for the Management Data Input Output (MDIO) Interface

45.5.3.2 PMA/PMD MMD options

Insert a new row at the end of the table in 45.5.3.2 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*BAU	Implementation of a BASE-AU PMA/PMD			PMA:O	Yes [] No []

45.5.3.3 PMA/PMD management functions

Insert items MM229 through MM231 at the end of the table in 45.5.3.3 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
...					
MM229	Writes to the BASE-AU PMA/PMD extended ability register 1.72 have no effect	45.2.1.60a		BAU:M	Yes [] N/A []
MM230	Bits 1.901.3:0 select only one of the modes of operation specified in Table 45–127a and advertised by register 1.72	45.2.1.158a		BAU:M	Yes [] N/A []
MM231	Writes of reserved values or modes of operation different of those advertised by register 1.72, to bits 1.901.3:0 have no effect	45.2.1.158a		BAU:M	Yes [] N/A []

⁷Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

45.5.3.6 PCS options

Insert a new row at the end of the table in 45.5.3.6 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
...					
*BUOAM	BASE-U OAM channel implementation	45.2.3.88 and 45.2.3.89		PCS:O	Yes [] No [] N/A []

45.5.3.7 PCS management functions

Insert items RM207 through RM232 at the end of the table in 45.5.3.7 as follows:

Item	Feature	Subclause	Value/Comment	Status	Support
...					
RM207	BASE-U OAM message format to be transmitted	45.2.3.88.5	TXO_DATA0[11:0] = OUI[23:12] and TXO_DATA1[15:4] = OUI[11:0] Content of TXO_DATA1[3:0] and TXO_DATA2 through TXO_DATA8 is vendor specific per OUI or CID	PCS *BUOAM :M	Yes [] N/A []
RM208	Default operation mode	45.2.3.90	Default value of bits 3.2348.15:13 is binary 000 (normal mode)	PCS:M	Yes [] N/A []
RM209	Test operation modes	45.2.3.90	Bits 3.2348.15:13 select test modes with behavior per 166.5 and mapped per Table 45–313c	PCS:M	Yes [] N/A []
RM210	Operation mode change effect	45.2.3.90.1	Changes in the value of bits 3.2348.15:13 only take effect after PMA reset	PCS:M	Yes [] N/A []
RM211	Default loopback mode	45.2.3.90.2	Default value of bits 3.2348.12:10 is binary 000 (no loopback)	PCS:M	Yes [] N/A []
RM212	Loopback modes	45.2.3.90.2	Bits 3.2348.12:10 select loopback modes with behavior per 166.8 and mapped per Table 45–313c	PCS:M	Yes [] N/A []
RM213	Loopback mode change effect	45.2.3.90.2	Changes in the value of bits 3.2348.12:10 only take effect after PMA reset	PCS:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
RM214	BASE-U OAM conditional enable	45.2.3.90.3	Write of one to bit 3.2348.1 only produces effect if bit 3.2349.1 is read as one	PCS *BUOAM :M	Yes [] N/A []
RM215	BASE-U OAM disable	45.2.3.90.3	Write of zero to bit 3.2348.1 prevents BASE-U OAM channel establishment with link partner	PCS *BUOAM :M	Yes [] N/A []
RM216	BASE-U OAM advertisement enable change effect	45.2.3.90.3	Changes in the value of bit 3.2348.1 only take effect after PMA reset	PCS *BUOAM :M	Yes [] N/A []
RM217	EEE advertisement enable	45.2.3.90.4	Write of one to bit 3.2348.0 enables the advertisement of local PHY EEE ability per 166.4	PCS *EEE:M	Yes [] N/A []
RM218	EEE conditional enable	45.2.3.90.4	Write of one to bit 3.2348.0 only enables EEE capability if bit 3.2349.0 is read as one	PCS *EEE:M	Yes [] N/A []
RM219	EEE disable	45.2.3.90.4	Write of zero to bit 3.2348.0 prevents EEE operation with link partner	PCS *EEE:M	Yes [] N/A []
RM220	EEE advertisement enable change effect	45.2.3.90.4	Changes in the value of bit 3.2348.0 only take effect after PMA reset	PCS *EEE:M	Yes [] N/A []
RM221	Link status behavior	45.2.3.91.3	Bit 3.2349.13 has latching low behavior	PCS:M	Yes [] N/A []
RM222	Link status mirroring	45.2.3.91.3	Bit 3.2349.13, bit 1.1.2, and bit 3.1.2 reflect the same state variable link_status and a read of any of them releases the latch for all the bits	PCS:M	Yes [] N/A []
RM223	PHD lock status behavior	45.2.3.91.6	Bit 3.2349.10 has latching low behavior	PCS:M	Yes [] N/A []
RM224	Tx Assert LPI received behavior	45.2.3.91.7	Bit 3.2349.8 has latching high behavior	PCS *EEE:M	Yes [] N/A []
RM225	Tx Assert LPI received mirroring	45.2.3.91.7	Bit 3.2349.8 is a copy of bit 3.1.11 and a read of either of them releases the latch for both of the bits	PCS *EEE:M	Yes [] N/A []
RM226	Rx Assert LPI generated behavior	45.2.3.91.8	Bit 3.2349.7 has latching high behavior	PCS *EEE:M	Yes [] N/A []
RM227	Rx Assert LPI generated mirroring	45.2.3.91.8	Bit 3.2349.7 is a copy of bit 3.1.10 and a read of either of them releases the latch for both of the bits	PCS *EEE:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
RM228	BASE-U OAM ability	45.2.3.91.13	Registers 3.2330 through 3.2347 are implemented and 3.2349.1 reads as one	PCS *BUOAM :M	Yes [] N/A []
RM229	BER test mode counter is reset to all zeros when read	45.2.3.94	Register 3.2352 is reset to all zeros when read	PCS:M	Yes [] N/A []
RM230	BER test mode counter holds all ones in case of overflow	45.2.3.94	Register 3.2352 holds all ones in case of overflow	PCS:M	Yes [] N/A []
RM231	RS-FEC codeword error counter is reset to all zeros when read	45.2.3.95	Register 3.2353 is reset to all zeros when read	PCS:M	Yes [] N/A []
RM232	RS-FEC codeword error counter holds all ones in case of overflow	45.2.3.95	Register 3.2353 holds all ones in case of overflow	PCS:M	Yes [] N/A []

78. Energy-Efficient Ethernet (EEE)

78.1 Overview

78.1.4 PHY types optionally supporting EEE

Insert new rows for “2.5GBASE-AU” after “2.5GBASE-T1”, “5GBASE-AU” after “5GBASE-T1”, “10GBASE-AU” after “10GBASE-T1”, “25GBASE-AU” after “25GBASE-T”, and “50GBASE-AU” after “50GBASE-CR” in Table 78–1 as follows (unchanged rows not shown):

Table 78–1—Clauses associated with each PHY or interface type

PHY or interface type	Clause
...	
2.5GBASE-AU ^b	166
...	
5GBASE-AU ^b	166
...	
10GBASE-AU ^b	166
...	
25GBASE-AU ^b	166
...	
50GBASE-AU ^b	166
...	

^a 25GAUI/XLAUI/CAUI-n shutdown is supported only when deep sleep is enabled for the associated PHY.

^b The deep sleep mode of EEE is not supported for this PHY.

78.5 Communication link access latency

Insert new rows for “2.5GBASE-AU fast wake” after “1000BASE-T”, “5GBASE-AU fast wake” after “2.5GBASE-T”, “10GBASE-AU fast wake” after “5GBASE-T”, “25GBASE-AU fast wake” after “10GBASE-T”, and “50GBASE-AU fast wake” after “40GBASE-T” in Table 78–4 as follows (unchanged rows not shown):

Table 78–4—Summary of the LPI timing parameters for supported PHYs or interfaces

PHY or interface type	Case	$T_{w_sys_tx}$ (min) (μs)	T_{w_phy} (min) (μs)	$T_{phy_shrink_tx}$ (max) (μs)	$T_{phy_shrink_rx}$ (max) (μs)	$T_{w_sys_rx}$ (min) (μs)
...						
2.5GBASE-AU fast wake		4.1	4.1	0	0	2.05
...						
5GBASE-AU fast wake		2.05	2.05	0	0	1.03
...						
10GBASE-AU fast wake		1.03	1.03	0	0	0.52
...						
25GBASE-AU fast wake		0.41	0.41	0	0	0.21
...						
50GBASE-AU fast wake		0.21	0.21	0	0	0.11
...						

105. Introduction to 25 Gb/s networks

105.1 Overview

105.1.1 Scope

Change the first paragraph of 105.1.1 as follows:

25 Gigabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 25 Gb/s, coupled with any IEEE 802.3 25GBASE Physical Layer implementation, connected through a 25 Gigabit Media Independent Interface (25GMII) to Physical Layer devices such as 25GBASE-CR, 25GBASE-CR-S, 25GBASE-KR, 25GBASE-KR-S, 25GBASE-SR, 25GBASE-T, 25GBASE-LR, and 25GBASE-ER.

105.1.2 Relationship of 25 Gigabit Ethernet to the ISO OSI reference model

Replace Figure 105–1 with the following figure, which adds 25GBASE-AU:

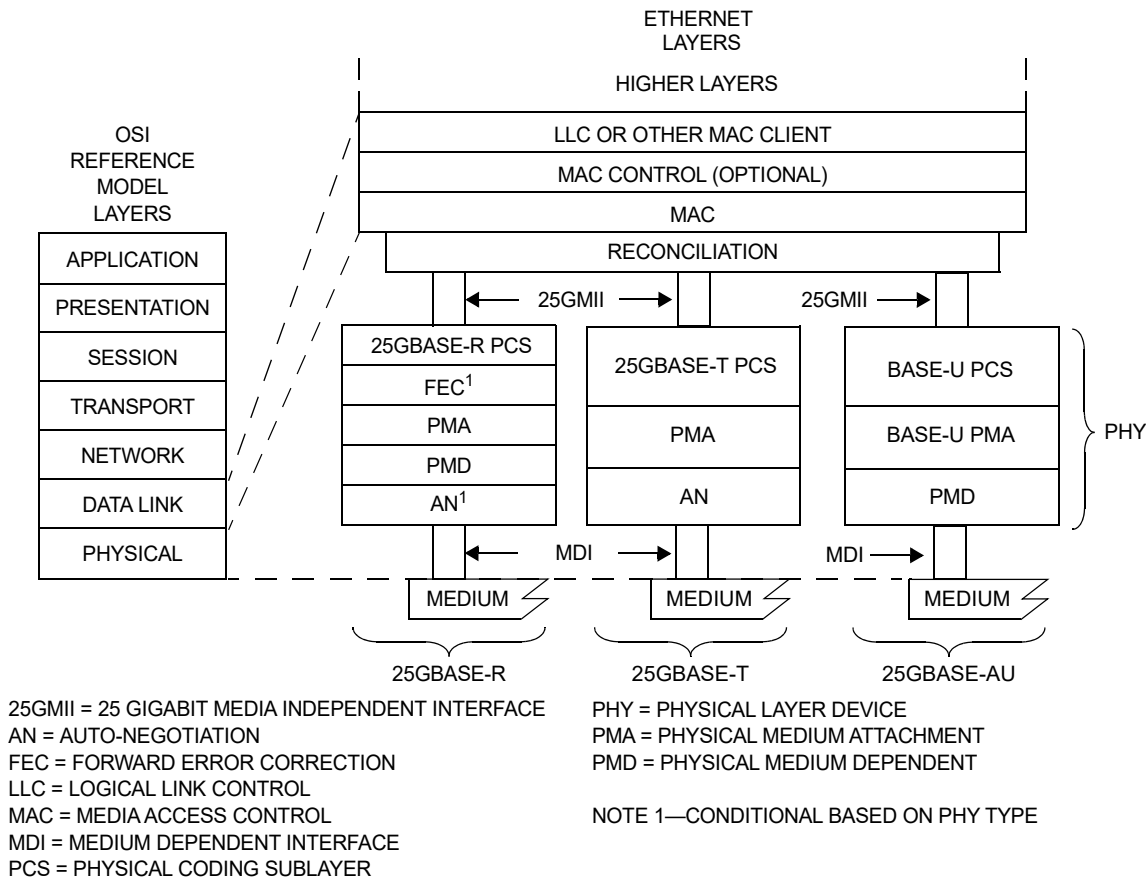


Figure 105–1—Architectural positioning of 25 Gigabit Ethernet

Change the text of list item c) in 105.1.2 as follows:

- c) The Media Dependent Interface (MDI) as specified in **Clause 110** for 25GBASE-CR and 25GBASE-CR-S, in **Clause 111** for 25GBASE-KR and 25GBASE-KR-S, in **Clause 112** for 25GBASE-SR, ~~or in Clause 114~~ for 25GBASE-LR and 25GBASE-ER, or in Clause 166 for 25GBASE-AU uses a single-lane data path.

105.1.3 Nomenclature

Insert a new paragraph after the third paragraph of 105.1.3 as follows:

25GBASE-AU represents Physical Layer devices using Clause 166 Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, for transmitting 25 Gb/s Ethernet over glass optical fiber in an automotive environment. 25GBASE-AU uses 64B/65B coding placed into RS-FEC codewords that are transmitted using NRZ modulation.

Insert a new row for “25GBASE-AU” after the row for “25GBASE-T” in Table 105–1 as follows (unchanged rows not shown):

Table 105–1—25 Gb/s PHYs

Name	Description
...	
25GBASE-AU	25 Gb/s PHY using 64B/65B and Reed-Solomon encoding with NRZ modulation over glass optical fiber in an automotive environment (see Clause 166).
...	

105.2 Physical Layer signaling systems

Change Table 105–2 as follows:

Table 105–2—Nomenclature and clause correlation, 25 Gb/s Ethernet PHYs

Nomenclature	Clause/Annex																			
	28	73	74	78	106		107	108	109	109A	109B	110		111		112	113	114		166
	Auto-Negotiation	Auto-Negotiation	BASE-R FEC	EEE	RS	25GMII	25GBASE-R PCS	25GBASE-R RS-FEC	PMA	25GAUI C2C	25GAUI C2M	25GBASE-CR PMD	25GBASE-CR-S PMD	25GBASE-KR PMD	25GBASE-KR-S PMD	25GBASE-SR PMD	25GBASE-T PCS/PMA	25GBASE-LR PMD	25GBASE-ER PMD	25 BASE-AU PCS/PMA/PMD
25GBASE-KR-S		M ^a	M	O ^a	M	O	M		M	O					M					
25GBASE-KR		M	M	O	M	O	M	M	M	O				M						
25GBASE-CR-S		M	M	O	M	O	M		M	O			M							
25GBASE-CR		M	M	O	M	O	M	M	M	O		M								
25GBASE-T	M			O	M	O											M			
<u>25GBASE-AU</u>				<u>O</u>	<u>M</u>	<u>O</u>														<u>M</u>
25GBASE-SR				O	M	O	M	M	M	O	O					M				
25GBASE-LR				O	M	O	M	M	M	O	O							M		
25GBASE-ER				O	M	O	M	M	M	O	O								M	

^aO = Optional, M = Mandatory

105.3 Summary of 25 Gigabit Ethernet sublayers

105.3.2 Physical Coding Sublayer (PCS)

Insert a new paragraph at the end of 105.3.2 as follows:

The 25GBASE-AU PCS is specified in Clause 166.

105.3.4 Physical Medium Attachment (PMA) sublayer

Insert a new paragraph at the end of 105.3.4 as follows:

The 25GBASE-AU PMA is specified in Clause 166.

105.3.5 Physical Medium Dependent (PMD) sublayer

Insert a new paragraph at the end of 105.3.5 as follows:

The 25GBASE-AU PMD and its corresponding media is specified in Clause 166.

105.5 Delay constraints

Change Table 105–3 as follows:

Table 105–3—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
25G RS, MAC, and MAC Control	8 192	16	327.68	See 106.1.4.
25GBASE-R PCS	3 584	7	143.36	See 107.4.
25G BASE-R FEC	6 144	12	245.76	See 74.6.
25GBASE-R RS-FEC	24 576	48	983.04	See 108.4.
25GBASE-R PMA ^d	4 096	8	163.84	See 109.5.
25GBASE-CR PMD	512	1	20.48	See 110.4.
25GBASE-CR-S PMD	512	1	20.48	See 110.4.
25GBASE-KR PMD	512	1	20.48	See 110.4.
25GBASE-KR-S PMD	512	1	20.48	See 110.4.
25GBASE-SR PMD	512	1	20.48	See 112.3.
25GBASE-LR PMD	512	1	20.48	See 114.3.
25GBASE-ER PMD	512	1	20.48	See 114.3.
25GBASE-T PHY	25 600	50	1024	See 113.11.
25GBASE-BR10 PMD	512	1	20.48	See 158.3.
25GBASE-BR20 PMD	512	1	20.48	See 158.3.
25GBASE-BR40 PMD	512	1	20.48	See 158.3.
<u>25GBASE-AU PHY</u>	<u>11 264</u>	<u>22</u>	<u>450.56</u>	<u>Does not include delay of medium. See 166.11.</u>

^a1 bit time (BT) is equal to 40 ps. (See 1.4.215 for the definition of bit time.)

^b1 pause_quantum is equal to 20.48 ns. (See 31B.2 for the definition of pause_quantum.)

^cShould there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^dCumulative round-trip delay contributed by up to four PMA stages in a PHY.

105.7 Protocol implementation conformance statement (PICS) proforma

Change the first paragraph of 105.7 as follows:

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 45, [Clause 73](#), [Clause 74](#), [Clause 106](#) through [Clause 112](#), [Clause 114](#), [Clause 166](#), and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

125. Introduction to 2.5 Gb/s and 5 Gb/s networks

125.1 Overview

125.1.2 Relationship of 2.5 Gigabit and 5 Gigabit Ethernet to the ISO OSI reference model

Insert new list item f) after item e) in 125.1.2 as follows:

- f) The MDI as specified in Clause 166 for 2.5GBASE-AU and 5GBASE-AU uses a single-lane data path.

125.1.3 Nomenclature

Replace Figure 125–1 with the figure on the following page, which adds 2.5GBASE-AU and 5GBASE-AU.

Insert two new paragraphs at the end of 125.1.3 as follows:

2.5GBASE-AU represents Physical Layer devices using Clause 166 Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, for transmitting 2.5 Gb/s Ethernet over glass optical fiber in an automotive environment. 2.5GBASE-AU uses 64B/65B coding placed into RS-FEC codewords that are transmitted using NRZ modulation.

5GBASE-AU represents Physical Layer devices using Clause 166 Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, for transmitting 5 Gb/s Ethernet over glass optical fiber in an automotive environment. 5GBASE-AU uses 64B/65B coding placed into RS-FEC codewords that are transmitted using NRZ modulation.

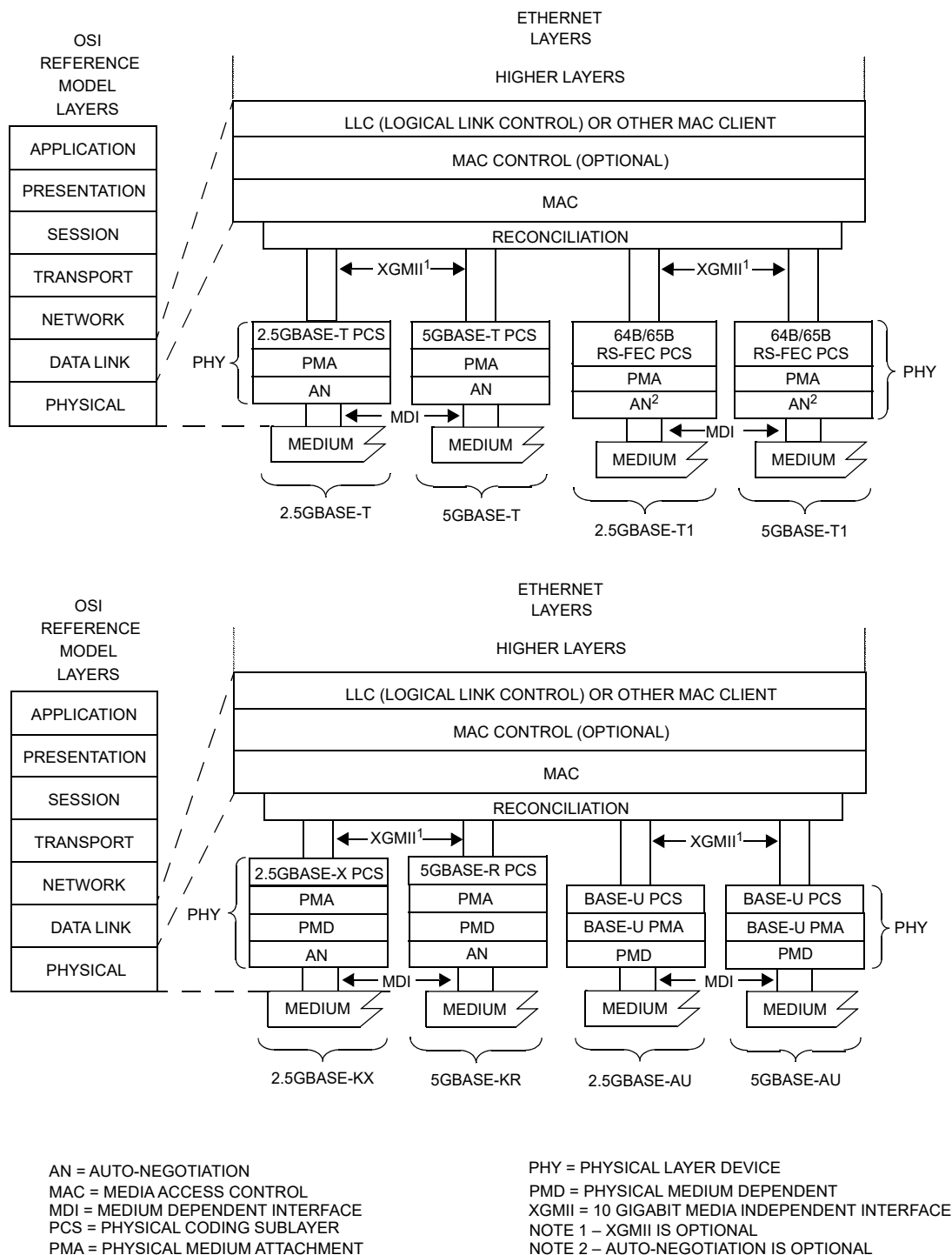


Figure 125–1—Architectural positioning of 2.5 Gigabit and 5 Gigabit Ethernet

125.1.4 Physical Layer signaling systems

Insert new rows for “2.5GBASE-AU” after “2.5GBASE-T1” and “5GBASE-AU” after “5GBASE-T1” in Table 125–1 as follows (unchanged rows not shown):

Table 125–1—2.5 Gb/s and 5 Gb/s PHYs

Name	Description
...	
2.5GBASE-AU	2.5 Gb/s PHY using 64B/65B and Reed-Solomon encoding with NRZ modulation over glass optical fiber in an automotive environment (see Clause 166)
...	
5GBASE-AU	5 Gb/s PHY using 64B/65B and Reed-Solomon encoding with NRZ modulation over glass optical fiber in an automotive environment (see Clause 166)

Change Table 125–2 as follows:

Table 125–2—Nomenclature and clause correlation

Nomenclature	Clause and annex ^a																	
	28	46		73	78	98	126		127	128	128A	129	130	130A	149		166	
	Auto-Negotiation	RS	XGMII	Auto-Negotiation	EEE	Auto-Negotiation	2.5GBASE-T PCS/PMA	5GBASE-T PCS/PMA	2.5GBASE-X PCS/PMA	2.5GBASE-KX PMD	2.5GSEI	5GBASE-R PCS/PMA	5GBASE-KR PMD	5GSEI	2.5GBASE-T1 PCS/PMA	5GBASE-T1 PCS/PMA	2.5GBASE-AU PCS/PMA/PMD	5GBASE-AU PCS/PMA/PMD
2.5GBASE-KX		M	O	M	O				M	M	O							
2.5GBASE-T	M	M	O		O		M											
2.5GBASE-T1		M	O		O	O									M			
2.5GBASE-AU		M	O		O												M	
5GBASE-KR		M	O	M	O							M	M	O				
5GBASE-T	M	M	O		O			M										
5GBASE-T1		M	O		O	O										M		
5GBASE-AU		M	O		O													M

^aO = Optional, M = Mandatory.

125.2 Summary of 2.5 Gigabit and 5 Gigabit Ethernet sublayers

125.2.2 Physical coding sublayer (PCS)

Insert a new paragraph at the end of 125.2.2 as follows:

2.5GBASE-AU and 5GBASE-AU use the PCS specified in Clause 166.

125.2.3 Physical Medium Attachment sublayer (PMA)

Insert a new paragraph at the end of 125.5.3 as follows:

2.5GBASE-AU and 5GBASE-AU use the PMA specified in Clause 166.

Insert 125.2.3a after 125.2.3 as follows:

125.2.3a Physical Medium Dependent sublayer (PMD)

2.5GBASE-AU and 5GBASE-AU use the PMD and its corresponding media specified in Clause 166.

125.3 Delay Constraints

Insert new rows for “2.5GBASE-AU PHY” after “2.5GBASE-T1 PHY” and “5GBASE-AU PHY” after “5GBASE-T1 PHY 2x interleave” in Table 125–3 and change table footnotes a and b as follows (unchanged rows not shown):

Table 125–3—Sublayer delay constraints

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
...				
2.5GBASE-AU PHY	11 264	22	4505.6	Does not include delay of medium. See 166.11.
...				
5GBASE-AU PHY	11 264	22	2252.8	Does not include delay of medium. See 166.11.

^a For ~~2.5GBASE-T, 2.5GBASE-X, and 2.5GBASE-T1~~ 2.5 Gigabit Ethernet, 1 bit time (BT) is equal to 400 ps and for ~~5GBASE-T, 5GBASE-R, and 5GBASE-T1~~ 5 Gigabit Ethernet, 1 bit time (BT) is equal to 200 ps. (See 1.4.215 for the definition of bit time.)

^b For ~~2.5GBASE-T, 2.5GBASE-X, and 2.5GBASE-T1~~ 2.5 Gigabit Ethernet, 1 pause_quantum is equal to 204.8 ns and for ~~5GBASE-T, 5GBASE-R, and 5GBASE-T1~~ 5 Gigabit Ethernet, 1 pause_quantum is equal to 102.4 ns. (See 31B.2 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

131. Introduction to 50 Gb/s networks

131.1 Overview

131.1.2 Relationship of 50 Gigabit Ethernet to the ISO OSI reference model

Replace Figure 131–1 with the following figure, which adds 50GBASE-AU:

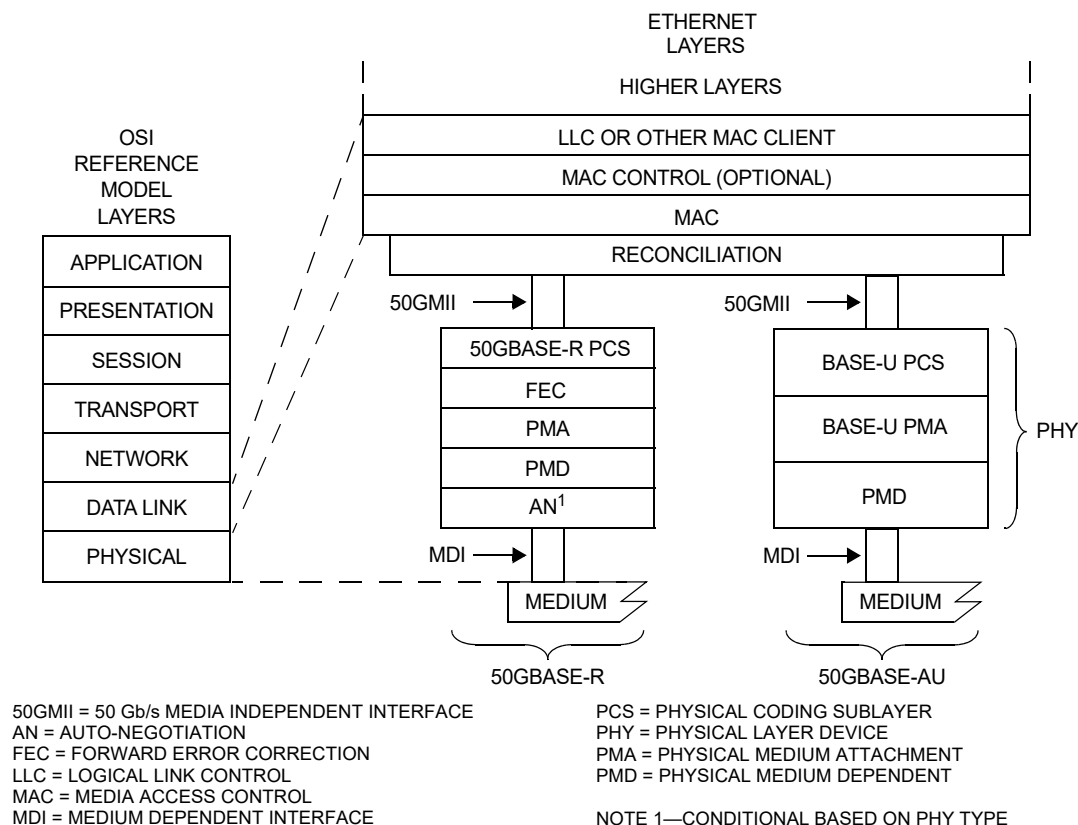


Figure 131–1—Architectural positioning of 50 Gigabit Ethernet

Change list item e) of 131.1.2 as follows:

- e) The MDI as specified in Clause 136 for 50GBASE-CR, Clause 137 for 50GBASE-KR, Clause 138 for 50GBASE-SR, and Clause 139 for 50GBASE-FR, 50GBASE-LR, and 50GBASE-ER, and Clause 166 for 50GBASE-AU uses a one-lane data path.

131.1.3 Nomenclature

Insert a new paragraph at the end of 131.1.3 as follows:

50GBASE-AU represents Physical Layer devices using Clause 166 Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, for transmitting 50 Gb/s Ethernet over glass optical fiber in an automotive environment. 50GBASE-AU uses 64B/65B coding placed into RS-FEC codewords that are transmitted using PAM4 modulation.

Insert a new row for “50GBASE-AU” after the row for “50GBASE-CR” in Table 131–1 as follows (unchanged rows not shown):

Table 131–1—50 Gb/s PHYs

Name	Description
...	
50GBASE-AU	50 Gb/s PHY using 64B/65B and Reed-Solomon encoding with PAM4 modulation over glass optical fiber in an automotive environment (see Clause 166).
...	

131.1.4 Physical Layer signaling systems

Change Table 131–3 as follows:

Table 131–3—PHY types and clause correlation (50GBASE optical)

PHY type	Clause ^a																
	EEE	78		132		133	134	135	135B	135C	135D	135E	135F	135G	138	139	
		RS	50GMII	50GBASE-R PCS	50GBASE-R RS-FEC	50GBASE-R PMA	LAUI-2 C2C	LAUI-2 C2M	50GAUI-2 C2C	50GAUI-2 C2M	50GAUI-1 C2C	50GAUI-1 C2M	50GBASE-SR PMD	50GBASE-FR PMD	50GBASE-LR PMD	50GBASE-ER PMD	50GBASE-AU PCS/PMA/PMD
50GBASE-AU	<u>O</u>	<u>M</u>	<u>O</u>														<u>M</u>
50GBASE-SR	O	M	O	M	M	M	O	O	O	O	O	O	M				
50GBASE-FR	O	M	O	M	M	M	O	O	O	O	O	O		M			
50GBASE-LR	O	M	O	M	M	M	O	O	O	O	O	O			M		
50GBASE-ER	O	M	O	M	M	M	O	O	O	O	O	O				M	

^aO = Optional, M = Mandatory.

131.2 Summary of 50 Gigabit Ethernet sublayers

131.2.2 Physical Coding Sublayer (PCS)

Insert a new paragraph at the end of 131.2.2 as follows:

The 50GBASE-AU PCS is specified in Clause 166.

131.2.4 Physical Medium Attachment (PMA) sublayer

Insert a new paragraph at the end of 131.2.4 as follows:

The 50GBASE-AU PMA is specified in Clause 166.

131.2.5 Physical Medium Dependent (PMD) sublayer

Insert a new paragraph at the end of 131.2.5 as follows:

The 50GBASE-AU PMD and its corresponding media is specified in Clause 166.

131.4 Delay constraints

Insert a new row for “50GBASE-AU PHY” after the row for “50GBASE-KR PMD” in Table 131–4 and change table footnotes a and b as follows (unchanged rows not shown):

Table 131–4—Sublayer delay constraints (50GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c
...				
50GBASE-AU PHY	14 848	29	296.96	Does not include delay of medium. See 166.11.
...				

^a For ~~50GBASE-R~~ 50 Gigabit Ethernet, 1 bit time is equal to 20 ps. (See 1.4.215 for the definition of bit time.)

^b For ~~50GBASE-R~~ 50 Gigabit Ethernet, 1 pause_quantum is equal to 10.24 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

Insert new Clause 166 as follows:

166. Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU

166.1 Overview

This clause defines the Physical Coding Sublayer (PCS) and Physical Medium Attachment (PMA) sublayer types 2.5GBASE-U, 5GBASE-U, 10GBASE-U, 25GBASE-U, and 50GBASE-U, as well as Physical Medium Dependent (PMD) sublayer types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU. Together, the corresponding PCS, PMA, and PMD sublayers comprise a 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, or 50GBASE-AU Physical Layer entity (PHY).

These PHY types are specified to support operation in automotive applications. This clause specifies the operation between link partners implementing the same BASE-AU PHY type. The link segment specifications were derived from automotive requirements, but may also be used for non-automotive applications. Each of the specified PHYs support full duplex operation for a BER of 10^{-12} or better.

The MDI connecting the PMD to the optical fiber medium is typically a receptacle and a mated plug. This clause assumes the fiber optic cabling characteristics in 166.6.6. The MDI connector, in-line connectors, and cable may have additional requirements for installation in a vehicle that are beyond the scope of this standard.

This clause also specifies an optional Energy-Efficient Ethernet (EEE) fast wake capability. A 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, or 50GBASE-AU PHY that supports this capability may enter a Low Power Idle (LPI) mode of operation during periods of low link utilization as described in Clause 78. The deep sleep mode of EEE is not supported.

Additionally, this clause specifies how a 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, or 50GBASE-AU PHY may use the optional BASE-U PCS-based operations, administration, and maintenance (OAM) channel for communication between link partners.

166.1.1 Conventions

The 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU PHYs described in this clause represent five distinct PHY types that share the same PCS, PMA, and PMD specifications subject to frequency scaling and modulation scheme. In order to efficiently describe the five PHYs, the nomenclature BASE-AU is used to describe specifications that apply to the 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU PHYs hereafter. For parameters that scale with the PHY's data rate, the parameter S is used for scaling. Additionally, for parameters that depend on the modulation scheme and therefore the bit grouping, the parameter G is used. See Table 166–1 for the definition of S and G for each BASE-AU PHY.

The nomenclature BASE-U is used to describe specifications that apply to the PCS or PMA sublayer of a BASE-AU PHY.

The notation used in the state diagrams in this clause follows the conventions in 21.5. Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails. The notation ++ after a counter or integer variable indicates that its value is to be incremented by one.

Table 166–1—Scaling parameter S and bit grouping parameter G

PHY type	S	G
50GBASE-AU	1	2
25GBASE-AU	1	1
10GBASE-AU	0.4	1
5GBASE-AU	0.2	1
2.5GBASE-AU	0.1	1

The notation (m, n) is used to represent an m -bit binary number, which is to be interpreted as a signed two's complement fixed-point number, where n is the number of bits used to represent the integer part (sign bit included) and $(m - n)$ bits are used to represent the fractional part. This fixed-point format is defined in 115.3.8.

166.1.2 Relationship to other Ethernet PHY types

The 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU PHY types are specified with the PCS interfacing to an XGMII (see Clause 46). The 25GBASE-AU PHY type is specified with the PCS interfacing to a 25GMII (see Clause 106). The 50GBASE-AU PHY type is specified with the PCS interfacing to a 50GMII (see Clause 132). Physical implementation of the XGMII, 25GMII, and 50GMII is optional. System operation from the perspective of signals at the MDI and management objects shall be identical whether the XGMII, 25GMII, or 50GMII is implemented or not.

For this clause, the term xMII defines a generic Media Independent Interface referring to XGMII, 25GMII, or 50GMII.

166.1.3 Relationship of BASE-AU PHYs to other standards

The relationship between the BASE-AU PHYs, the ISO Open Systems Interconnection (OSI) reference model, and the IEEE 802.3 Ethernet model is shown in Figure 166–1. The PHY sublayers in Figure 166–1 connect one Media Access Control (MAC) layer (see Clause 4 and Annex 4A) to the medium. The XGMII, 25GMII, and 50GMII are specified in Clause 46, Clause 106, and Clause 132, respectively.

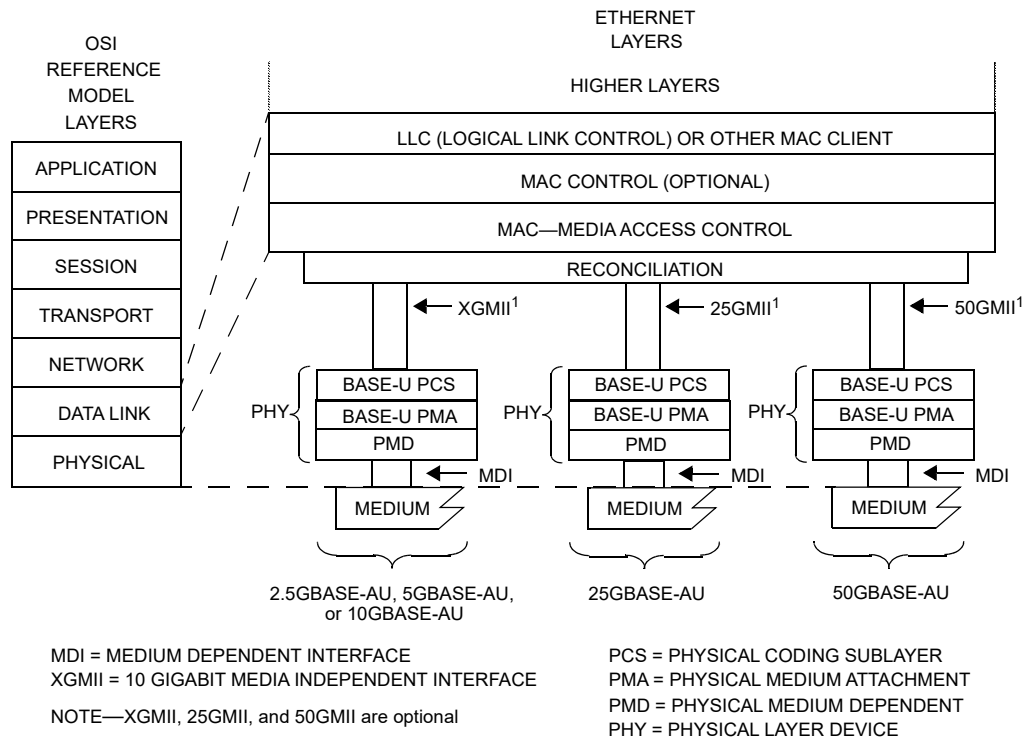


Figure 166-1—Relationship of BASE-AU PHYs to the ISO/IEC OSI reference model and the IEEE 802.3 Ethernet model

166.1.4 Operation of BASE-AU PHYs

BASE-AU PHY types support full duplex operation only, using two optical fibers as the medium. Each fiber is used for unidirectional transmission with the BASE-AU port on one end of the link segment transmitting on one fiber and receiving on the other fiber. The local PMD transmitter and PMD receiver are connected to the link partner using the fiber optic cabling. The PMD transmitter and PMD receiver compose the PMD sublayer.

A BASE-AU PHY transmitter shall be composed of PCS, PMA, and PMD sublayers specified for the same data rate. A BASE-AU PHY receiver shall be composed of PCS, PMA, and PMD sublayers specified for the same data rate.

The topology of BASE-AU is illustrated in Figure 166-2.

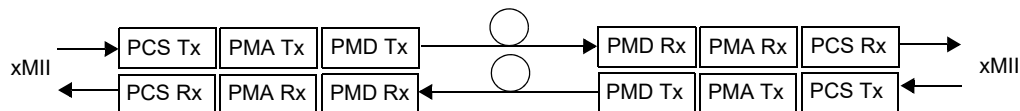


Figure 166-2—BASE-AU topology

The BASE-U PCS manages interleaving of xMII data streams with physical layer control information. A fixed-length Transmit Block provides the structure for time division multiplexing of these two streams of information. A frame from the xMII can be contained in one or more Transmit Blocks, and xMII frame boundaries have no correlation to Transmit Block boundaries. On the transmit path, the PCS encodes the

xMII data stream using 64B/65B encoding (see 166.2.2.7). The encoded xMII data stream is also referred to as the payload.

The physical layer control is organized into a block called physical header data (PHD) (see Table 166–2). The 224-bit PHD is followed by the result of CRC calculation. The resulting block is encoded using an interleaved repetition code called three repetition code (TRC, see 166.2.2.1.5), which generates a sequence called encoded PHD. The encoded PHD is divided into a series of 20-bit encoded PHD sub-blocks. Each encoded PHD sub-block is placed into the Transmit Block after a portion of payload composed of 80×65 -bit blocks.

The resulting information blocks are encoded by a systematic RS-FEC code. A Transmit Block holds 36 RS-FEC codewords. A Transmit Block is scrambled before transmission. The scrambler is initialized to a predefined value at the beginning of each Transmit Block.

On the receive path, the BASE-U PCS performs the descrambling, decodes received RS-FEC codewords, and separates the payload from the control information. The received payload is 64B/65B decoded to create the xMII receive data stream. A series of received PHD sub-blocks are concatenated and TRC decoded to reconstruct the PHD followed by a CRC (see 166.2.2.1). PHD information reliability is checked by CRC calculation and, if it is correct, it is used by the PCS and PMA sublayers. PHD information exchange with the link partner provides bidirectional link monitoring, PHY control, capabilities announcement, and BASE-U OAM message communication.

A BASE-AU PHY may optionally support Energy-Efficient Ethernet (see Clause 78) and advertise the EEE capability using the PHD. The EEE capability is a mechanism by which a BASE-AU PHY is able to reduce power consumption during periods of low link utilization. BASE-AU EEE operation is specified in 166.4.

A BASE-AU PHY may optionally support the BASE-U PCS-based operations, administration, and maintenance (OAM). The BASE-U OAM information is exchanged between two BASE-AU PHYs as part of the PHD. The BASE-U OAM is specified in 166.7, and the BASE-AU PHY advertises its BASE-U OAM capability using the PHD.

The BASE-U PMA couples messages from the PCS to the PMD and provides clock and data recovery, link management, and PHY control functions. The PMA provides full duplex communications over two optical fibers, with nominal signaling rates of 2.65625 GBd for 2.5GBASE-AU, 5.3125 GBd for 5GBASE-AU, 10.625 GBd for 10GBASE-AU, and 26.5625 GBd for 25GBASE-AU and 50GBASE-AU. The PMA functionality is specified in 166.3. The PMD functionality is specified in 166.6. Figure 166–3 shows the BASE-AU PHY functional block diagram.

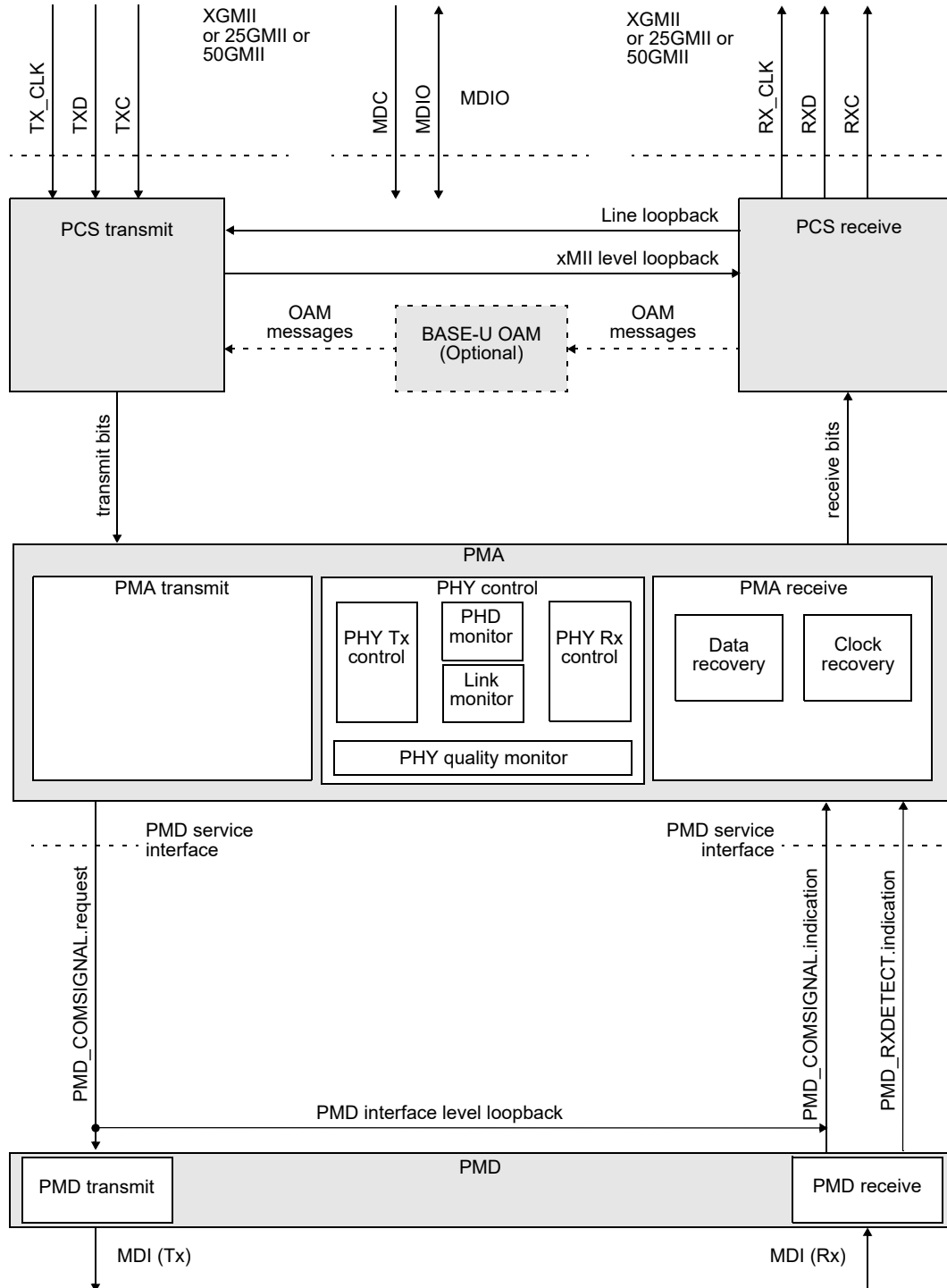


Figure 166-3—BASE-AU PHY functional block diagram

166.2 Physical Coding Sublayer (PCS)

166.2.1 PCS functions

The PCS service interface of 2.5GBASE-AU, 5GBASE-AU, or 10GBASE-AU is the 10 Gigabit Media Independent Interface (XGMII), which is defined in [Clause 46](#). The 2.5GBASE-AU, 5GBASE-AU, or 10GBASE-AU PCS provides all services required by the XGMII and couple it to the 2.5GBASE-AU, 5GBASE-AU, or 10GBASE-AU Physical Medium Attachment (PMA) sublayer.

The 25GBASE-AU PCS service interface is the Media Independent Interface for 25 Gb/s operation (25GMII), which is defined in [Clause 106](#). The 25GBASE-AU PCS provides all services required by the 25GMII and couples it to the 25GBASE-AU PMA sublayer.

The 50GBASE-AU PCS service interface is the Media Independent Interface for 50 Gb/s operation (50GMII), which is defined in [Clause 132](#). The 50GBASE-AU PCS provides all services required by the 50GMII and couples it to the 50GBASE-AU PMA sublayer.

A block diagram of the PCS transmit function is shown in Figure 166–4.

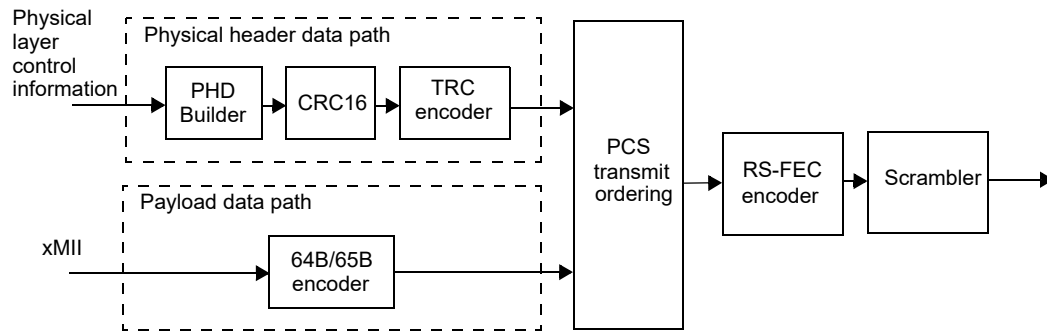


Figure 166–4—PCS transmit function

The PCS transmit function has two paths: the physical header data path (see 166.2.2.1) and the payload data path (see 166.2.2.2). The result of each path is combined by the PCS transmit ordering (see 166.2.2.3), encoded using an RS-FEC code (see 166.2.2.4), and then scrambled (see 166.2.2.5).

The physical header data path has three stages. In the first stage, the physical layer control information is organized into a 224-bit PHD as specified in 166.2.2.1.1. The second stage calculates the CRC16 checksum, which is concatenated at the end of the PHD. The third stage encodes the resulting 240-bit block using an interleaved TRC (see 166.2.2.1.4), generating the 720-bit encoded PHD.

The payload data path has one stage that encapsulates and encodes the xMII transmit data stream into payload 65-bit blocks using the 64B/65B encoding function specified in 166.2.2.7.

The PCS transmit ordering divides the 720-bit encoded PHD, generated by the physical header data path, into 36×20 -bit encoded PHD sub-blocks, and places each 20-bit encoded PHD sub-block after a group of 80 payload 65-bit blocks (see 166.2.2.3). The 5220-bit RS-FEC message is composed of the group of 80 payload 65-bit blocks followed by the 20-bit encoded PHD sub-block. See Figure 166–10 for details on PCS physical header bit ordering. See Figure 166–11 for details on PCS bit ordering.

The 5220-bit RS-FEC message is encoded by a systematic RS-FEC encoder as specified in 166.2.2.4. The resulting 5440-bit RS-FEC codeword is composed of the 5220-bit RS-FEC message followed by the 220 parity bits calculated by the RS-FEC encoder.

The 36 codewords that are needed to transmit an encoded PHD are scrambled as specified in 166.2.2.5. The 195 840-bit block generated by the scrambler is a Transmit Block. The scrambler uses a linear feedback shift register that is initialized to a predefined value at the beginning of each Transmit Block. See Figure 166–11 for details on PCS transmit bit ordering.

The PCS receive function includes binary descrambling, RS-FEC decoding of the received Transmit Block, 64B/65B decoding of payload portion to extract the xMII receive data stream, and TRC decoding and CRC16 checking of the PHD. The decoded PHD is also provided to the PMA sublayer for coordinated control of local and remote PHYs.

The relationship of 65-bit block bit positions to PHD, RS-FEC messages, codewords, and other PCS constructs is illustrated in Figure 166–10 and Figure 166–11 for transmit, and Figure 166–18 for receive. The mapping of xMII data transaction(s) to a 65-bit block is shown in Figure 166–12 for XGMII or 25GMII and Figure 166–13 for 50GMII. The mapping of a 65-bit block to xMII data transaction(s) is shown in Figure 166–19 for XGMII or 25GMII and Figure 166–20 for 50GMII. For values shown as binary, the leftmost bit is the first transmitted bit. These figures illustrate the processing of 65-bit blocks containing eight data octets. See 166.2.2.7 for information on how 65-bit blocks containing control characters are generated.

166.2.2 PCS transmit function

166.2.2.1 Physical header data path

166.2.2.1.1 Physical header data (PHD) structure

The PHD shall consist of the fields detailed in Table 166–2.

Table 166–2—PHD structure

Field name	Description	Number of bits	Valid values
PHD.TX.NEXT.MODE	Transmission mode of the next Transmit Block, indicated to link partner to align its reception (see 166.5.1).	3	0: normal transmission 1: BER test mode transmission 2 through 7: reserved
PHD.RX.LINKSTATUS	Indicates whether the local PHY is able to receive 65-bit blocks with reliability. The value of this field is determined by the PHY quality monitor state diagram (see 166.3.4.6.4). The local PHY uses this received PHD field to determine the value of the variable <code>rem_rcvr_status</code> (see 166.3.4.1).	1	0: NOT_OK 1: OK

Table 166–2—PHD structure (*continued*)

Field name	Description	Number of bits	Valid values
PHD.RX.HDRSTATUS	Indicates whether the local PHY is able to receive the PHD from its link partner with reliability. The value of this field is determined by the local PHD reception monitor state diagram (see 166.3.4.5). The local PHY uses this received PHD field to determine the value of the variable <code>rem_rcvr_hdr_lock</code> (see 166.3.4.1).	1	0: NOT_OK 1: OK
PHD.RX.LINKMARGIN	The value of this field is determined by the PHY quality monitor state diagram (see 166.3.4.6.4) in response to link margin estimation as defined in 166.3.4.6.2. Upon reception of a valid PHD, the field is stored in bits 3.2351.7:0 (see 45.2.3.93).	8	This field is fixed-point formatted (8, 3) and is provided in log ₂ units (see 166.3.4.6.2). See 166.1.1 for fixed-point format.
PHD.CAP.LPI	This field indicates whether the PHY supports EEE and has enabled the announcement of this ability (see 166.4).	1	0: EEE is not supported or it is not announced 1: EEE is supported and it is announced
PHD.CAP.OAM	This field indicates whether the PHY supports BASE-U OAM and has enabled the announcement of this ability (see 166.7).	1	0: BASE-U OAM is not supported or it is not announced 1: BASE-U OAM is supported and it is announced
	Reserved.	65	0
PHD.OAM.DATA0	BASE-U OAM message data field 0 (see 166.7).	12	0x000 through 0xFFFF
PHD.OAM.MSGT	BASE-U OAM message identification bit (see 166.7).	1	0 or 1
PHD.OAM.MERT	BASE-U OAM STA read identification bit (see 166.7).	1	0 or 1
PHD.OAM.PHYT	BASE-U OAM PHY reception identification bit (see 166.7).	1	0 or 1
	Reserved.	1	0
PHD.OAM.DATA1	BASE-U OAM message data field 1 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA2	BASE-U OAM message data field 2 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA3	BASE-U OAM message data field 3 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA4	BASE-U OAM message data field 4 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA5	BASE-U OAM message data field 5 (see 166.7).	16	0x0000 through 0xFFFF

Table 166–2—PHD structure (*continued*)

Field name	Description	Number of bits	Valid values
PHD.OAM.DATA6	BASE-U OAM message data field 6 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA7	BASE-U OAM message data field 7 (see 166.7).	16	0x0000 through 0xFFFF
PHD.OAM.DATA8	BASE-U OAM message data field 8 (see 166.7).	16	0x0000 through 0xFFFF

The field PHD.TX.NEXT.MODE is used by the local PHY to provide the transmission mode of the next Transmit Block to the remote PHY, so that the remote PHY can align its reception.

The fields PHD.RX.* are transmitted to inform the link partner of the status of the local reception. Specifically, PHD.RX.LINKSTATUS indicates whether 65-bit blocks are being received with reliability, and PHD.RX.HDRSTATUS signals the correct/incorrect reception of PHDs. The field PHD.RX.LINKMARGIN is used to report the link margin. The PHD.RX.LINKMARGIN field is fixed-point formatted (8, 3). See 166.1.1 for fixed-point format definition.

PHD.CAP.* fields indicate if the local PHY is using optional features. PHD.CAP.LPI is used by the PHY to advertise that Energy-Efficient Ethernet (EEE) is supported and that it is enabled, and the field PHD.CAP.OAM signals that the PHY implements the capability to run the BASE-U OAM message exchange protocol and that it is enabled.

PHD.OAM.* fields are used to transmit and receive BASE-U OAM messages.

All the PHD fields are transmitted from the least to the most significant bit and are transmitted from top to bottom of Table 166–2. The PHD transmission shall conform to the physical header data transmit bit ordering in Figure 166–10.

Each PHY has to deal with transmit and receive PHDs simultaneously. The prefix LOCPHD refers to the fields of the PHD to be included in the next Transmit Block transmitted to the link partner from the local PHY. LOCPHD fields assigned by the state diagrams shall be sampled at the beginning of a Transmit Block by the PHD Builder to create the PHD included in that current Transmit Block.

The prefix REMPHD refers to the fields of the most recent PHD received, decoded, and validated from the link partner (from the remote PHY). The new values of REMPHD fields shall be available to the state diagrams and registers immediately after reception, decoding, and validation of the entire PHD and before decoding the first RS-FEC codeword of the next received Transmit Block.

166.2.2.1.2 Physical header encoding

The PHD data path shall be functionally equivalent to the result of the following process:

- 1) The 224-bit PHD is built from the physical layer control information as specified in 166.2.2.1.1.
- 2) A 16-bit cyclic redundancy check (CRC16) is calculated from the PHD as specified in 166.2.2.1.3 for error detection capability after decoding. The PHD is followed by the resulting 16-bit redundancy check to compose the 240-bit concatenation of the PHD and the CRC16.
- 3) The PHD splitting function is applied to the concatenation of the PHD and the CRC16 as specified in 166.2.2.1.4 to split the incoming 240 bits into 12×20 -bit sub-blocks.

- 4) The three repetition code (TRC) specified in 166.2.2.1.5 shall be used to code each one of the 20-bit sub-blocks, obtaining a total of 36×20 -bit PHD encoded sub-blocks per PHD.

Figure 166–5 shows the four PHD data path stages.

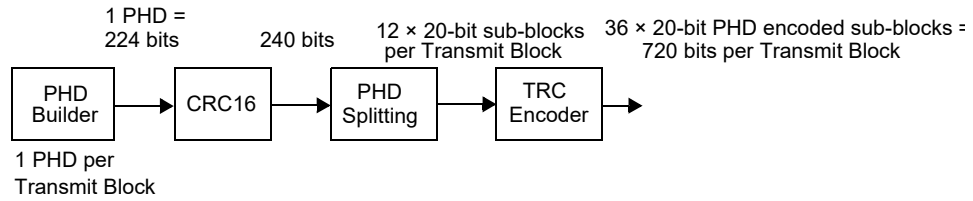


Figure 166–5—Physical header encoding

166.2.2.1.3 Physical header CRC16

The 224 PHD bits from the PHD Builder are followed by 16 cyclic redundancy check bits (CRC16) for error detection capability after TRC decoding. The CRC16 shall be computed from the PHD bits and shall produce the same result as the implementation shown in Figure 166–6.

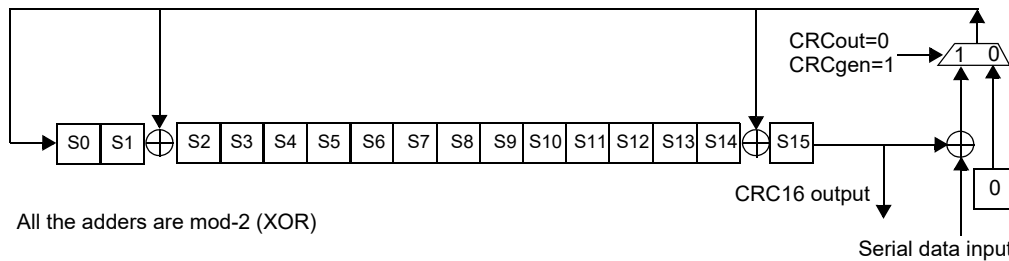


Figure 166–6—Physical header CRC16

The generator polynomial is $(x+1)(x^{15}+x+1)$. The shift register elements $S_0, \dots, S_{15}$, shall be initialized with the value of 0x0000 for each PHD. The 224 PHD bits, in transmit bit order, are then used to compute the CRC16 with the multiplexer configured to CRCgen. After the 224 bits have been serially processed, the multiplexer is configured to CRCout and the 16 stored values are the CRC16. The CRC16 is transmitted in order from S_{15} to S_0 .

166.2.2.1.4 Physical header splitting function

Each group of 240 information bits composed of the concatenation of a PHD and the CRC16 is split into 12×20 -bit sub-blocks before TRC coding. See Figure 166–10.

166.2.2.1.5 Physical header three repetition code (TRC)

Each 20-bit sub-block incoming from the PHD splitting function is encoded into three 20-bit PHD encoded sub-blocks by means of a three repetition code (TRC) encoder. The TRC encoder repeats each 20-bit sub-block three times. The process is repeated until the 12×20 -bit sub-blocks are processed, generating 36×20 -bit sub-blocks per PHD, called 20-bit encoded PHD sub-blocks. See Figure 166–10.

166.2.2.2 Payload data path

The incoming data from the xMII shall be encapsulated and encoded into 65-bit blocks (64B/65B encoder in Figure 166–7) for transmission as specified by the PCS 64B/65B transmit state diagram (see 166.2.2.8 and Figure 166–16). The 64B/65B encoder generates 187 200 bits per Transmit Block equivalent to 2880×65 -bit blocks. 5760 XGMII/25GMII or 2880 50GMII data transfers are encoded into a Transmit Block in normal operation mode, when the link has been established.

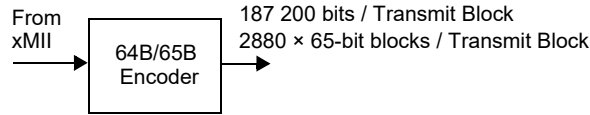


Figure 166–7—xMII data stream encoding

166.2.2.3 PCS transmit ordering

The PCS transmit ordering shall follow each sequence of 80×65 -bit blocks, called tx_group80x65b, coming from the payload data path, with a 20-bit encoded PHD sub-block. See Figure 166–11 for details on PCS bit ordering.

A tx_group80x65b followed by a 20-bit encoded PHD sub-block is called an RS-FEC message. The RS-FEC message is 5220 bits long. Therefore, the PCS transmit ordering generates 36 RS-FEC messages per each Transmit Block.

The RS-FEC message is formed as follows:

RS-FEC_message<5199:0> = tx_group80x65b<5199:0>.
RS-FEC_message<5219:5200> = PHD_sub-block<19:0>.

A complete PHD, coded and chunked into 36×20 -bit encoded PHD sub-blocks, is transmitted every Transmit Block. A Transmit Block starts with the first bit of the first RS-FEC message that carries the first 20-bit encoded PHD sub-block of a new PHD.

166.2.2.4 RS-FEC encoder

The RS-FEC encoder takes the 5220-bit RS-FEC message and shall generate the 22×10 -bit parity symbols (220 parity bits total) as specified in this subclause.

The resulting RS-FEC codeword is called a codeword. The codeword consists of the RS-FEC message, followed by 220 parity bits. The codeword is 5440 bits long. See Figure 166–11 for details on PCS bit ordering.

The RS-FEC message is encoded using a Reed-Solomon encoder operating over the Galois Field $GF(2^{10})$ where the symbol size is 10 bits. The encoder processes 522×10 -bit RS-FEC message symbols to generate 22×10 -bit RS-FEC parity symbols. A codeword of 544×10 -bit RS-FEC symbols is composed of 522×10 -bit RS-FEC message symbols followed by the resulting 22×10 -bit RS-FEC parity symbols. For the purposes of this clause, the particular Reed-Solomon code is denoted as RS-FEC(544,522).

The code is based on the generating polynomial given by Equation (166–1).

$$g(x) = \prod_{j=1}^{22} (x - \alpha^j) = g_{22}x^{22} + g_{21}x^{21} + \dots + g_4x^4 + g_3x^3 + g_2x^2 + g_1x + g_0 \quad (166-1)$$

In this specification of the RS-FEC encoder, x is used in general as the indeterminate variable of any polynomial in mathematical expressions. Polynomial operations are used to specify the parity calculation carried out by the RS-FEC encoder. In Equation (166-1), α is a primitive element of the finite Galois field $GF(2^m)$, therefore α is the root of a primitive polynomial of degree m in $GF(2)$. The primitive polynomial is $x^{10} + x^3 + 1$.

The coefficients of the generator polynomial for the RS-FEC(544,522) code are presented in Table 166-3.

Table 166-3—Coefficients of the generator polynomial $g(x)$ (decimal)

i	g_i	i	g_i
0	807	12	374
1	280	13	544
2	944	14	374
3	621	15	482
4	3	16	555
5	177	17	976
6	365	18	452
7	657	19	899
8	813	20	783
9	1010	21	513
10	712	22	1
11	466	—	—

The contents of the RS-FEC message are represented by a polynomial $m(x)$ whose coefficients are the message symbols m_{521} to m_0 as shown in Equation (166-2).

$$m(x) = m_{521}x^{543} + m_{520}x^{542} + \dots + m_1x^{23} + m_0x^{22} \quad (166-2)$$

Each message symbol m_i is the bit vector $(m_{i,9}, m_{i,8}, \dots, m_{i,1}, m_{i,0})$, which is identified with the element of the finite field. $m_{i,0}$ is the first bit transmitted. The message symbols are composed of the bits in RS-FEC_message<5219:0> where $m_{i,j} = \text{RS-FEC_message} \langle (521 - i) \times 10 + j \rangle$, for $i = 0$ to 521, and $j = 0$ to 9.

The first symbol input to the encoder is m_{521} .

The parity polynomial $p(x)$ is calculated as the remainder of polynomial division of $m(x)$ by $g(x)$. Its coefficients p_{21} to p_0 , as shown in Equation (166-3), are the parity symbols.

$$p(x) = p_{21}x^{21} + p_{20}x^{20} + \dots + p_2x^2 + p_1x + p_0 \quad (166-3)$$

Each parity symbol p_i is the bit vector $(p_{i,9}, p_{i,8}, \dots, p_{i,1}, p_{i,0})$, which is identified with the element of the finite field. $p_{i,0}$ is the first bit transmitted.

The calculation of the coefficients of $p(x)$ is illustrated in Figure 166–8. The outputs of the delay elements are initialized to zero prior to the computation of the parity for a given message. After the last message symbol, m_0 , is processed by the encoder, the outputs of the delay elements are the parity symbols for that message.

The codeword polynomial $c(x)$ is then the sum of $m(x)$ and $p(x)$ where the coefficient of the highest power of x , $c_{543} = m_{521}$ is transmitted first and the coefficient of the lowest power of x , $c_0 = p_0$ is transmitted last. The first bit transmitted from each symbol is bit 0.

The Reed-Solomon encoder functional model is shown in Figure 166–8.

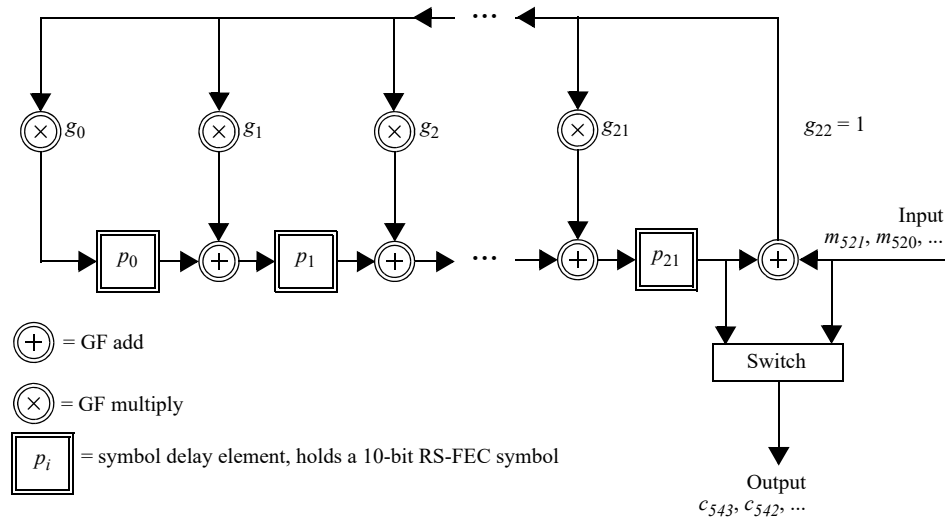


Figure 166–8—Reed-Solomon encoder functional model

Annex 166B provides an example RS-FEC codeword produced by the Reed-Solomon encoding.

166.2.2.5 Scrambler

The 195 840 bits that compose the aggregation of 36 codewords from the output of the RS-FEC encoder shall be scrambled prior to transmission using a scrambler that produces the same result as the implementation shown in Figure 166–9.

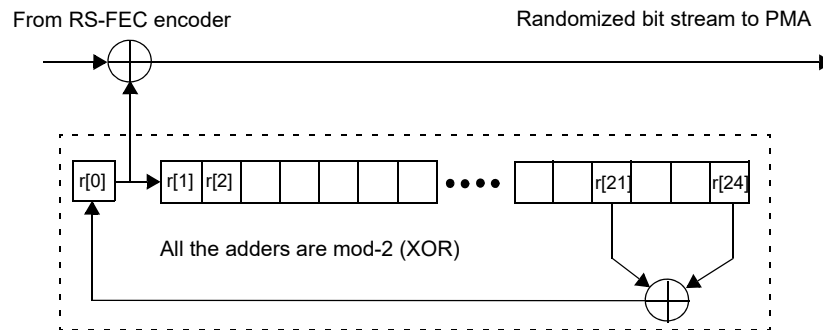


Figure 166–9—Scrambler

The block of 195 840 bits generated at the output of the scrambler is called a Transmit Block.

The shift register of the scrambler in Figure 166–9 $r[0:24]$ shall be initialized with 0x0FB9659 for BASE-AU PHYs with parameter $G = 1$ and with 0x020492C for BASE-AU PHYs with parameter $G = 2$ (see Table 166–1), where the leftmost digit corresponds to the initial value of register element $r[0]$. Therefore, the least significant bit of the rightmost digit corresponds to the initial value of register element $r[24]$.

The initial value of $r[0]$ is exclusive OR'd with the first bit from the RS-FEC encoder to generate the first input bit to the PMA.

The initialization shall be performed at the beginning of each Transmit Block.

Annex 166A provides partial listings of sequences produced by the BASE-U scrambler shift register for $G = 1$ and $G = 2$.

166.2.2.6 PCS transmit bit order

The PCS transmit function shall conform to the PCS physical header data transmit bit order in Figure 166–10.

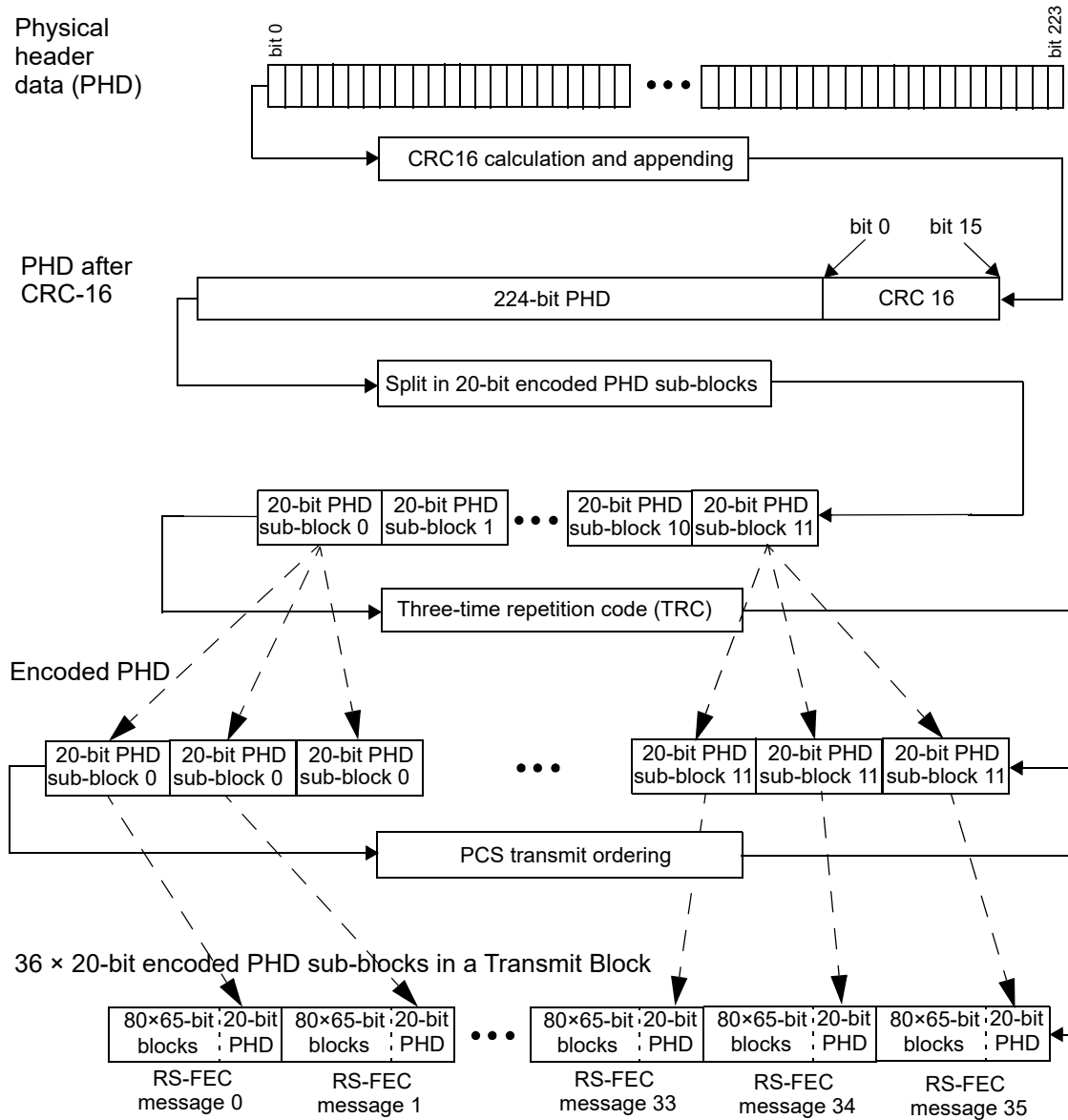
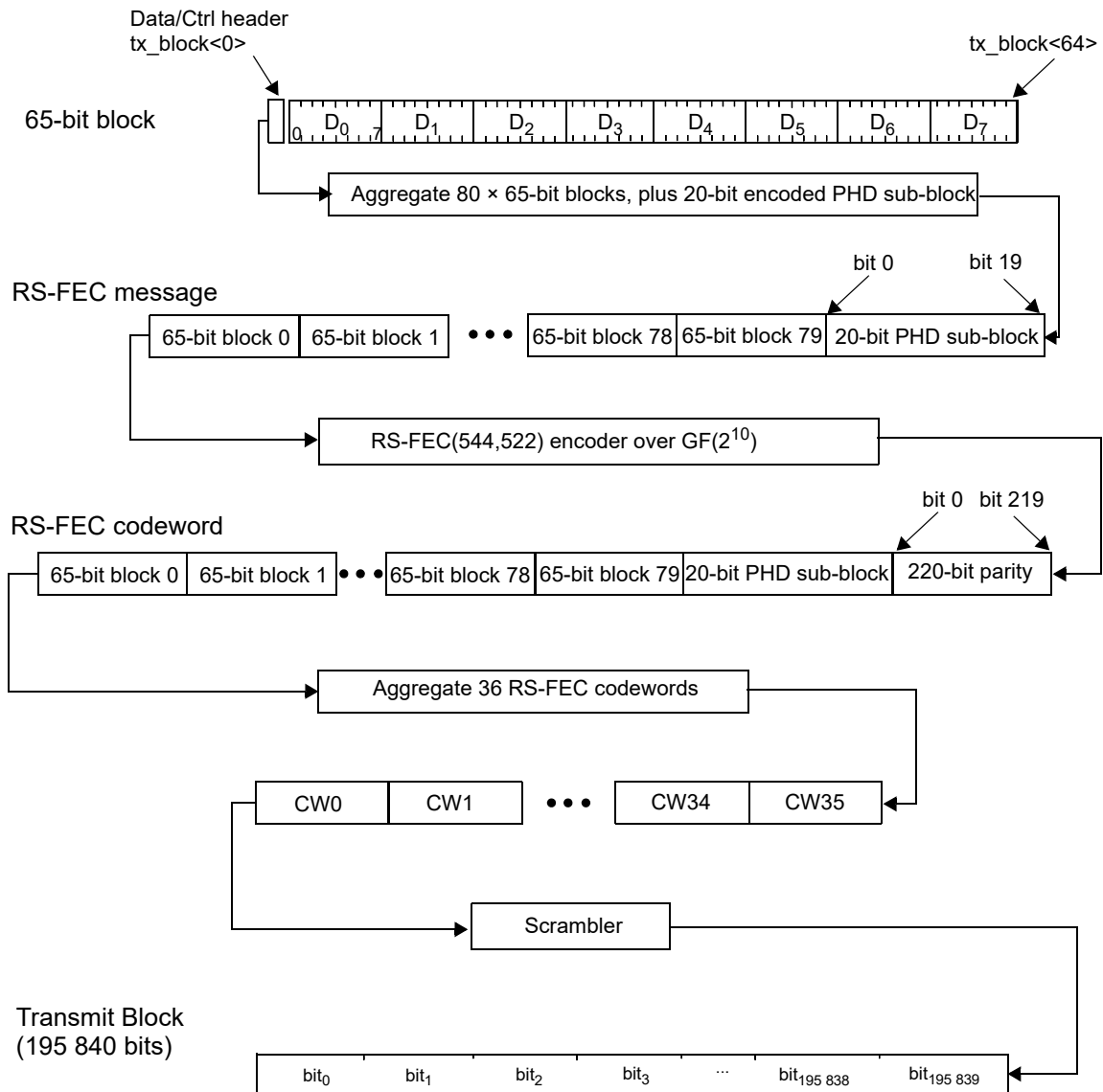


Figure 166–10—Physical header data transmit bit order

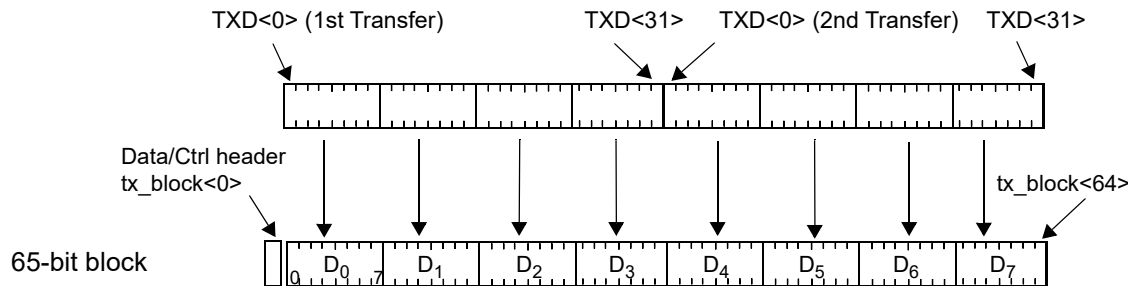
The PCS transmit function shall conform to the PCS transmit bit order in Figure 166–11. The mapping from the XGMII or 25GMII to a 65-bit block, shall conform to Figure 166–12. The mapping from the 50GMII to a 65-bit block, shall conform to Figure 166–13.



NOTE—CW0, CW1, ... CW35 stands for codeword 0, 1, ... 35.

Figure 166–11—PCS transmit bit order

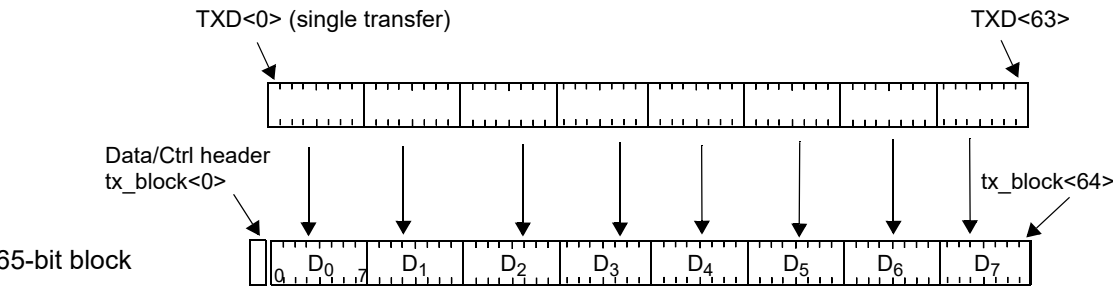
XGMII or 25GMII



NOTE—This figure shows the mapping from the XGMII or 25GMII to a 65-bit block for a block containing eight data characters.

Figure 166–12—PCS mapping from XGMII or 25GMII to a 65-bit block

50GMII



NOTE—This figure shows the mapping from the 50GMII to a 65-bit block for a block containing eight data characters.

Figure 166–13—PCS mapping from 50GMII to a 65-bit block

166.2.2.7 PCS 64B/65B encoding

166.2.2.7.1 Notation conventions

The 64B/65B encoder encodes eight data octets or control characters into a 65-bit block. 65-bit blocks containing control characters also contain a block type field. Data octets are labeled D_0 to D_7 . See 166.2.2.7.4 for BASE-U control character definitions.

For a BASE-U PCS connected to an XGMII or 25GMII, control characters other than /O/, /S/, and /T/ are labeled C_0 to C_7 . The control character for an ordered set is labeled as O_0 or O_4 since it is only valid on the first octet of the XGMII or 25GMII. The control character for start is labeled as S_0 or S_4 for the same reason. The control character for terminate is labeled as T_0 to T_7 .

For a BASE-U PCS connected to a 50GMII, the control characters /I/ and /E/ are labeled C_0 to C_7 . The control character /Q/ for a sequence ordered set is labeled as O_0 since it is only valid on the first octet of the 50GMII. The control character /S/ is labeled as S_0 for the same reason. The control character /T/ is labeled as T_0 to T_7 . The four trailing zero data octets in ordered sets are labeled as Z_4 to Z_7 .

For a BASE-U PCS connected to an XGMII or 25GMII, two XGMII or 25GMII transfers provide eight characters that are encoded into one 65-bit block (see Figure 166–12).

For a BASE-U PCS connected to a 50GMII, only one 50GMII transfer provides eight characters that are encoded into one 65-bit block (see Figure 166–13).

The subscript in the labels defined in the previous paragraphs indicates the position of the character in the eight characters from the XGMII, 25GMII, or 50GMII transfer(s).

Contents of block type fields, data octets, and control characters are shown as hexadecimal values. The LSB of the hexadecimal value represents the first transmitted bit. For instance, the block type field 0x1E is sent from left to right as 01111000. The bits of a transmitted 65-bit block are labeled $tx_block<64:0>$ where $tx_block<0>$ represents the first transmitted bit. The value of the data/ctrl header is shown as a binary value. Binary values are shown with the first transmitted bit (the LSB) on the left.

166.2.2.7.2 Transmit process

The transmit process shall generate 65-bit blocks as specified in the PCS 64B/65B transmit state diagram (Figure 166–16). The contents of each 65-bit block are contained in a vector $tx_block<64:0>$.

166.2.2.7.3 65-bit block structure

The first bit $tx_block<0>$ of a 65-bit block is the data/ctrl header. 65-bit blocks are either data blocks or control blocks. The data/ctrl header is 0 for data blocks and 1 for control blocks. The remainder of the 65-bit block contains the payload.

Data blocks contain eight data characters. Control blocks begin with an 8-bit block type field that indicates the format of the remainder of the 65-bit block. For control blocks containing a Start or Terminate character, that character is implied by the block type field. Other control characters are encoded in a 7-bit control code or a 4-bit O Code. Each control block contains eight characters.

The format of the 65-bit blocks for a BASE-U PCS connected to an XGMII or 25GMII is as shown in Figure 166–14.

The format of the 65-bit blocks for a BASE-U PCS connected to a 50GMII is as shown in Figure 166–15.

Input Data	data ctrl header	Block Payload										
Bit Position:	0	164										
Data Block Format:	0											
D ₀ D ₁ D ₂ D ₃ /D ₄ D ₅ D ₆ D ₇	0	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇			
Control Block Formats:		Block Type Field										
C ₀ C ₁ C ₂ C ₃ /C ₄ C ₅ C ₆ C ₇	1	0x1E	C ₀	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇		
C ₀ C ₁ C ₂ C ₃ /O ₄ D ₅ D ₆ D ₇	1	0x2D	C ₀	C ₁	C ₂	C ₃	O ₄	D ₅	D ₆	D ₇		
C ₀ C ₁ C ₂ C ₃ /S ₄ D ₅ D ₆ D ₇	1	0x33	C ₀	C ₁	C ₂	C ₃			D ₅	D ₆	D ₇	
O ₀ D ₁ D ₂ D ₃ /S ₄ D ₅ D ₆ D ₇	1	0x66	D ₁	D ₂	D ₃	O ₀			D ₅	D ₆	D ₇	
O ₀ D ₁ D ₂ D ₃ /O ₄ D ₅ D ₆ D ₇	1	0x55	D ₁	D ₂	D ₃	O ₀	O ₄	D ₅	D ₆	D ₇		
S ₀ D ₁ D ₂ D ₃ /D ₄ D ₅ D ₆ D ₇	1	0x78	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇			
O ₀ D ₁ D ₂ D ₃ /C ₄ C ₅ C ₆ C ₇	1	0x4B	D ₁	D ₂	D ₃	O ₀	C ₄	C ₅	C ₆	C ₇		
T ₀ C ₁ C ₂ C ₃ /C ₄ C ₅ C ₆ C ₇	1	0x87			C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	
D ₀ T ₁ C ₂ C ₃ /C ₄ C ₅ C ₆ C ₇	1	0x99	D ₀			C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	
D ₀ D ₁ T ₂ C ₃ /C ₄ C ₅ C ₆ C ₇	1	0xAA	D ₀	D ₁			C ₃	C ₄	C ₅	C ₆	C ₇	
D ₀ D ₁ D ₂ T ₃ /C ₄ C ₅ C ₆ C ₇	1	0xB4	D ₀	D ₁	D ₂			C ₄	C ₅	C ₆	C ₇	
D ₀ D ₁ D ₂ D ₃ /T ₄ C ₅ C ₆ C ₇	1	0xCC	D ₀	D ₁	D ₂	D ₃			C ₅	C ₆	C ₇	
D ₀ D ₁ D ₂ D ₃ /D ₄ T ₅ C ₆ C ₇	1	0xD2	D ₀	D ₁	D ₂	D ₃	D ₄			C ₆	C ₇	
D ₀ D ₁ D ₂ D ₃ /D ₄ D ₅ T ₆ C ₇	1	0xE1	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅			C ₇	
D ₀ D ₁ D ₂ D ₃ /D ₄ D ₅ D ₆ T ₇	1	0xFF	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆			

NOTE—This figure is the same as Figure 55–9, Figure 113–9, Figure 126–8, and Figure 149–8 in IEEE Std 802.3-2022.

Figure 166–14—65-bit block format for a BASE-U PCS connected to an XGMII or 25GMII

Note that Figure 166–14 and Figure 166–15 are the same, except that block type field 0x4B is encoded differently, and block type field values 0x55, 0x66, 0x33, and 0x2D are only in Figure 166–14.

In both figures, the column labeled Input Data shows, in abbreviated form, the eight characters used to create the 65-bit block. These characters are either data characters or control characters and, when transferred across the xMII, the corresponding TXC or RXC bit is set accordingly. Within the Input Data column, D₀ through D₇ are data octets and are transferred with the corresponding TXC or RXC bit set to zero. All other characters are control octets and are transferred with the corresponding TXC or RXC bit set to one. The single bit fields (thin rectangles with no label in the figure) are sent as zero and ignored upon receipt.

Bits and field positions are shown with the least significant bit on the left. Hexadecimal numbers are shown prepended with ‘0x’, and with the least significant digit on the right. For example the block type field 0x1E is sent as 01111000 representing bits 1 through 8 of the 65-bit block. The least significant bit for each field is placed in the lowest numbered position of the field.

All unused values of block type field⁸ are reserved.

⁸The block type field values have been chosen to have a four-bit Hamming distance between them.

Input Data	data ctrl header	Block Payload											
Bit Position:	0	1 64											
Data Block Format:	0												
D ₀ D ₁ D ₂ D ₃ D ₄ D ₅ D ₆ D ₇	0	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇				
Control Block Formats:		Block Type Field											
C ₀ C ₁ C ₂ C ₃ C ₄ C ₅ C ₆ C ₇	1	0x1E	C ₀	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇			
S ₀ D ₁ D ₂ D ₃ D ₄ D ₅ D ₆ D ₇	1	0x78	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇				
O ₀ D ₁ D ₂ D ₃ Z ₄ Z ₅ Z ₆ Z ₇	1	0x4B	D ₁	D ₂	D ₃	O ₀	0x000_0000						
T ₀ C ₁ C ₂ C ₃ C ₄ C ₅ C ₆ C ₇	1	0x87			C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇		
D ₀ T ₁ C ₂ C ₃ C ₄ C ₅ C ₆ C ₇	1	0x99	D ₀		C ₂	C ₃	C ₄	C ₅	C ₆	C ₇			
D ₀ D ₁ T ₂ C ₃ C ₄ C ₅ C ₆ C ₇	1	0xAA	D ₀	D ₁		C ₃	C ₄	C ₅	C ₆	C ₇			
D ₀ D ₁ D ₂ T ₃ C ₄ C ₅ C ₆ C ₇	1	0xB4	D ₀	D ₁	D ₂		C ₄	C ₅	C ₆	C ₇			
D ₀ D ₁ D ₂ D ₃ T ₄ C ₅ C ₆ C ₇	1	0xCC	D ₀	D ₁	D ₂	D ₃		C ₅	C ₆	C ₇			
D ₀ D ₁ D ₂ D ₃ D ₄ T ₅ C ₆ C ₇	1	0xD2	D ₀	D ₁	D ₂	D ₃	D ₄		C ₆	C ₇			
D ₀ D ₁ D ₂ D ₃ D ₄ D ₅ T ₆ C ₇	1	0xE1	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅		C ₇			
D ₀ D ₁ D ₂ D ₃ D ₄ D ₅ D ₆ T ₇	1	0xFF	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆				

NOTE—This figure is the same as Figure 113–10 in IEEE Std 802.3-2022.

Figure 166–15—65-bit block format for a BASE-U PCS connected to a 50GMII

166.2.2.7.4 Control codes

The same set of control characters are supported by the XGMII, 25GMII, 50GMII, and the BASE-U PCS. The representations of the control characters are the control codes. The xMII encodes a control character into an octet (an eight-bit value). The BASE-U PCS encodes the start and terminate control characters implicitly by the block type field. The BASE-U PCS encodes the ordered set control codes using a combination of the block type field and a four-bit O code for each ordered set. The BASE-U PCS encodes each of the other control characters into a seven-bit C code.

The control characters and their mappings to BASE-U control codes and xMII control codes are specified in Table 166–4 for a BASE-U PCS connected to an XGMII or 25GMII, and in Table 166–5 for a BASE-U PCS connected to a 50GMII.

For a BASE-U PCS connected to an XGMII or 25GMII, all control code values that do not appear in Table 166–4 shall not be transmitted and shall be treated as an error if received. For a BASE-U PCS connected to a 50GMII, all control code values that do not appear in Table 166–5 shall not be transmitted and shall be treated as an error if received.

Note that Table 166–4 and Table 166–5 are the same, except that reserved control characters are only in Table 166–4.

The ability to transmit or receive Low Power Idle (LPI) is required for PHYs with EEE capability (see Clause 78). If EEE capability is not enabled, LPI shall not be transmitted and shall be treated as an error if received.

Table 166–4—Control codes for a BASE-U PCS connected to an XGMII or 25GMII

Control character	Notation	XGMII or 25GMII control code	BASE-U PCS control code	BASE-U PCS O code
idle	/I/	0x07	0x00	
LPI	/LI/	0x06	0x06	
start	/S/	0xFB	Encoded by block type field	
terminate	/T/	0xFD	Encoded by block type field	
error	/E/	0xFE	0x1E	
Sequence ordered set	/Q/	0x9C	Encoded by block type field plus O code	0x0
reserved0		0x1C	0x2D	
reserved1		0x3C	0x33	
reserved2		0x7C	0x4B	
reserved3		0xBC	0x55	
reserved4		0xDC	0x66	
reserved5		0xF7	0x78	
Signal ordered set ^a	/Fsig/	0x5C	Encoded by block type field plus O code	0xF

^a Reserved for INCITS T11 Fibre Channel use.

Table 166–5—Control codes for a BASE-U PCS connected to a 50GMII

Control character	Notation	50GMII control code	BASE-U PCS control code	BASE-U PCS O code
idle	/I/	0x07	0x00	
LPI	/LI/	0x06	0x06	
start	/S/	0xFB	Encoded by block type field	
terminate	/T/	0xFD	Encoded by block type field	
error	/E/	0xFE	0x1E	
Sequence ordered set	/Q/	0x9C	Encoded by block type field plus O code	0x0
Signal ordered set ^a	/Fsig/	0x5C	Encoded by block type field plus O code	0xF

^a Reserved for INCITS T11 Fibre Channel use.

166.2.2.7.5 Idle (/I/)

Idle control characters (/I/) are transmitted when idle control characters are received from the xMII. Idle characters may be added or deleted by the PCS to adapt between clock rates. For a BASE-U PCS connected to an XGMII or 25GMII, /I/ insertion and deletion shall occur in groups of four. For a BASE-U PCS connected to a 50GMII, /I/ insertion and deletion shall occur in groups of eight. /I/s may be added following idle or ordered sets. /I/ shall not be added while data is being received. For a BASE-U PCS connected to an XGMII or 25GMII, when deleting /I/s, the first four characters after a /T/ shall not be deleted.

166.2.2.7.6 LPI (/LI/)

Low Power Idle (LPI) control characters (/LI/) on the xMII indicate that the LPI client is requesting operation in the LPI transmit mode. A continuous stream of LPI control characters (/LI/) is used to maintain a link in the LPI transmit mode. Idle control characters (/I/) are used to transition from the LPI transmit mode to the normal mode. PHYs with EEE capability respond to the LPI xMII control characters using the procedure specified in 166.4. LPI characters may be added or deleted by the PCS to adapt between clock rates. For a BASE-U PCS connected to an XGMII or 25GMII, /LI/ insertion and deletion shall occur in groups of four. For a BASE-U PCS connected to a 50GMII, /LI/ insertion and deletion shall occur in groups of eight. /LI/s may be added following Low Power Idle characters. /LI/ shall not be added while data is being received.

If EEE is not supported, then /LI/ is not a valid control character.

166.2.2.7.7 Start (/S/)

The start control character (/S/) indicates the start of a packet. This delimiter is only valid on the first octet of the xMII transaction (TXD<7:0> and RXD<7:0>). Receipt of an /S/ on any other octet of TXD indicates an error. For a BASE-U PCS connected to an XGMII or 25GMII, block type field values implicitly encode an /S/ as the fifth or first character of the 65-bit block. For a BASE-U PCS connected to a 50GMII, block type field values implicitly encode an /S/ as the first character of the 65-bit block. These are the only characters of a 65-bit block on which a start can occur.

166.2.2.7.8 Terminate (/T/)

The terminate control character (/T/) indicates the end of a packet. Since packets may be any length, the /T/ can occur on any octet of the xMII transaction and within any character of the 65-bit block. The location of the /T/ in the 65-bit block is implicitly encoded in the block type field. A valid end of packet occurs when a 65-bit block containing a /T/ is followed by a control block that does not contain a /T/ or an /E/.

166.2.2.7.9 Ordered set (/O/)

Ordered sets are used to extend the ability to send control and status information over the link such as remote fault and local fault status. For a BASE-U PCS connected to an XGMII or 25GMII, ordered sets consist of a control character followed by three data characters. For a BASE-U PCS connected to a 50GMII, ordered sets consist of a control character followed by three data characters followed by four zero data characters. Ordered sets always begin on the first octet of the xMII transaction.

The BASE-U PCS uses one kind of ordered set: the sequence ordered set (see 46.3.4). The sequence ordered set control character is denoted /Q/. An additional ordered set, the signal ordered set, has been reserved and it begins with another control code. The four-bit O field encodes the control code. See Table 166–4 and Table 166–5 for the mappings.

The ordered set control characters (/O/) indicate the start of an ordered set. When it is necessary to designate the control character for the sequence ordered set specifically, /Q/ is used. /O/ is only valid on the first octet

of the xMII transaction. Receipt of an /O/ on any other octet of TXD indicates an error. For a BASE-U PCS connected to an XGMII or 25GMII, block type field values implicitly encode an /O/ as the first or fifth character of the block. For a BASE-U PCS connected to a 50GMII, block type field values implicitly encode an /O/ as the first character of the block. The 4-bit O code encodes the specific /O/ character for the ordered set.

Continuous LBLOCK_T are transmitted when link has not been established yet (see 166.2.2.8 and Figure 166–16). For a BASE-U PCS connected to an XGMII or 25GMII, LBLOCK_T contains two Local Fault ordered sets. For a BASE-U PCS connected to a 50GMII, LBLOCK_T contains only one Local Fault ordered set. The Local Fault ordered set for XGMII and 25GMII is defined in 46.3.4. The Local Fault ordered set for 50GMII is defined in 81.3.4.

Sequence ordered sets may be deleted by the PCS to adapt between clock rates. Such deletion shall only occur when two consecutive sequence ordered sets have been received and shall delete only one of the two. Only Idles may be inserted for clock compensation. Signal ordered sets are not deleted for clock compensation.

166.2.2.7.10 Error (/E/)

The /E/ is sent whenever an /E/ is received. The /E/ allows physical sublayers such as the PCS to propagate received errors.

166.2.2.8 PCS 64B/65B transmit state diagram parameters

166.2.2.8.1 Constants

EBLOCK_T<64:0>

65-bit vector to be sent to the RS-FEC encoder containing /E/ in all the eight character locations.

LBLOCK_T<64:0>

For a BASE-U PCS connected to an XGMII or 25GMII, a 65-bit vector to be sent to the RS-FEC encoder containing two Local Fault ordered sets.

For a BASE-U PCS connected to a 50GMII, a 65-bit vector to be sent to the RS-FEC encoder containing one Local Fault ordered set.

166.2.2.8.2 Variables

pcs_reset

Boolean variable that controls the resetting of the PCS. It is TRUE whenever a reset is necessary including when reset is initiated from the MDIO, during power on, and when the MDIO has put the PCS into Low Power = 1 (see Table 166–21).

tx_block<64:0>

Vector containing the output from the 64B/65B encoder. The format for this vector is shown in Figure 166–14 for a BASE-U PCS connected to an XGMII or 25GMII, and in Figure 166–15 for a BASE-U PCS connected to a 50GMII. The leftmost bit in the figure is tx_block<0> and the rightmost bit is tx_block<64>.

tx_raw<71:0>

For a BASE-U PCS connected to an XGMII or 25GMII, a vector containing two successive XGMII or 25GMII transfers. TXC<3:0> for the first transfer are placed in tx_raw<3:0>.

TXC<3:0> for the second transfer are placed in tx_raw<7:4>. TXD<31:0> for the first transfer are placed in tx_raw<39:8>. TXD<31:0> for the second transfer are placed in tx_raw<71:40>.

For a BASE-U PCS connected to a 50GMII, a vector containing a single 50GMII transfer.

TXC<7:0> for the transfer are placed in tx_raw<7:0>. TXD<63:0> for the transfer are placed in tx_raw<71:8>.

tx_xmii_enable

Variable set by the PHY Tx control state diagram to control the 64B/65B encoder operation (see 166.3.4.2).

Values:

TRUE: The 64B/65B encoder encodes the xMII transmit data stream transfers into 65-bit blocks.

FALSE: The 64B/65B encoder does not encode the xMII transmit data stream. Local Fault ordered set(s) are encoded in transmitted 65-bit blocks, which may be used by the link partner receiver as a pre-known sequence to support clock and data recovery.

166.2.2.8.3 Functions

ENCODE(tx_raw<71:0>)

Encodes the 72-bit vector received from the xMII, returning 65-bit vector tx_block. The ENCODE function shall encode the block as specified in 166.2.2.7.

T_BLOCK_TYPE = {C, S, T, D, E, LI}

This function classifies each 72-bit tx_raw vector as belonging to one of the six types {C, S, T, D, E, LI} depending on its contents.

Values for a BASE-U PCS connected to an XGMII or 25GMII:

C; The vector contains a data/ctrl header of 1 and one of the following:

- a) Eight valid control characters other than /O/, /S/, /T/, and /E/; and, if the EEE capability is supported, zero or four of the characters are /LI/.
- b) One valid ordered set and four valid control characters other than /O/, /S/ and /T/.
- c) Two valid ordered sets.

S; The vector contains an /S/ in its first or fifth character, any characters before the /S/ character are valid control characters other than /O/, /S/, and /T/ or form a valid ordered set, and all characters following the /S/ are data characters.

T; The vector contains a /T/ in one of its characters, all characters before the /T/ are data characters, and all characters following the /T/ are valid control characters other than /O/, /S/, and /T/.

D; The vector contains eight data characters.

E; The vector does not meet the criteria for any other value.

LI; For EEE capability, this vector contains eight /LI/ characters.

Values for a BASE-U PCS connected to a 50GMII:

C; The vector contains one of the following:

- a) Eight valid control characters other than /O/, /S/, /T/, /LI/, and /E/.
- b) One valid ordered set.

S; The vector contains an /S/ in its first character, and all characters following the /S/ are data characters.

T; The vector contains a /T/ in one of its characters, all characters before the /T/ are data characters, and all characters following the /T/ are valid control characters other than /O/, /S/ and /T/.

D; The vector contains eight data characters.

E; The vector does not meet the criteria for any other value.

LI; For EEE capability, this vector contains eight /LI/ characters.

A tx_raw character is a control character if its associated TXC bit is asserted. When a BASE-U PCS is connected to an XGMII or 25GMII, a valid control character is one containing a control code specified in Table 166–4. When a BASE-U PCS is connected to a 50GMII, a valid control character is one containing a control code specified in Table 166–5. For a BASE-U PCS connected to an XGMII or 25GMII, a valid ordered set consists of a valid /O/ character in the first or fifth characters and data characters in the three characters following the /O/. A valid /O/ is any character with a value for O code in Table 166–4. For a BASE-U PCS connected to a 50GMII, a valid

ordered set consists of a valid /O/ character in the first character and data characters in the seven characters following the /O/. A valid /O/ is any character with a value for O code in Table 166–5.

NOTE—A BASE-U PHY without EEE capability classifies vectors containing one or more /LI/ control characters as type E.

T_TYPE(tx_raw<71:0>)

Returns the T_BLOCK_TYPE of the tx_raw<71:0> bit vector.

166.2.2.9 PCS 64B/65B transmit state diagram

The PCS 64B/65B transmit state diagram shown in Figure 166–16 controls the encoding of 65-bit blocks. It makes exactly one transition for each 65-bit block processed.

166.2.3 PCS receive function

The PCS receive function shall conform to the PCS 64B/65B receive state diagram in Figure 166–21 and Figure 166–22.

The PCS receive function accepts detected bits provided by the PMA receive function. The PCS receive function knows to which part of the received Transmit Block the bits belong, based on the Transmit Block alignment information provided by the PMA receive function.

The PCS receive function shall carry out the binary descrambling, RS-FEC decoding, PHD decoding, and 64B/65B decoding.

Following descrambling, the codewords are decoded with the Reed-Solomon decoder. Codewords that cannot be corrected are marked as erroneous by the decoder. Each RS-FEC message obtained after decoding is then separated into a 20-bit encoded PHD sub-block and 80×65 -bit block vectors.

The bits of a received 65-bit block are labeled rx_block<64:0> where rx_block<0> represents the first received bit.

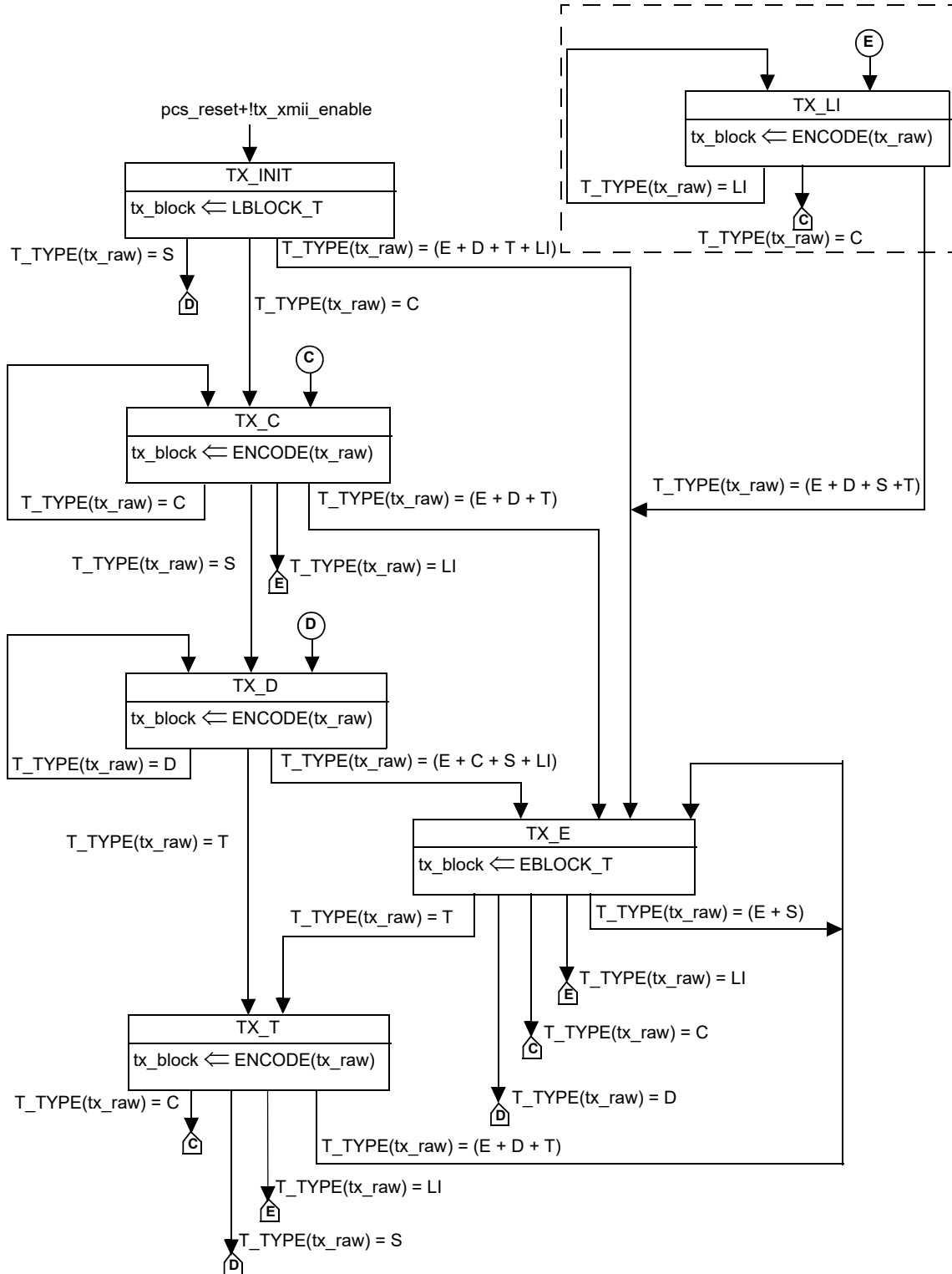
For a BASE-U PCS connected to an XGMII or 25GMII, each 65-bit block vector rx_block<64:0> is decoded to form the XGMII or 25GMII signals RXD<31:0> and RXC<3:0> as specified in 166.2.3.7 with the mapping of Figure 166–19. Two XGMII or 25GMII transfers are decoded from each 65-bit block in this case.

For a BASE-U PCS connected to a 50GMII, each 65-bit block vector rx_block<64:0> is decoded to form the 50GMII signals RXD<63:0> and RXC<7:0> as specified in 166.2.3.7 with the mapping of Figure 166–20. Only one 50GMII transfer is decoded from each 65-bit block in this case.

When the PHY Rx control state diagram sets rx_xmii_enable to TRUE (see Figure 166–24), the PCS receive process decodes the received 65-bit blocks and generates the corresponding xMII signals.

The 36×20 -bit encoded PHD sub-blocks that are in a Transmit Block are TRC decoded by majority voting for error correction and CRC16 checked to generate a received PHD.

Only received PHDs that have been marked as correct by the CRC16 check are used by the PHY.



NOTE—Optional state (inside the dotted box) and transition E are mandatory for PHYs with the EEE capability.

Figure 166–16—PCS 64B/65B transmit state diagram

166.2.3.1 Descrambler

The descrambler shall process the 195 840 bits of a received Transmit Block to reverse the effect of the scrambler using the same polynomial and the same initialization value as specified in 166.2.2.5. The PCS descrambles the data stream and returns the proper sequence of bits to the RS-FEC decoder.

NOTE—The timing of descrambler initialization is established during link establishment, using knowledge of the sequence (LBLOCK_T) sent by the link partner (see 166.3.4.3).

166.2.3.2 RS-FEC decoder

The descrambled bits shall be RS-FEC decoded into RS-FEC messages, with error correction and error detection. The RS-FEC decoder shall correct any combination of up to $t = 11$ symbol errors in a codeword and detect any combination of up to $2t = 22$ symbol errors in a codeword. If during RS-FEC decoding it is detected that a codeword contains errors that could not be corrected, the resulting bits belonging to that codeword shall be marked as corrupt by setting the R_BLOCK_TYPE of the affected 65-bit blocks equal to E (see 166.2.3.7.3), and the codeword shall be counted as an RS-FEC codeword error and shall be reflected in the RS-FEC codeword error counter (see 45.2.3.95).

166.2.3.3 PCS receiver ordering

The PCS receiver shall separate each RS-FEC message into the group of 80×65 -bit blocks and the 20-bit encoded PHD sub-block as specified in Figure 166–18.

166.2.3.4 PHD decoding

The PHD decoding includes TRC decoding by majority voting for error correction and CRC16 checking for each received PHD. Only when the CRC16 computation indicates that the received PHD is correct shall the contents of the different PHD fields be available to the PMA and PCS receive sublayers. The PHD decoding function shall conform to the PHD receive bit order in Figure 166–17.

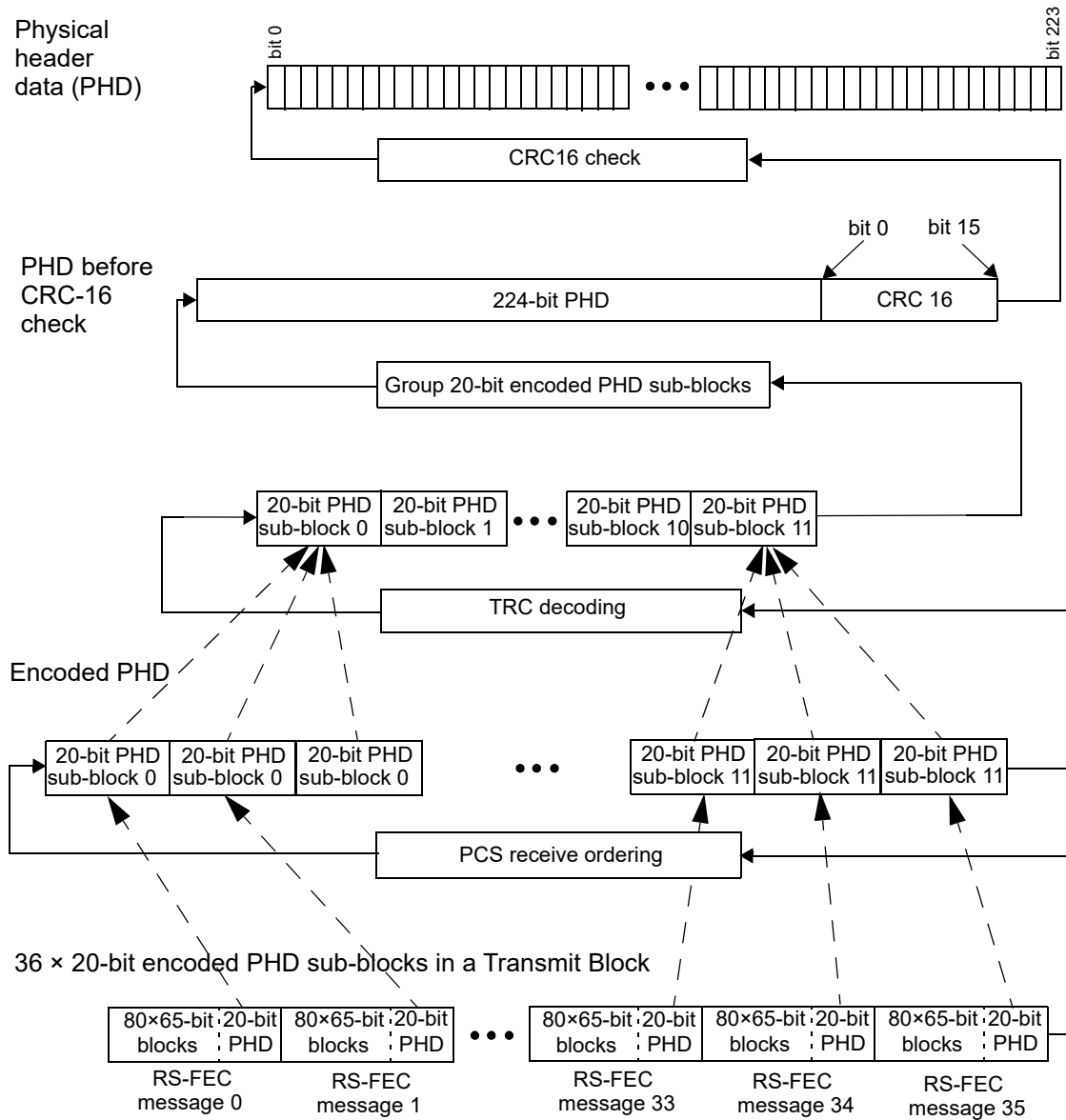


Figure 166-17—Physical header data receive bit order

166.2.3.5 Invalid 65-bit blocks

A 65-bit block is invalid and its R_BLOCK_TYPE set to E if any of the following conditions exist:

- The block type field contains a reserved value (see 166.2.2.7.3).
- Any control character contains a value not in Table 166-4 or Table 166-5 (see 166.2.2.7.4).
- Any O code contains a value not in Table 166-4 or Table 166-5 (see 166.2.2.7.4).
- The 65-bit block contains information from an invalid RS-FEC codeword (see 166.2.3.2).

166.2.3.6 PCS receive bit order

The PCS receive function shall conform to the PCS receive bit order in Figure 166–18.

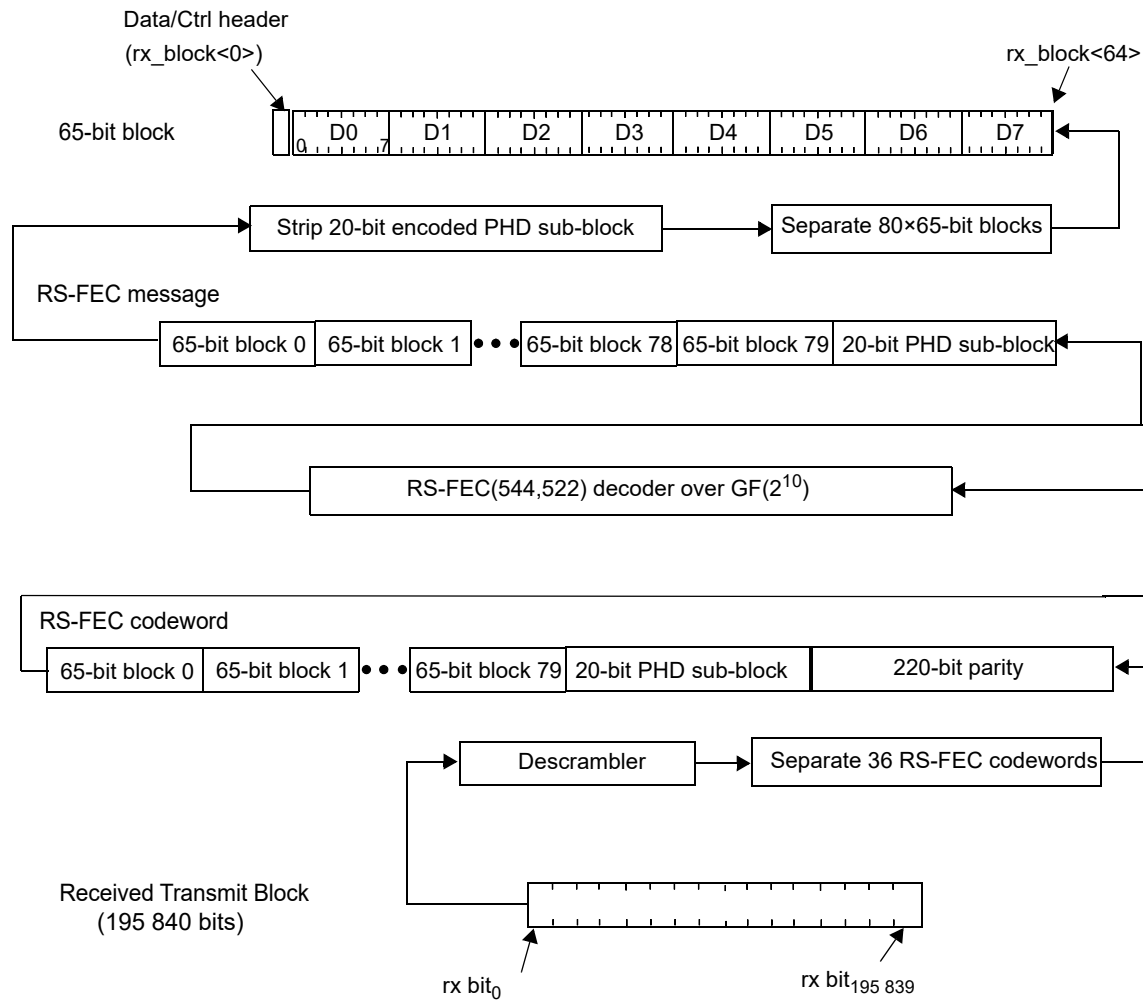
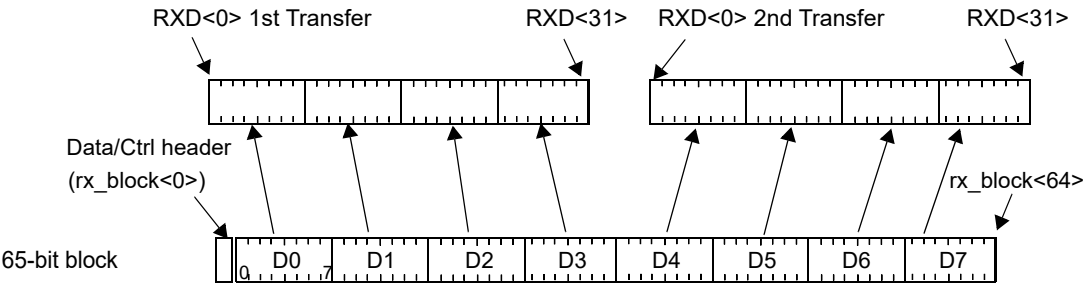


Figure 166–18—PCS receive bit ordering

The mapping from 65-bit blocks to XGMII or 25GMII shall conform to Figure 166–19.

XGMII or
25GMII

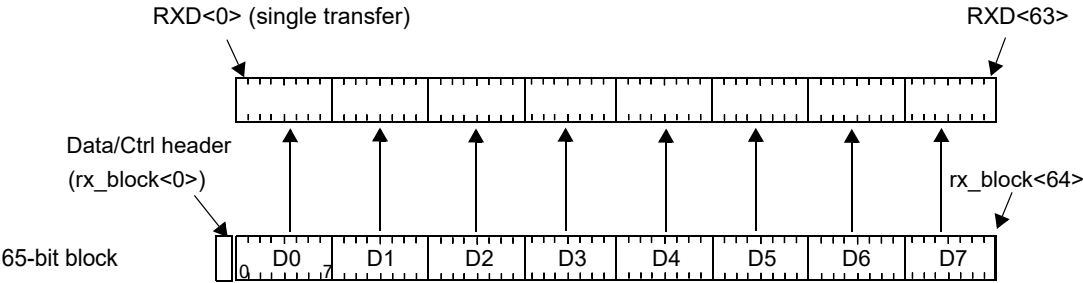


NOTE—This figure shows the mapping from a 65-bit block containing eight data characters to the XGMII or 25GMII.

Figure 166–19—PCS mapping from a 65-bit block to the XGMII or 25GMII

The mapping from 65-bit blocks to 50GMII shall conform to Figure 166–20.

50GMII



NOTE—This figure shows the mapping from a 65-bit block containing eight data characters to the 50GMII.

Figure 166–20—PCS mapping from a 65-bit block to the 50GMII

166.2.3.7 PCS 64B/65B receive state diagram parameters

166.2.3.7.1 Constants

EBLOCK_R<71:0>

72-bit vector to be sent to the xMII interface containing /E/ in all the eight character locations.

LBLOCK_R<71:0>

For a BASE-U PCS connected to an XGMII or 25GMII, a 72-bit vector to be sent to the xMII interface containing two Local Fault ordered sets. The Local Fault ordered set for XGMII and 25GMII is defined in 46.3.4.

For a BASE-U PCS connected to a 50GMII, a 72-bit vector to be sent to the xMII interface containing one Local Fault ordered set. The Local Fault ordered set for 50GMII is defined in 81.3.4.

LPBLOCK_R<71:0>

72-bit vector to be sent to the xMII containing /LI/ in all the eight character locations.

IBLOCK_R<71:0>

72-bit vector to be sent to the xMII containing /I/ in all the eight character locations.

166.2.3.7.2 Variables

All the variables used in the PCS 64B/65B receive state diagram that have not been previously introduced follow:

rx_block<64:0>

Vector containing the input to the 64B/65B decoder. The format for this vector is shown in Figure 166–14 for a BASE-U PCS connected to an XGMII or 25GMII, and Figure 166–15 for a BASE-U PCS connected to a 50GMII. The leftmost bit in the figures is rx_block<0> and the rightmost bit is rx_block<64>.

rx_raw<71:0>

For a BASE-U PCS connected to an XGMII or 25GMII, a vector containing two successive XGMII or 25GMII output transfers. RXC<3:0> for the first transfer are taken from rx_raw<3:0>. RXC<3:0> for the second transfer are taken from rx_raw<7:4>. RXD<31:0> for the first transfer are taken from rx_raw<39:8>. RXD<31:0> for the second transfer are taken from rx_raw<71:40>. For a BASE-U PCS connected to a 50GMII, a vector containing a single 50GMII output transfer. RXC<7:0> for the transfer are taken from rx_raw<7:0>. RXD<64:0> for the transfer are taken from rx_raw<71:8>.

sotxb_synch

Variable set by the PMA Transmit Block synchronization function to indicate synchronization with the beginning of Transmit Blocks.

Values:

OK: Transmit Block synchronization has been achieved.

NOT_OK: Transmit Block synchronization has not been achieved.

166.2.3.7.3 Functions

DECODE(rx_block<64:0>)

In the PCS receive process, this function takes as its argument the 65-bit rx_block<64:0> from the RS-FEC decoder and decodes it returning a vector rx_raw<71:0>, which is sent to the xMII. The DECODE function shall decode the rx_block based on the specification in 166.2.2.7.

R_BLOCK_TYPE = {C, S, T, D, I, LI, W, E}

When the EEE capability is not supported, this function classifies each 65-bit rx_block vector as belonging to one of the five types {C, S, T, D, E} depending on its contents.

When the EEE capability is supported, this function classifies each 65-bit rx_block vector as belonging to one of the eight types {C, S, T, D, I, LI, W, E} depending on its contents. A vector

may simultaneously belong to C and I types when it contains eight valid control characters that are all /I/, but in every other case the vector belongs to only one type.

Values for a BASE-U PCS connected to an XGMII or 25GMII:

- C; The vector contains a data/ctrl header of 1 and one of the following:
 - a) A block type field of 0x1E and eight valid control characters other than /E/ and, if the EEE capability is supported, zero or four of the characters are /LI/.
 - b) A block type field of 0x2D or 0x4B, a valid O code, and four valid control characters.
 - c) A block type field of 0x55 and two valid O codes.
- S; The vector contains a data/ctrl header of 1 and one of the following:
 - a) A block type field of 0x33 and four valid control characters.
 - b) A block type field of 0x66 and a valid O code.
 - c) A block type field of 0x78.
- T; The vector contains a data/ctrl header of 1, a block type field of 0x87, 0x99, 0xAA, 0xB4, 0xCC, 0xD2, 0xE1, or 0xFF and all control characters are valid.
- D; The vector contains a data/ctrl header of 0.
- I; If the optional EEE capability is supported, then the I type is a special case of the C type where the vector contains a data/ctrl header of 1, a block type field of 0x1E, and eight control characters of /I/.
- LI; If the optional EEE capability is supported, then the LI type occurs when the vector contains a data/ctrl header of 1, a block type field of 0x1E, and eight control characters of /LI/.
- W; If the optional EEE capability is supported, then the W type occurs when the vector contains an LPI wake 65-bit block as defined in 166.4.3.2.
- E; The vector does not meet the criteria for any other value.

A valid control character is one containing a BASE-U control code in Table 166–4. A valid O code is one containing an O code specified in Table 166–4.

Values for a BASE-U PCS connected to a 50GMII:

- C; The vector contains a data/ctrl header of 1 and one of the following:
 - a) A block type field of 0x1E and eight valid control characters other than /E/ or /LI/.
 - b) A block type field of 0x4B.
- S; The vector contains a data/ctrl header of 1 and the following:
 - a) A block type field of 0x78.
- T; The vector contains a data/ctrl header of 1, a block type field of 0x87, 0x99, 0xAA, 0xB4, 0xCC, 0xD2, 0xE1, or 0xFF and all control characters are valid.
- D; The vector contains a data/ctrl header of 0.
- LI; If the optional EEE capability is supported, the LI type is supported where the vector contains a data/ctrl header of 1, a block type field of 0x1E, and eight control characters of /LI/.
- W; If the optional EEE capability is supported, then the W type occurs when the vector contains an LPI wake 65-bit block as defined in 166.4.3.2.
- E; The vector does not meet the criteria for any other value.

A valid control character is one containing a BASE-U control code in Table 166–5. A valid O code is one containing an O code specified in Table 166–5.

NOTE—A BASE-U PHY without EEE capability classifies vectors containing one or more /LI/ control characters as type E.

R_TYPE(rx_block<64:0>)

Returns the R_BLOCK_TYPE of the rx_block<64:0> bit vector.

R_TYPE_NEXT

Prescient end of packet check function. It returns the R_BLOCK_TYPE of the rx_block vector immediately following the current rx_block vector.

166.2.3.7.4 Counters

The following counters are required for PHYs that support the EEE capability:

lpi_rxw_err_cnt

An integer value that counts the number of received wake on error conditions. lpi_rxw_err_cnt is reset to zero when pcs_reset is set. The counter is reflected in register 3.22 (see [45.2.3.12](#)).

lpi_rx_wake_cnt

An integer value that counts the number of 65-bit blocks rx_block<64:0> of type W.

166.2.3.8 PCS 64B/65B receive state diagram

The PCS 64B/65B receive state diagram is shown in Figure 166–21 and Figure 166–22 and controls the decoding of received 65-bit blocks. It makes exactly one transition for each received 65-bit block processed.

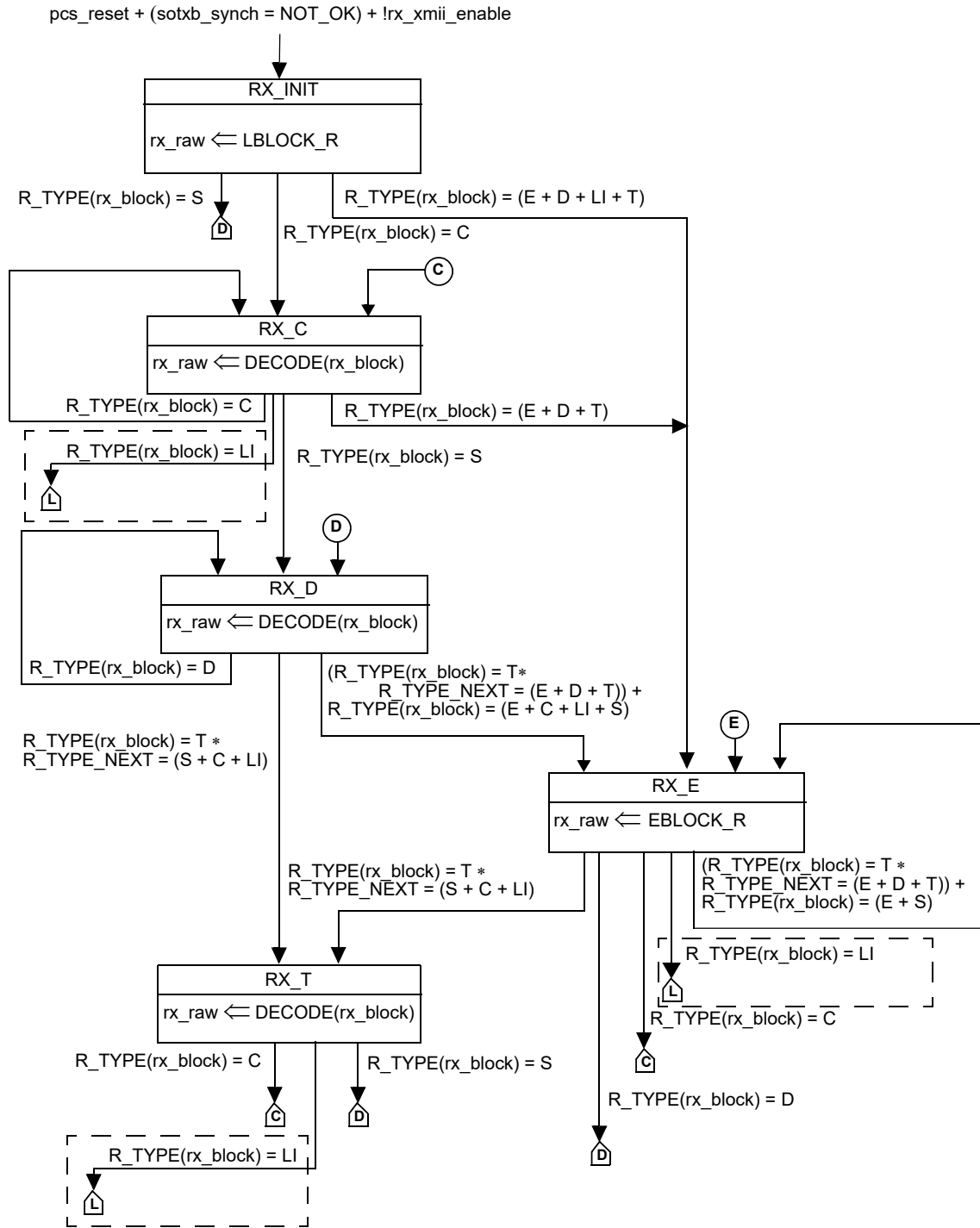
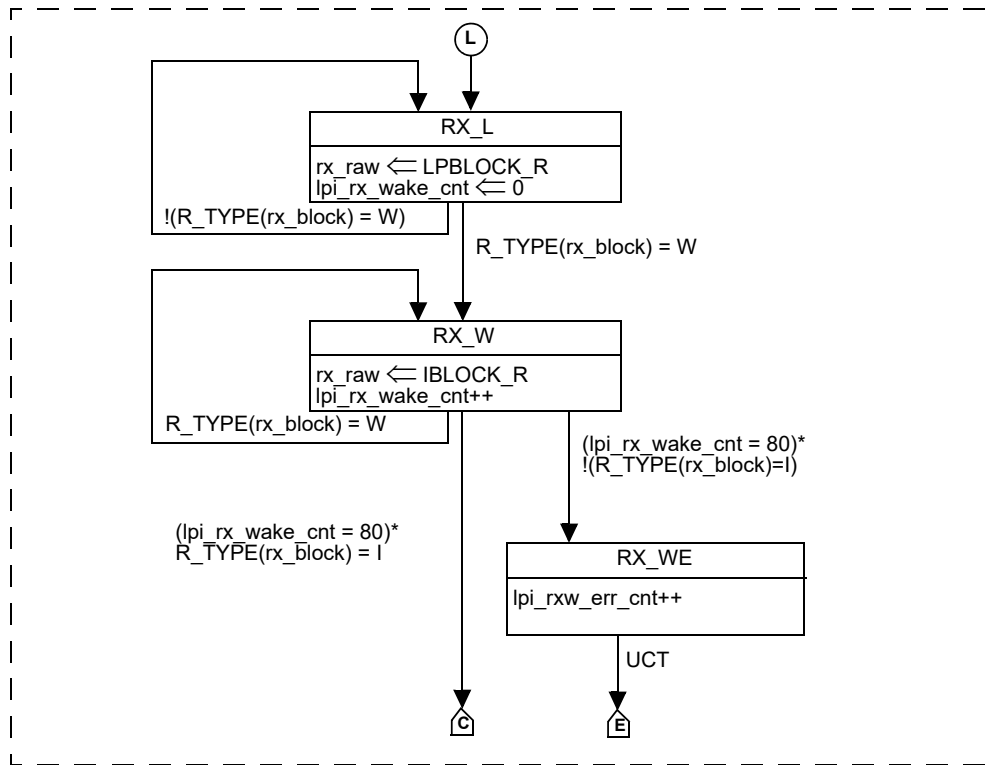


Figure 166–21—PCS 64B/65B receive state diagram, part a



NOTE—This figure is mandatory for PHYs with the EEE capability.

Figure 166-22—PCS 64B/65B receive state diagram, part b

166.3 Physical Medium Attachment (PMA) sublayer

The Physical Medium Attachment sublayer includes the transmit and receive functions, protocols, and state diagrams for PHY Tx control (see 166.3.4.2), PHY Rx control (see 166.3.4.3), link monitoring (see 166.3.4.4), PHD monitoring (see 166.3.4.5), and PHY quality monitor (see 166.3.4.6). These protocols include exchange of information that is encoded within the PHD and the state diagrams that control both the local and remote PHYs. PMA operational information is gathered and organized for transmission by the PHD Builder (Figure 166-5).

166.3.1 PMA transmit function

The PMA transmit function depends on the bit grouping parameter G defined for each BASE-AU PHY in Table 166-1.

The PMA transmit function shall map each k group of G Transmit Block bits into a single symbol denoted as s_k chosen from a set of 2^G possible values following the scheme specified in Table 166-6.

For BASE-AU PHYs with $G = 1$, the Transmit Block shall be taken in 195 840 single-bit groups, $\{\text{bit}_k\}$, starting from $\{\text{bit}_0\}$ in Figure 166-11. Each bit group shall be mapped into a symbol s_k as specified in Table 166-6.

For BASE-AU PHYs with $G = 2$, the Transmit Block shall be taken in 97 920 bit groups of two bits, $\{\text{bit}_{2k}, \text{bit}_{2k+1}\}$, starting from the bit group $\{\text{bit}_0, \text{bit}_1\}$ in Figure 166–11. Each bit group shall be mapped into a symbol s_k as specified in Table 166–6.

Table 166–6—Bit mapping

G	Bit group value	Symbol s_k
1	{0}	{−1}
	{1}	{+1}
2	{0,0}	{−1}
	{1,0}	{−1/3}
	{1,1}	{+1/3}
	{0,1}	{+1}

Symbols are transmitted to the PMD with a nominal symbol period of $1000/(26.5625 \times S)$ ps, which depends on the BASE-AU PHY. See Table 166–1 for the definition of S for each BASE-AU PHY.

166.3.2 PMA receive function

The PMA receive function includes Transmit Block synchronization and the clock and data recovery from the signal received from the PMD receive function.

The PMA performs clock recovery on the received signal. The clock recovery includes coarse timing recovery for synchronization with the beginning of the received Transmit Block, clock frequency deviation estimation, and fine timing recovery to provide a stable clock to sample the received signal from the PMD with a suitable phase for reliable reception (see 166.3.4.6.1).

A stream of symbols is extracted from the PMD input signal (PMD_COMSIGNAL.indication(rx_signal) in 166.6.1.2) and converted to bits (per Table 166–6) that are delivered to the PCS sublayer.

The PMA receiver may implement channel equalization.

166.3.3 Interface to the PMD

The interface between the PMA and the PMD is specified in an abstract manner in 166.6.1.

166.3.4 PHY control

166.3.4.1 PHY control state variables

hdr_crc16_status

Result of the CRC16 evaluation for a received PHD from the link partner; this variable is assigned for each received PHD.

Values: OK: The most recent received PHD is correct as determined by CRC16 verification.

NOT_OK: The most recent received PHD is not correct as determined by CRC16 verification.

hdr_fail_count
Variable used to count the reception of contiguous erroneous PHDs.

link_control
Variable that controls PMA functional operation. All state diagrams respond to the open-ended `link_control = DISABLE`.
Values: `DISABLE`: prevent operation of PMA sublayer.
`ENABLE`: permit operation of PMA sublayer.

link_status
Variable that is set by the link monitor state diagram and that reflects the current link reliability.
Values: `OK`: the link has been established between link partners with data reliability in both communication directions.
`FAIL`: link is not established (one or both directions are not providing reliable payload data decoding).

loc_rcvr_hdr_lock
Variable set by the local PHD reception monitor state diagram to indicate the reliability of PHD reception.
Values: `OK`: Local PHD reception is reliable.
`NOT_OK`: Local PHD reception is unreliable.

loc_rcvr_status
Variable set by the PHY quality monitor state diagram to indicate correct or incorrect payload data decoding by the local PHY receiver.
Values: `OK`: Payload data reception by the local PHY is reliable.
`NOT_OK`: Payload data reception by the local PHY is unreliable.

new_rxblock_event
Signal sent by the PHY receiver to indicate the start of reception of a new Transmit Block. This event persists only long enough to cause one state diagram transition.

new_rxphd_event
Signal sent by the PHY receiver to indicate the complete reception (including data decoding and CRC16 checking) of a new PHD from the link partner. This event persists only long enough to cause one state diagram transition.

pma_reset
Variable that causes reset of all PMA functions. PMA reset occurs with power on or the PHY reset being set to one (bit 1.0.15). All state diagrams respond to the open-ended `pma_reset = ON`.
Values: `ON`: reset is asserted.
`OFF`: reset is deasserted.

rcvr_clock_lock
Variable set by the PMA receive clock recovery function to indicate that the clock has been properly recovered from the received signal.
Values: `OK`: clock is stable and phase adjusted for sampling the received symbols.
`NOT_OK`: clock has not been recovered from the received signal and/or it is not stable.

rcvr_hdr_lock
Variable set by the PHD monitor state diagram to indicate the reliability of both the PHD transmission from local to remote PHY and the PHD reception from remote to local PHY.
Values: `OK`: PHD transmission and reception are reliable.
`NOT_OK`: PHD transmission or reception is unreliable.

rem_rcvr_hdr_lock
Variable set by the remote PHD reception monitor state diagram to indicate the reliability of PHD reception in the remote PHY (link partner).
Values: `OK`: Link partner PHD reception is reliable.
`NOT_OK`: Link partner PHD reception is unreliable.

rem_rcvr_status

Variable set by the link monitor state diagram. It indicates the receiver status of the remote (link partner) PHY payload data decoding.

Values: OK: Payload data reception by the remote PHY is reliable.

NOT_OK: Payload data reception by the remote PHY is unreliable.

rx_xmii_enable

Variable set by the PHY Rx control state diagram to control the 64B/65B decoder operation (see 166.2.3.8).

Values: TRUE: The 64B/65B decoder receives 65-bit blocks from the link partner and decodes them into xMII receive data stream transfers.

FALSE: The 64B/65B decoder does not decode received 65-bit blocks from the link partner and Local Fault ordered set(s) are signaled in xMII.

sotxb_synch

See 166.2.3.7.2.

tx_xmii_enable

See 166.2.2.8.2.

tx_enable

Variable set by the PHY Tx control state diagram to enable PCS and PMA transmission.

Values: TRUE: PCS and PMA transmission is enabled.

FALSE: PCS and PMA transmission is disabled.

166.3.4.2 PHY Tx control state diagram

Transmit operation shall conform to the PHY Tx control state diagram in Figure 166–23.

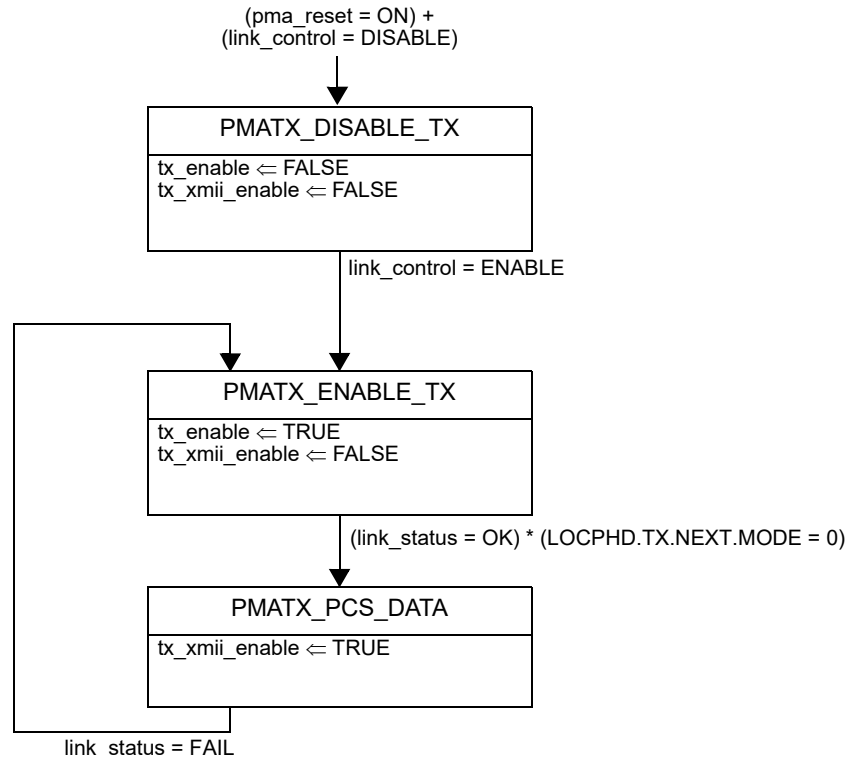


Figure 166–23—PHY Tx control state diagram

Upon reset or disable of the PMA, PHY transmit operation is disabled (state PMATX_DISABLE_TX). Once the PMA is enabled (link_control = ENABLE), the local PHY starts sending Transmit Blocks as specified in 166.2.2 (state PMATX_ENABLE_TX), so that the remote PHY can perform Transmit Block synchronization, clock recovery, and equalizer training (tx_enable $\Leftarrow$ TRUE).

While establishing the bidirectional link, the 64B/65B PCS encoder does not encode the xMII transmit stream (tx_xmii_enable $\Leftarrow$ FALSE). Instead of this, the 64B/65B PCS encoder encodes Local Fault sequence ordered sets to be used for the link partner alignment (see Figure 166–16). Once the bidirectional link is established and the transmitter is not in BER test mode, the 64B/65B encoder begins encoding the xMII transmit stream into 65-bit blocks (tx_xmii_enable $\Leftarrow$ TRUE in state PMATX_PCS_DATA). If one of the link partners fails to receive Transmit Blocks with reliability (link_status = FAIL), the 64B/65B PCS encoder does not encode the xMII transmit stream until the bidirectional link is re-established.

166.3.4.3 PHY Rx control state diagram

Receive operation shall conform to the PHY Rx control state diagram in Figure 166–24.

Upon reset or disable of the PMA, PHY receive operation is disabled. Once the PMA is enabled (link_control = ENABLE), the PHY receiver begins link establishment by synchronizing the Transmit Block and recovering clock from the received signal. This is accomplished in two steps. The first step is coarse timing recovery in PMARX_TIMING_COARSE, where Transmit Block synchronization is performed. After synchronization with the beginning of the received Transmit Block is achieved (sotxb_synch = OK), fine timing recovery is carried out in order to provide a stable clock that samples the received signal with a suitable phase for reliable reception.

When the clock is stable (rcvr_clock_lock = OK), the PHY receiver may train the equalizers (if implemented) to compensate for the inter-symbol interference caused by the communication channel.

Once the equalizers have been properly estimated, the PHY receiver processes each PHD from the link partner to determine if PHD reception is reliable in both directions. The state diagrams that monitor the reliability of PHD reception are described in 166.3.4.5.

As soon as both link partners detect reliable PHD reception (rcvr_hdr_lock = OK), the PHY receiver determines according to the state diagram described in 166.3.4.6.4 whether the 65-bit block reception is reliable.

Once a bidirectional link is established (link_status = OK), and after checking that the remote PHY is not in BER test mode (REMPHD.TX.NEXT.MODE = 0), the 64B/65B PCS decoder begins to decode the received 65-bit blocks onto the xMII receive data stream (rx_xmii_enable $\Leftarrow$ TRUE).

If one of the link partners fails to receive Transmit Blocks with reliability (link_status = FAIL), the 64B/65B PCS decoder does not decode 65-bit blocks received from link partner into the xMII receive stream (rx_xmii_enable $\Leftarrow$ FALSE) and generates LBLOCK_R (see Figure 166–21).

In a similar way, if one of the link partners fails to receive valid PHDs (rcvd_hdr_lock = NOT_OK), the state diagram transitions to the initial state (PMARX_DISABLE) and the rx_xmii_enable variable is set to FALSE.

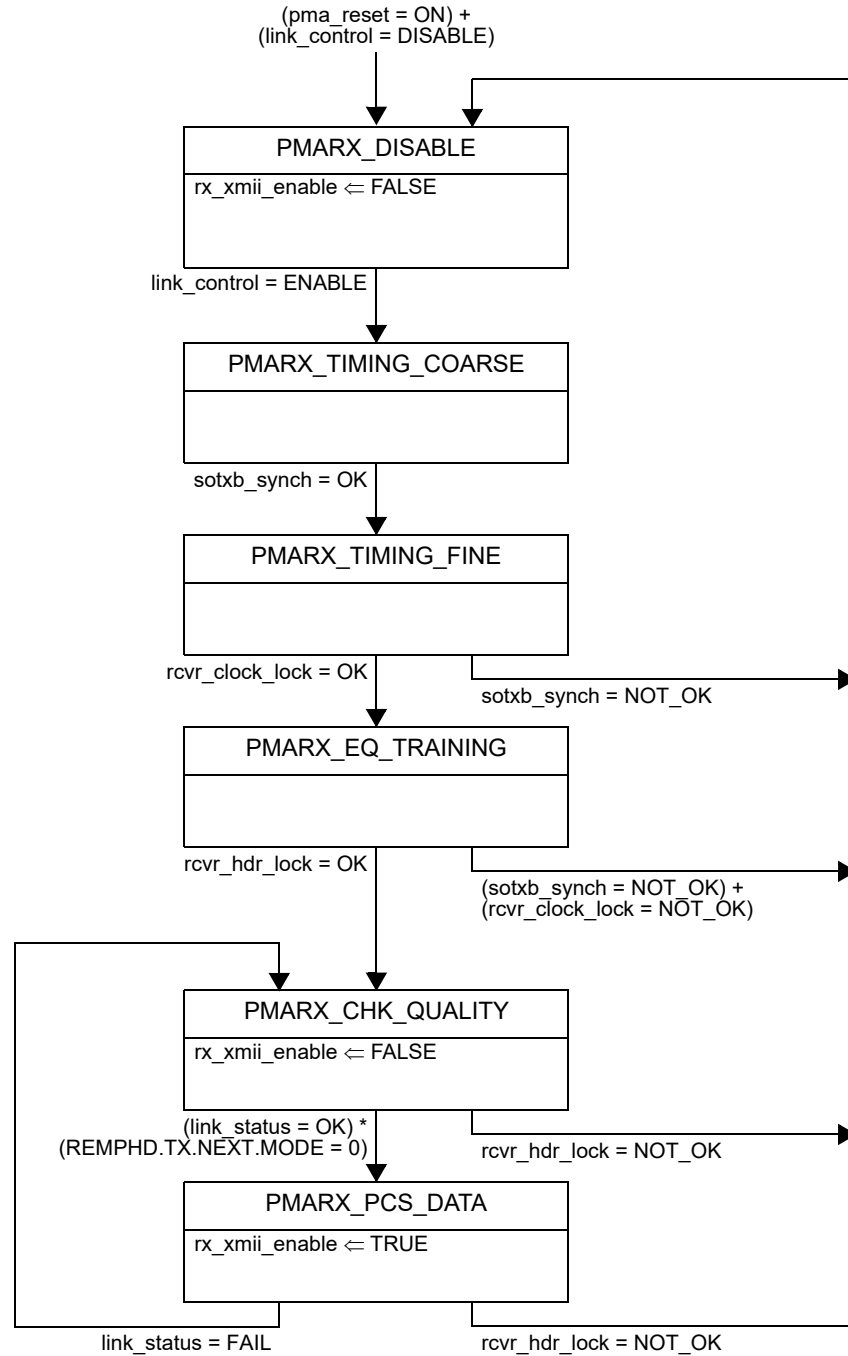


Figure 166–24—PHY Rx control state diagram

166.3.4.4 Link monitor state diagram

Link status shall be determined as specified by the link monitor state diagram in Figure 166–25. The state diagram determines the value of the link_status and rem_rcvr_status state variables.

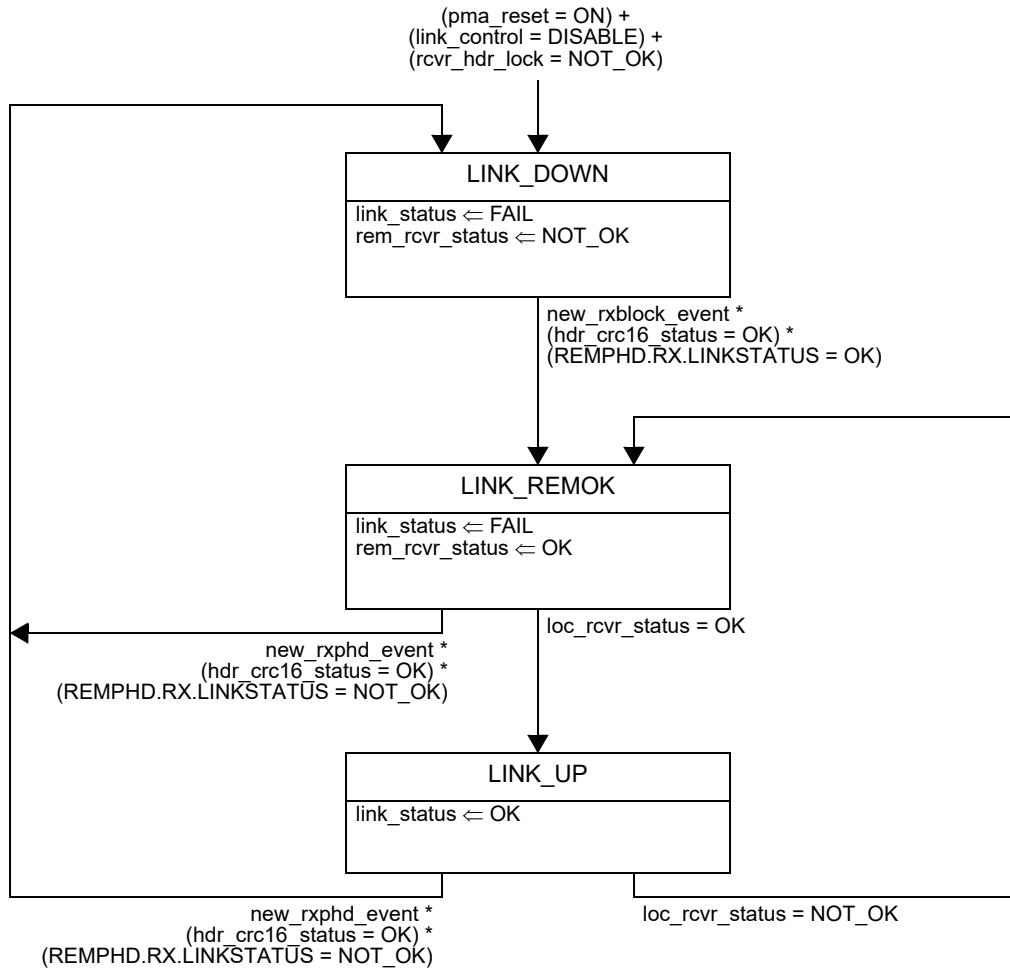


Figure 166–25—Link monitor state diagram

In order to support a link-up time of less than 25 ms in a communication system composed of two connected link partners as shown in Figure 166–2, the following requirements apply:

- a) The time from the fulfillment of the following conditions together:
 - pma_reset is equal to OFF
 - pcs_reset is equal to FALSE
 until a compliant optical signal is presented at TP2 shall be less than 5 ms.
- b) The time from the fulfillment of the following conditions together:
 - pma_reset is equal to OFF
 - pcs_reset is equal to FALSE
 - a compliant optical signal is presented at TP3
 until the assertion of the link_status variable to OK shall be less than 20 ms.

Upon reset, disable of the PMA, or unreliable PHD communication, the link is lost ($\text{link_status} \Leftarrow \text{FAIL}$) and the remote receiver status is unknown ($\text{rem_rcvr_status} \Leftarrow \text{NOT_OK}$). When the remote PHY indicates reliable reception ($\text{REMPHD.RX.LINKSTATUS} = \text{OK}$), the state diagram transitions to the LINK_REMOK state synchronously on the event of receiving a new Transmit Block (new_rxblock_event). Once in this state, the bidirectional link is established ($\text{link_status} \Leftarrow \text{OK}$) when local reliable reception is signaled by the PHY quality monitor ($\text{loc_rcvr_status} = \text{OK}$).

If the local PHY fails to receive 65-bit blocks with reliability ($\text{loc_rcvr_status} = \text{NOT_OK}$) but the remote PHY reception is indicated reliable ($\text{REMPHD.RX.LINKSTATUS} = \text{OK}$), the bidirectional link is lost and the state diagram transitions to LINK_REMOK state that waits for the local receiver reliability to be recovered. If the remote PHY receiver indicates that it does not operate with reliability ($\text{REMPHD.RX.LINKSTATUS} = \text{NOT_OK}$), the bidirectional link is lost and the state diagram transitions to LINK_DOWN state in any case.

166.3.4.5 PHD monitor state diagrams

The reliability of PHD reception and transmission shall be determined by the PHD monitor state diagrams in Figure 166–26, Figure 166–27, and Figure 166–28.

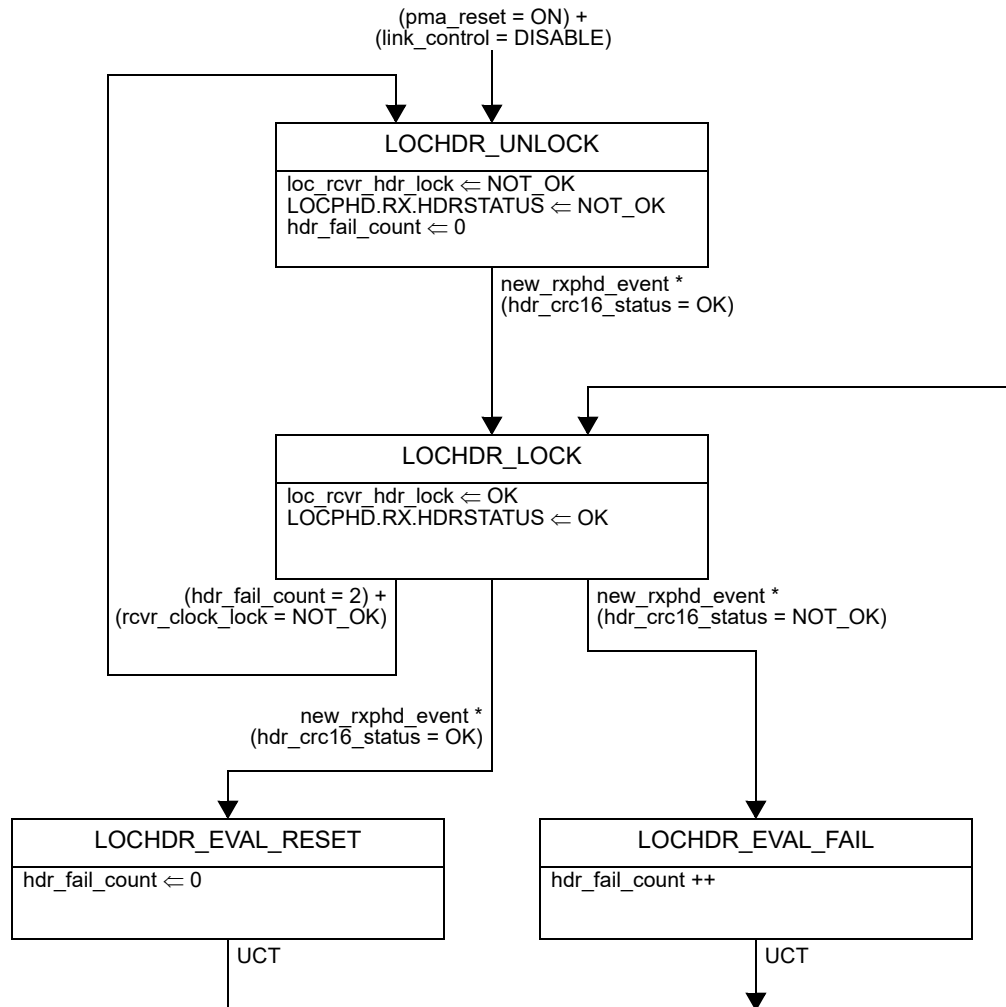


Figure 166–26—Local PHD reception monitor state diagram

The local PHD reception monitor state diagram of Figure 166–26 monitors the PHD reception reliability for the local PHY and indicates it via the variable `loc_rcvr_hdr_lock`. Upon reset or disable of the PMA, or if the local PHY fails to reliably receive PHDs (`loc_rcvr_hdr_lock` $\Leftarrow$ NOT_OK), this is indicated to the link partner by assigning NOT_OK to the field `LOCPHD.RX.HDRSTATUS`. In the state `LOCHDR_UNLOCK`, the receiver waits for reception of a valid PHD (i.e., one with correct CRC16), and the variable `hdr_fail_count` (the count of contiguous PHDs received with errors) is reset (`hdr_fail_count` $\Leftarrow$ 0).

In `LOCHDR_UNLOCK`, reception of one correct PHD triggers the transition to the state `LOCHDR_LOCK`. On entry to `LOCHDR_LOCK` the variable `loc_rcvr_hdr_lock` and the field `LOCPHD.RX.HDRSTATUS` are assigned the value OK. The PHY keeps checking the CRC16 of received PHDs, incrementing `hdr_fail_count` with each erroneous PHD (in state `LOCHDR_EVAL_FAIL`) and resetting `hdr_fail_count` with each valid PHD (in state `LOCHDR_EVAL_RESET`). If `hdr_fail_count` reaches the limit of two, or the clock recovery function detects that the PHY has lost synchronization (`rcvr_clock_lock` = NOT_OK), then the state transitions back to `LOCHDR_UNLOCK`.

The remote PHD reception monitor state diagram of Figure 166–27 monitors PHD reception reliability reports from the remote PHY and indicates the status via the variable `rem_rcvr_hdr_lock`. Upon reset, disable of the PMA, or unreliable local PHD reception, the status variable `rem_rcvr_hdr_lock` is assigned the value NOT_OK pending the arrival of a new PHD from the link partner (`REMHDR_WAIT` state). From then on, the value of the variable `rem_rcvr_hdr_lock` is updated with the value of the field `REMPHD.RX.HDRSTATUS` (`rem_rcvr_hdr_lock` $\Leftarrow$ `REMPHD.RX.HDRSTATUS`).

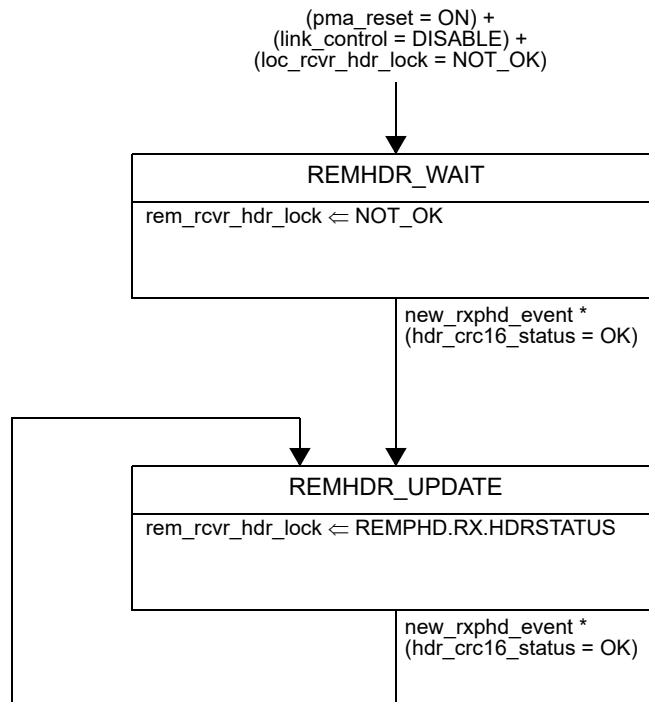


Figure 166–27—Remote PHD reception monitor state diagram

The PHD monitor state diagram of Figure 166–28 indicates via the variable `rcvr_hdr_lock` if both link partners are reliably receiving PHDs. Upon reset or disable of the PMA, the PHD communication is unreliable (`rcvr_hdr_lock` $\Leftarrow$ NOT_OK). Whenever a new PHD is received from the link partner, the status of both the local (variable `loc_rcvr_hdr_lock`) and remote (variable `rem_rcvr_hdr_lock`) PHD reception are updated as indicated in Figure 166–26 and Figure 166–27, respectively.

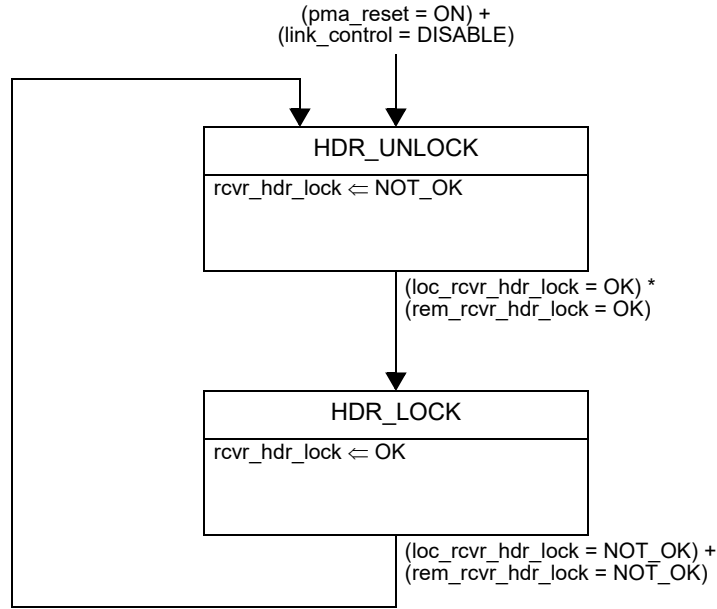


Figure 166–28—PHD monitor state diagram

A transition from the HDR_UNLOCK state to the HDR_LOCK state takes place when both the local and the remote PHYs are reliably receiving PHDs ($\text{loc_rcvr_hdr_lock} = \text{OK}$ and $\text{rem_rcvr_hdr_lock} = \text{OK}$). The value OK is then assigned to the variable rcvr_hdr_lock to signal the reliable transmission and reception of PHDs. Subsequently, if either of the link partners fail to receive PHDs reliably ($\text{loc_rcvr_hdr_lock} = \text{NOT_OK}$ or $\text{rem_rcvr_hdr_lock} = \text{NOT_OK}$), a transition to HDR_UNLOCK occurs resulting in assigning NOT_OK to the rcvr_hdr_lock variable.

166.3.4.6 PHY quality monitor

166.3.4.6.1 PHY quality criterion

Reception of 65-bit blocks is reliable when the RS-FEC codeword error ratio is less than 4.5×10^{-10} after the RS-FEC decoding function. The RS-FEC codeword error ratio is defined as the ratio of the RS-FEC codeword error counter (register 3.2353, see 45.2.3.95) and the number of RS-FEC codewords received since the last reset of RS-FEC codeword error counter.

166.3.4.6.2 PHY quality assessment

The assessment of the above defined PHY quality criterion may be based on estimation of the noise variance at the symbol detector decision points $E[n_d^2]$, which expressed in base-2 logarithmic units has to be lower than an implementation dependent threshold T_{LM} to have reliable local reception of 65-bit blocks. The variable loc_rcvr_status is assigned the value OK if the condition $\log_2(E[n_d^2]) \leq T_{LM}$ holds. Otherwise, it is assigned the value NOT_OK. In normal operation mode, the noise variance at the symbol detector can be estimated either by measuring the modulation error ratio (MER) at the decision points or measuring the ratio of symbols corrected by the RS-FEC decoder per codeword. In LPI mode, it can be estimated by measuring the MER or the corrected bits in the reception of the 12-time repeated 20-bit encoded PHD sub-block belonging to each LPI refresh codeword (see 166.4). The value of the threshold and the information used to estimate the RS-FEC decoder noise variance is implementation dependent.

As defined in 166.2.2.1.1, the local PHY reports the link margin to the link partner by using the field LOCPHD.RX.LINKMARGIN. The link margin (LM) is defined as the SNR margin relative to the SNR required for reception of RS-FEC codewords with a quality specified in 166.3.4.6.1. Link margin is formally defined per Equation (166–4).

$$LM = T_{LM} - \log_2(E[n_d^2]) \quad (166-4)$$

166.3.4.6.3 PHY quality monitor state variables

All the variables used in the PHY quality monitor state diagram that have not been previously introduced follow.

new_link_margin_event

Signal sent by the PHY receiver to indicate a new estimation of detector link margin is available.

This event persists only long enough to cause one state diagram transition.

link_margin

Variable set by the PHY receiver containing the value of the last link margin estimation.

Values: Any value determined per Equation (166–4) with $E[n_d^2]$ and T_{LM} as defined in 166.3.4.6.2.

166.3.4.6.4 PHY quality monitor state diagram

The variable `loc_rcvr_status` (see 166.3.4.1), which indicates if the local PHY is reliably receiving payload data per the specifications of 166.3.4.6.1, shall be determined as specified by the PHY quality monitor state diagram in Figure 166–29.

Upon reset, disable of the PMA, or loss of HDR lock, the PHY quality monitor functionality is disabled (PMAMON_DISABLE state), and the state variable `loc_rcvr_status` is assigned the value NOT_OK. This is indicated to the link partner by assigning the value NOT_OK to the field LOCPHD.RX.LINKSTATUS. Additionally, the reported link margin is set to 0x80 (the smallest value representable in 8-bit signed format, see 166.1.1).

The state diagram remains in the PMAMON_DISABLE state until the first estimate of the link margin is available. From then on, whenever a new estimate of the link margin is available (`new_link_margin_event`), it is evaluated. If the link margin is larger than or equal to zero, the local reception of payload data sub-blocks is reliable.

Reliable local reception is indicated to the link partner by assigning OK to the field LOCPHD.RX.LINKSTATUS (state PMAMON_OK). After at least one locally transmitted Transmit Block (PMAMON_SYNCH state), the local receiver status is signaled to the link monitor state diagram (`loc_rcvr_status` $\leftarrow$ OK) synchronously on the event of the beginning of a new Transmit Block (PMAMON_UPDATE state).

Otherwise, when the link margin is smaller than zero, unreliable receive link status is signaled (`loc_rcvr_status` $\leftarrow$ NOT_OK). This corresponds to the state PMAMON_FAIL. The local receiver status is always indicated to the link partner by using the field LOCPHD.RX.LINKSTATUS. The measured link margin is also reported via the field LOCPHD.RX.LINKMARGIN.

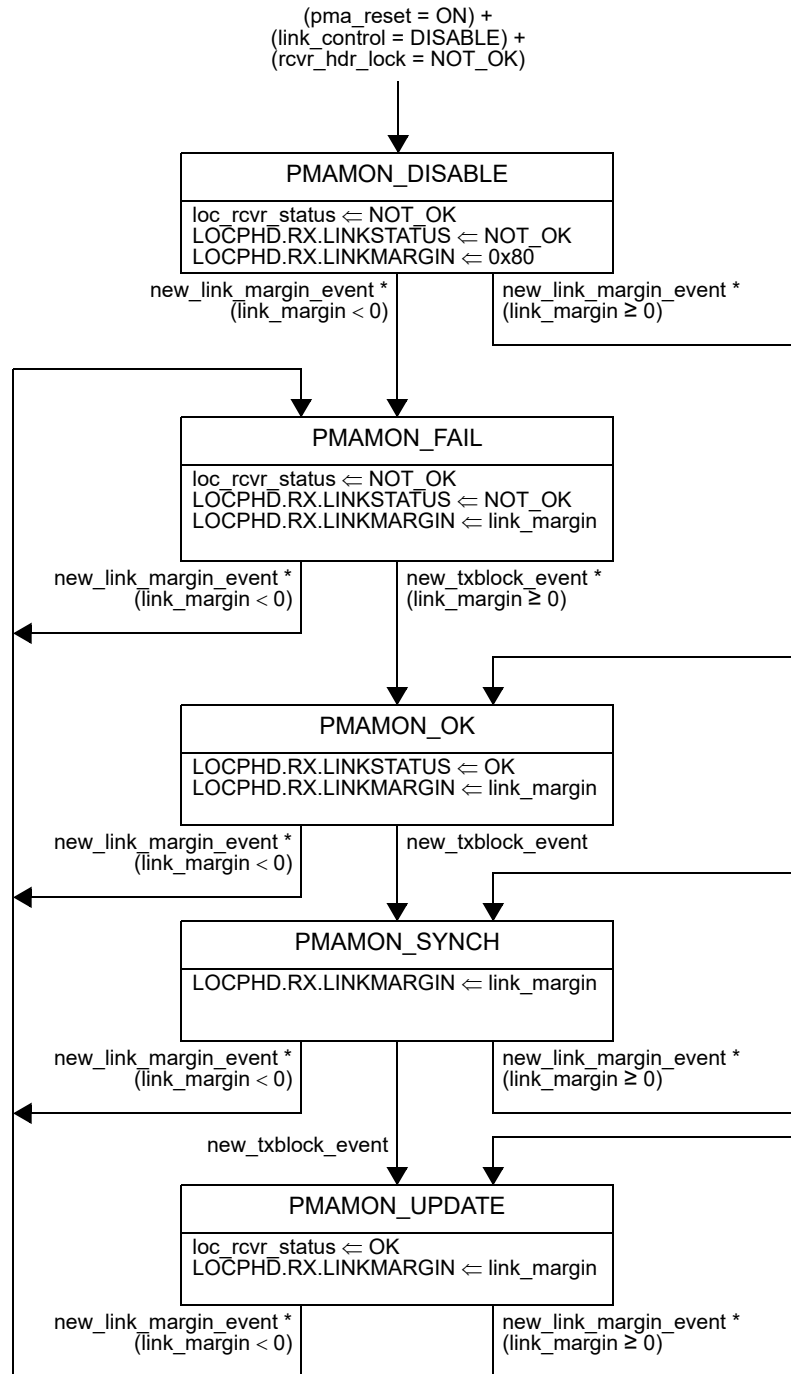


Figure 166–29—PHY quality monitor state diagram

166.4 Energy-Efficient Ethernet (EEE)

166.4.1 EEE capability enable

The optional EEE capability is announced by setting the field PHD.CAP.LPI (see Table 166–2) on the transmitted PHD to one. EEE functionality shall be enabled when both the transmitted and received fields PHD.CAP.LPI are equal to one, and disabled otherwise.

166.4.2 Fast wake mode LPI operation

A BASE-AU PHY that implements the optional EEE capability follows fast wake mode of LPI operation as specified in 78.1.3.3.1 in the sense that the PHY transmitter remains transmitting signals during LPI (same symbol rate and modulation as normal mode). The codewords generated by the PCS sublayer during LPI consist of specific characters without RS-FEC encoding, which allow power saving, robust OAM side communication channel, and robust wake signal detection in the receiver.

166.4.3 LPI transmit operation mode

The BASE-U PCS transmit function of an EEE capable PHY in normal operation mode shall monitor 65-bit blocks generated by the PCS 64B/65B transmit state diagram (see 166.2.2.9) to detect the condition to enter to LPI operation mode as specified in 166.4.3.3.

The BASE-U PCS transmit function in LPI operation mode shall monitor 65-bit blocks generated by the PCS 64B/65B transmit state diagram (see 166.2.2.9) to detect the condition to resume to normal operation mode as specified in 166.4.3.4.

The BASE-U PCS transmit function in LPI operation mode shall transmit LPI codewords as specified in Figure 166–30, instead of RS-FEC codewords specified for normal operation mode in 166.2.2.4. The BASE-U PCS PHD transmit function shall operate as specified for normal operation mode (see 166.2.2.1).

There are two types of LPI codewords: LPI refresh codewords and LPI wake codewords.

166.4.3.1 LPI refresh codewords

LPI refresh codewords shall be generated as specified in Figure 166–30, aggregating 80 LPI refresh 65-bit blocks.

The LPI refresh 65-bit block has the data/control header bit equal to 1, the block type field octet equal to 0x00, and each of the control characters C_0 , C_1 , C_2 , C_3 , C_4 , C_5 , C_6 , and C_7 equal to 0x35.

166.4.3.2 LPI wake codewords

A LPI wake codeword shall be generated as specified in Figure 166–30, aggregating 80 LPI wake 65-bit blocks.

The LPI wake 65-bit block has the data/control header bit equal to 1, the block type field octet equal to 0x00, and each of the control characters C_0 , C_1 , C_2 , C_3 , C_4 , C_5 , C_6 , and C_7 equal to 0x6A.

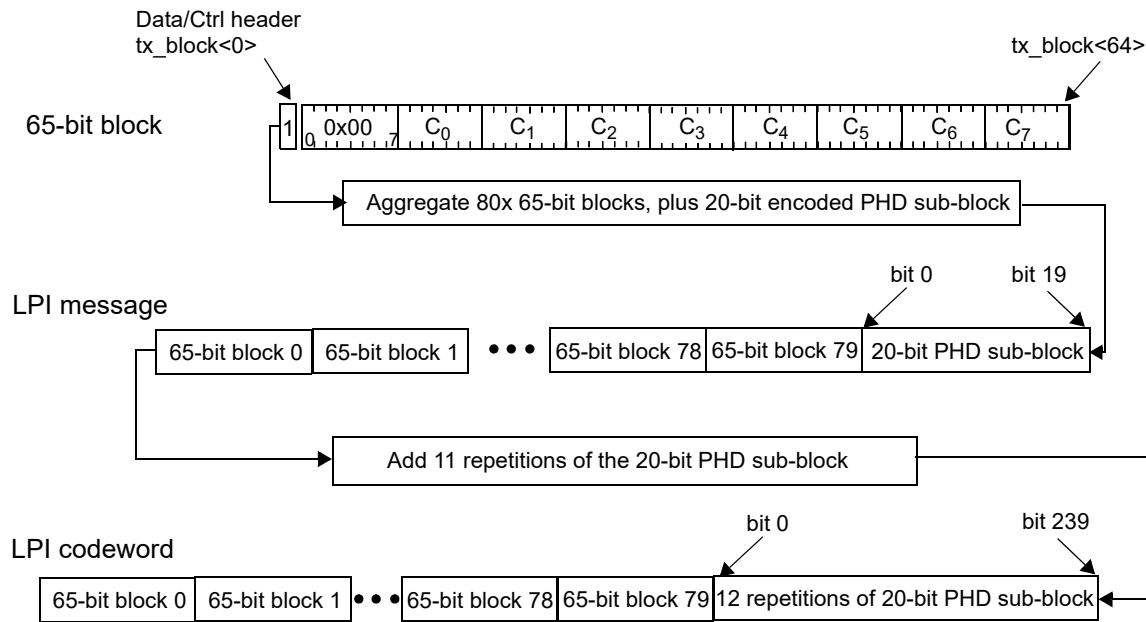


Figure 166–30—LPI codeword transmission

166.4.3.3 Transition of PCS transmit function from normal to LPI operation mode

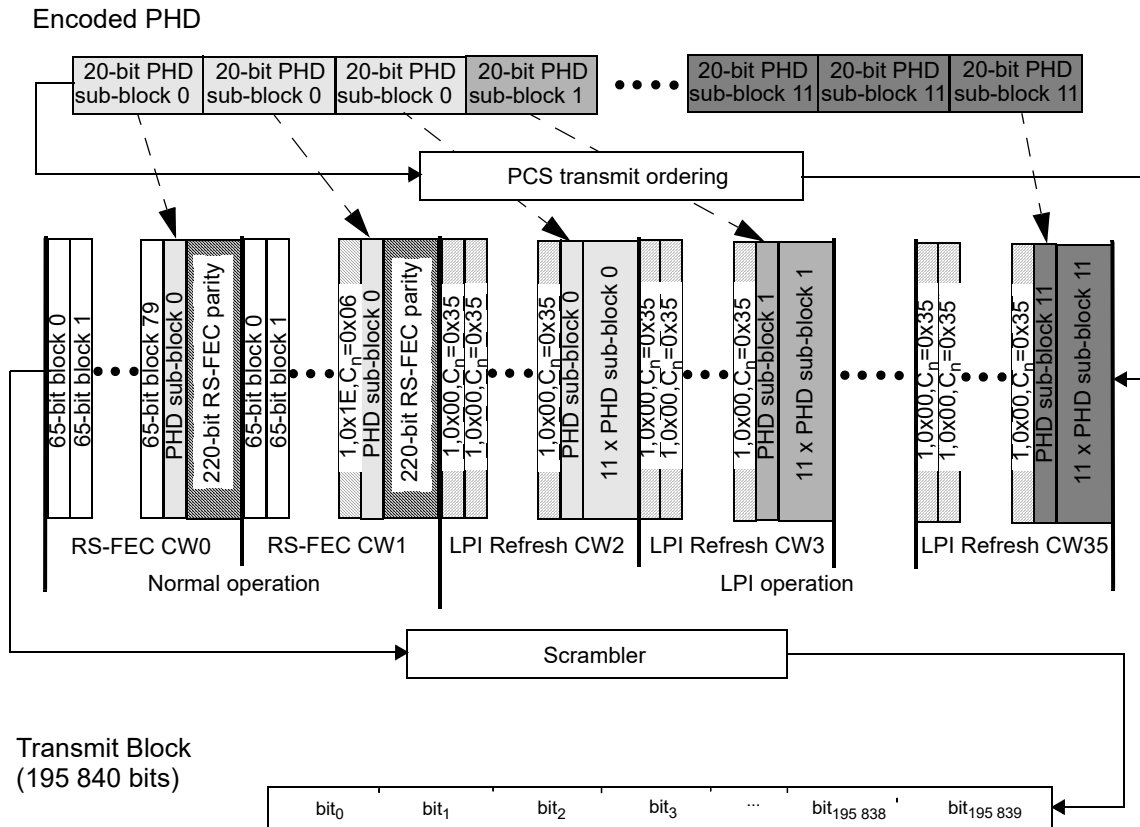
The PCS transmit function shall enter LPI operation mode when the last 65-bit block (i.e., the 65-bit block #79) of an RS-FEC codeword encodes eight control characters with value /LI/ (see 166.2.2.7.6).

Following the specification of 65-bit block coding of Figure 166–14 and Figure 166–15, this last 65-bit block of an RS-FEC codeword has the data/control header bit equal to 1, the block type field octet equal to 0x1E and each of the control characters C₀, C₁, C₂, C₃, C₄, C₅, C₆, and C₇ equal to 0x06.

The PCS transmit function shall enter LPI operation mode aligned with the next RS-FEC codeword boundary after transmission of the RS-FEC codeword with the last 65-bit block encoding eight control characters with value /LI/.

The BASE-U PCS transmit function in LPI operation mode shall transmit LPI refresh codewords.

Figure 166–31 shows an example of transition from normal to LPI operation mode.



NOTE—CW0, CW1, ... CW35 stands for codeword 0, 1, ... 35.

Figure 166–31—Example of transition from normal to LPI operation

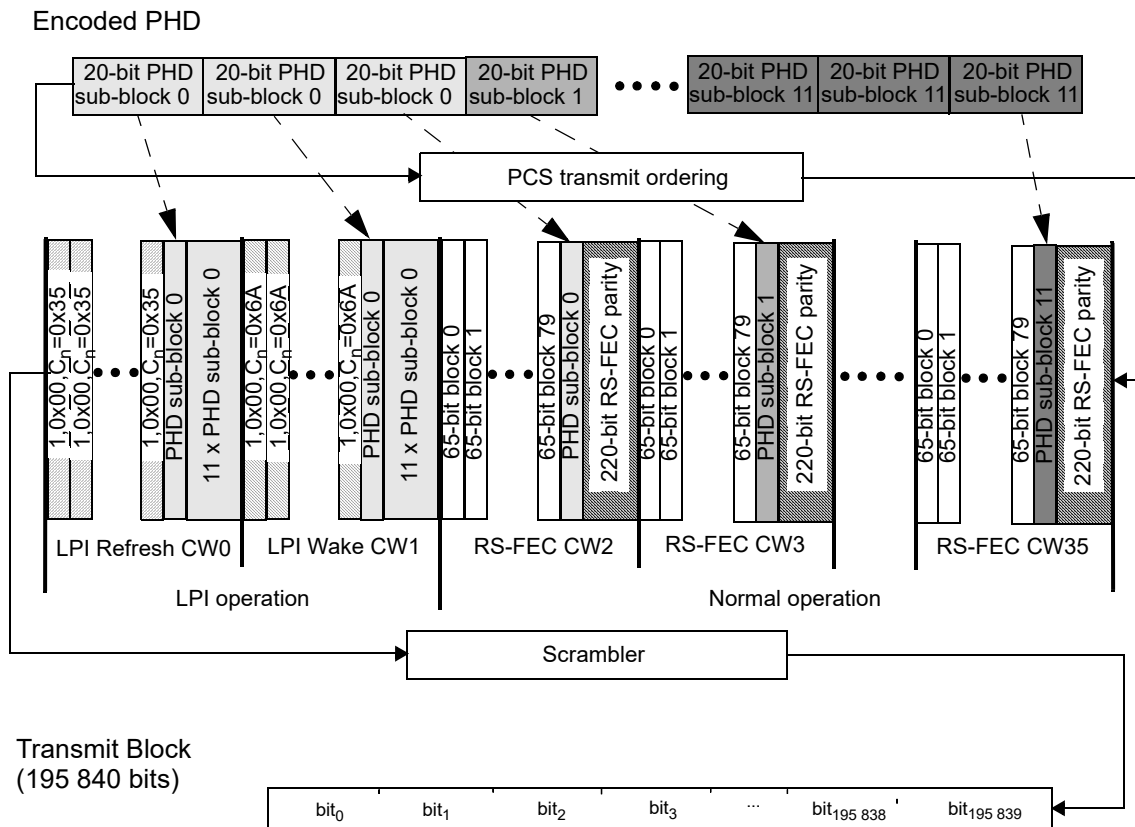
166.4.3.4 Transition of PCS transmit function from LPI to normal operation mode

The PCS transmit function shall transmit an LPI wake codeword when the last 65-bit block (i.e., the 65-bit block #79) of an RS-FEC codeword is replaced with an LPI refresh codeword that encodes eight control characters with value /I/ (see 166.2.2.7.5).

Following the specification of 65-bit block coding of Figure 166–14 and Figure 166–15, this last 65-bit block of an RS-FEC codeword has the data/control header bit equal to 1, the block type field octet equal to 0x1E and each of the control characters C₀, C₁, C₂, C₃, C₄, C₅, C₆, and C₇ equal to 0x07.

After transmission of an LPI wake codeword, the next codeword shall be an RS-FEC codeword resulting from normal operation of the PCS transmit function, that shall resume its normal operation.

Figure 166–32 shows an example of transition from LPI to normal operation.



NOTE—CW0, CW1, ... CW35 stands for codeword 0, 1, ... 35.

Figure 166–32—Example of transition from LPI to normal operation

166.4.4 LPI receive operation mode

The PCS receive function shall be in LPI operation mode when the PCS 64B/65B receive function is in any of the states described in Figure 166–22. Otherwise, the PCS receive function shall be in normal operation mode.

The BASE-U PCS receive function in LPI operation mode shall process LPI codewords as specified in Figure 166–33, rather than decode RS-FEC codewords specified for normal operation mode in 166.2.2.4. The BASE-U PCS PHD receive function shall operate as specified for normal operation mode (see 166.2.3.4).

The PCS receive function in LPI operation mode replaces the Reed-Solomon decoder functionality by the LPI wake codeword detection functionality, which is to detect the reception of an LPI wake codeword as specified in 166.4.3.2.

When the PCS receive function in LPI operation mode detects an LPI wake codeword, the corresponding 80×65 -bit blocks shall be set as LPI wake 65-bit blocks (see 166.4.3.2).

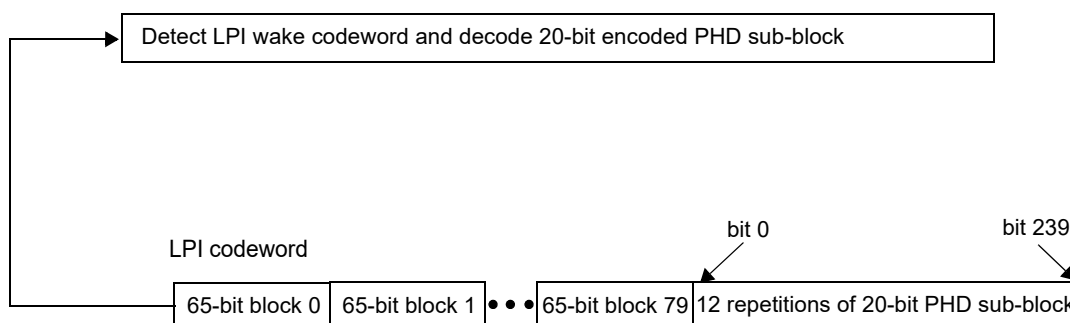


Figure 166–33— LPI codeword reception.

From each LPI codeword received, the 20-bit encoded PHD sub-block shall be decoded by majority voting using a number of repetitions equal to or less than 11. The number of repetitions to be used is implementation dependent.

166.5 Test Modes

The BER test mode shall be provided to allow BER testing.

The SSQWP, FSQWP, SSPR-NRZ, SSPR-PAM4 test patterns, and the test pattern for stressed receiver sensitivity measurement shall be provided to allow optical parameter measurements (see 166.6.4.1).

The test modes and patterns shall be configured by setting the BASE-U PCS control register, operation mode bits defined in 45.2.3.90. The test modes only change the symbols provided to the BASE-AU PMD and do not alter the optical and jitter characteristics at the MDI from those of normal (non-test mode) operation.

166.5.1 BER test mode

BER test mode is for measurement of the bit error ratio (BER) of the link including the PCS, PMA, and PMD sublayers of two BASE-AU PHYs and a fiber optic cable connected to them. A BER test is run between the transmitter of a PHY and the receiver of its link partner. The BER test mode can be configured independently for each of the unidirectional transmissions.

A PCS transmitter in BER test mode operation mode shall continuously generate LBLOCK_T 65-bit blocks (see 166.2.2.8.1) regardless of the link status, and ignore xMII transactions. In the absence of errors, a continuous sequence of LBLOCK_T 65-bit blocks is expected after the RS-FEC decoder of the link partner receiver. Any data bit difference with respect to a LBLOCK_T binary sequence shall be counted as a bit error in the BER test mode counter (see 45.2.3.94). Any RS-FEC codeword containing bit errors shall be counted as an RS-FEC codeword error and reflected in the RS-FEC codeword error counter (see 45.2.3.95).

When the link partner receiver is in BER test mode, its PCS receiver shall generate LBLOCK_R (see 166.2.3.7.1) at the xMII output.

The PMA and PMD functions of a BASE-AU PHY in BER test mode shall operate as in normal mode (non-test). A BASE-AU PHY shall establish link (link_status = OK) to allow a BER test in either traffic direction.

NOTE—BER test mode is not used to confirm compliance with any requirement, but it is useful within the supply chain for subsystem testing.

The slow square wave pattern of n_{sq} symbols is a repeating sequence of $n_{\text{sq}} \{+1\}$ symbols followed by $n_{\text{sq}} \{-1\}$ symbols, where n_{sq} depends on the BASE-AU PMD under test, with $n_{\text{sq}} = 16$ for 50GBASE-AU and 25GBASE-AU, $n_{\text{sq}} = 8$ for 10GBASE-AU, $n_{\text{sq}} = 4$ for 5GBASE-AU, and $n_{\text{sq}} = 2$ for 2.5BASE-AU.

166.5.3 FSQWP test pattern

The PMA generates this pattern for the service interface below the PMA via the PMD COMSIGNAL.request primitive (see 166.6.1.1).

The short stress pattern random for NRZ (SSPR-NRZ) test pattern is a repeating 2^{15} symbol, two level (NRZ) sequence constructed as follows:

- 108

- NOTE—A CSV file, Clause_166_SSPR-NRZ_pattern.csv, containing the entire NRZ symbol sequence for the SSPR-NRZ test pattern is available at <https://standards.ieee.org/downloads/802.3/>.

The diagram shows a 28-bit shift register. The cells are labeled $r[0]$, $r[1]$, ..., $r[24]$, ..., $r[27]$. The output is $r[27]$. A feedback loop is shown where the outputs of $r[24]$ and $r[27]$ are combined in a mod-2 (XOR) adder. The result of the XOR operation is fed back into $r[0]$. The text "The adder is mod-2 (XOR)" is placed near the adder.

166.5.5 SSPR-PAM4 test pattern

- Bit sequence A1 is a 10 924-bit sequence generated as if produced by the shift register implementation shown in Figure 166–35 and the seed used is 0x20000000, which is a 31-bit hexadecimal value used to preset r[30] through r[0] (r[30] is set to the LSB and r[0] is set to the MSB) prior to the generation of the sequence for the indicated length of bits.
- The first 85 bits of the bit sequence A1 are a 29-bit sequence of 0s, a single bit of 1, a 27-bit sequence of 0s, a single bit of 1, a single bit of 0, a single bit of 0, a single bit of 1, and a 24-bit sequence of 0s.
- Bit sequence A2 is a 10 922-bit sequence generated as if produced by the shift register implementation shown in Figure 166–35 and the seed used is 0x77FE4016, which is a 31-bit hexadecimal value used to preset r[30] through r[0] (r[30] is set to the LSB and r[0] is set to the MSB) prior to the generation of the sequence for the indicated length of bits.
- The first 27 bits of the bit sequence A2 are a single bit of 0, a 2-bit sequence of 1s, a single bit of 0, a single bit of 1, a 9-bit sequence of 0s, a single bit of 1, a 2-bit sequence of 0s, and a 10-bit sequence of 1s.
- Bit sequence A3 is a 10 922-bit sequence generated as if produced by the shift register implementation shown in Figure 166–35 and the seed used is 0x19999998, which is a 31-bit hexadecimal value used to preset r[30] through r[0] (r[30] is set to the LSB and r[0] is set to the MSB) prior to the generation of the sequence for the indicated length of bits.
- The first 28 bits of the bit sequence A3 are 0, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, 1, 1, 0, 0, and 1.
- Bit sequence A is formed by binary inverting the concatenation of bit sequences A1, A2, and A3.

- Bit sequence B is a 65 636-bit sequence formed by removing the first bit from the concatenation of two bit sequences A and appending a 0 to the end.
- PAM4 sequence 1 is a 16 384-symbol sequence formed by mapping bit sequence A according to Table 166–6 for $G = 2$.
- PAM4 sequence 2 is PAM4 sequence 1 inverting the sign for each symbol.
- PAM4 sequence 3 is a 16 383-symbol sequence formed by mapping the first 32 766 bits of sequence B according to Table 166–6 for $G = 2$.
- PAM4 sequence 4 is a 16 385-symbol sequence formed by mapping the last 32 770 bits of sequence B according to Table 166–6 for $G = 2$, and inverting the sign for each symbol.
- The repeating SSPR-PAM4 pattern is formed by concatenating PAM4 sequences 1, 2, 3, and 4.

NOTE 1—A CSV file, Clause_166_SSPR-PAM4_pattern.csv, containing the entire PAM4 symbol sequence for the SSPR-PAM4 test pattern is available at <https://standards.ieee.org/downloads/802.3/>.

NOTE 2—The SSPR-PAM4 test pattern is different from SSPRQ test pattern specified in 120.5.11.2.3.

Figure 166–35 shows the shift register used for generation of the SSPR-PAM4 test pattern.

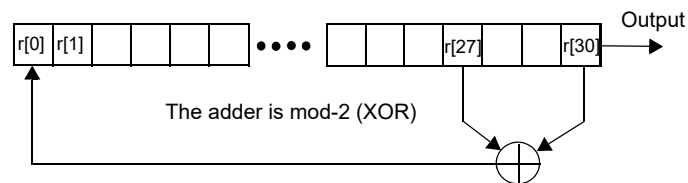


Figure 166–35—Shift register used for generation of the SSPR-PAM4 test pattern

166.5.6 Pattern for stressed receiver sensitivity measurement

The pattern for stressed receiver sensitivity measurement is a repeating normal Transmit Block encoding data in BER test mode (see 166.5.1), mapped according to Table 166–6 for the G parameter of the PMD under test, with the following PHD content (see 166.2.2.1.1):

- PHD.TX.NEXT.MODE = 1.
- PHD.RX.LINKSTATUS = 1.
- PHD.RX.HDRSTATUS = 1.
- PHD.RX.LINKMARGIN = 0x7F.
- PHD.CAP.LPI = 0.
- PHD.CAP.OAM = 0.
- PHD.OAM.DATA0 = 0x000.
- PHD.OAM.MSGT = 0.
- PHD.OAM.MERT = 0.
- PHD.OAM.PHYT = 0.
- PHD.OAM.DATA1 = 0x0000.
- PHD.OAM.DATA2 = 0x0000.
- PHD.OAM.DATA3 = 0x0000.
- PHD.OAM.DATA4 = 0x0000.
- PHD.OAM.DATA5 = 0x0000.
- PHD.OAM.DATA6 = 0x0000.
- PHD.OAM.DATA7 = 0x0000.
- PHD.OAM.DATA8 = 0x0000.

NOTE—Two CSV files, Clause_166_G1_SRS_pattern.csv and Clause_166_G2_SRS_pattern.csv, containing the entire symbol sequence for stressed receiver sensitivity pattern for $G=1$ and $G=2$, respectively are available at <https://standards.ieee.org/downloads/802.3/>.

166.6 Physical Medium Dependent (PMD) sublayer

166.6.1 PMD service interface

The following specifies the services provided by a BASE-AU PMD connected to a BASE-U PMA. The PMD sublayer service interface is described in an abstract manner and does not imply any particular implementation.

The PMD service interface supports the exchange of communication signals between PMA and PMD sublayers. The PMD translates the transmit and receive PMA signals to and from optical signals suitable for the specified medium.

The following primitives are defined:

PMD_COMSIGNAL.request
PMD_COMSIGNAL.indication
PMD_RXDETECT.indication

166.6.1.1 PMD_COMSIGNAL.request

This primitive defines the transfer of symbol values $\{-1\}$, $\{-1/3\}$, $\{+1/3\}$, or $\{+1\}$ from the PMA to the PMD.

166.6.1.1.1 Semantics of the primitive

PMD_COMSIGNAL.request(tx_signal)

During transmission, this primitive conveys to the PMD, via the parameter tx_signal, the amplitude of the output signal to be produced by the PMD transmit function.

166.6.1.1.2 When generated

The PMA generates PMD_COMSIGNAL.request(tx_signal) synchronously with every transmit symbol period.

166.6.1.1.3 Effect of receipt

Upon receipt of this primitive, the PMD converts tx_signal into the MDI optical signal (see 166.6.2.2).

166.6.1.2 PMD_COMSIGNAL.indication

This primitive defines the transfer of a signal amplitude from the PMD to the PMA.

166.6.1.2.1 Semantics of the primitive

PMD_COMSIGNAL.indication(rx_signal).

This primitive conveys to the PMA, via the parameter rx_signal, the amplitude relative to the instantaneous power of the optical signal received by the PMD at the MDI (see 166.6.2.3).

166.6.1.2.2 When generated

PMD_COMSIGNAL.indication(rx_signal) is continuously generated by the PMD.

166.6.1.2.3 Effect of receipt

Upon receipt of this primitive the PMA performs clock and data recovery (see 166.3.2).

166.6.1.3 PMD_RXDETECT.indication

This primitive is generated by the PMD signal detect function to indicate the status of the receive optical signal from the MDI as specified in 166.6.2.4.

166.6.1.3.1 Semantics of the service primitive

PMD_RXDETECT.indication(signal_detect)

The signal_detect parameter can take one of two values: OK or FAIL, indicating whether the PMD is detecting average optical power over a threshold (see 166.6.2.4) at the receiver (OK) or not (FAIL). When signal_detect = FAIL, then the rx_signal is undefined.

NOTE—A signal_detect = OK indication with enough average optical power to allow the PHY partners to start establishing the link, but not enough to meet the RS-FEC codeword error ratio target (see 166.3.4.6.1), results in the PHY indicating link_status = FAIL.

166.6.1.3.2 When generated

The PMD signal detect function generates this primitive to indicate a change in the value of signal_detect.

166.6.1.3.3 Effect of receipt

PMD_RXDETECT.indication(OK) indicates to the PMA receive function that an optical signal is being received from the MDI and that the link may be established with a link partner. PMD_RXDETECT.indication(OK) may be used to wake up from a low power state in a system that includes a BASE-AU PHY.

PMD_RXDETECT.indication(FAIL) indicates to the PMA receive function that an optical signal is not being received so link establishment cannot be initiated. PMD_RXDETECT.indication(FAIL) may be used to transition a system that includes a BASE-AU PHY into a low power state.

166.6.2 PMD functional specifications

166.6.2.1 PMD block diagram

For purposes of system conformance, the PMD sublayer is standardized at the test points TP2 and TP3, depicted in Figure 166–36. The optical transmit signal is defined at the output end of a multimode fiber patch cord (TP2), of 1 m to 3 m in length consistent with the PHY type under test (see 166.6.6.1). Unless specified otherwise, all transmitter measurements and defined tests are made at TP2. The optical receive signal is defined at the output of the fiber optic cabling (TP3) at the MDI. Unless specified otherwise, all receiver measurements and defined tests are made at TP3.

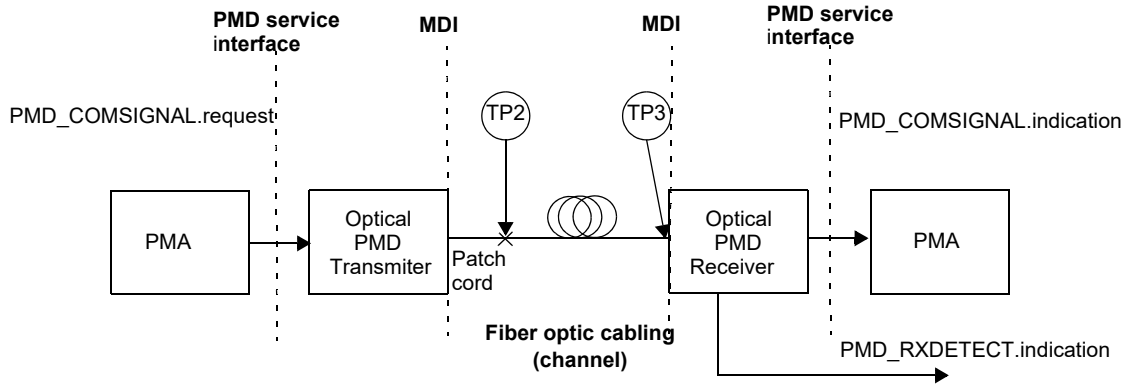


Figure 166-36—Test points for BASE-AU PHYs

166.6.2.2 PMD transmit function.

The PMD transmit function shall convert the communication signal amplitude given by the `tx_signal` parameter requested by the PMD service interface primitive `PMD_COMSIGNAL.request` (see 166.6.1.1) into optical signal p at TP2 according to the following function:

$$p = \frac{P_{\max} - P_{\min}}{2} \times \text{tx_signal} + \frac{P_{\max} + P_{\min}}{2} \quad (166-5)$$

where $P_{\min}$ and $P_{\max}$ are respectively the minimum and maximum optical power at TP2, and the parameter `tx_signal` fits $-1 \leq \text{tx_signal} \leq +1$ (see 166.3.4).

Optical signal p at TP2 shall comply with the transmitter optical characteristics in 166.6.3.2.

According to Equation (166-5), the maximum amplitude of `tx_signal` is translated into the highest optical power ($P_{\max}$) at TP2 and the minimum amplitude of `tx_signal` is translated into the lowest optical power ($P_{\min}$) at TP2.

166.6.2.3 PMD receive function

The PMD receive function shall convert the optical signal received at the MDI into the communication signal amplitude given by the `rx_signal` parameter of the PMD service interface primitive `PMD_COMSIGNAL.indication` (see 166.6.1.2), according to the receive optical characteristics in 166.6.3.3.

166.6.2.4 PMD signal detect function

The PMD signal detect function determines the value of the `signal_detect` parameter of the `PMD_RXDETECT.indication` primitive, which is signaled when the value of `signal_detect` changes.

The value of the `signal_detect` parameter shall be generated in response to the average optical power (AOP) present at TP3 according to the conditions defined in Table 166-7. The PMD signal detect function is not required to verify whether a compliant BASE-AU signal is being received. This standard imposes no response time requirements on the generation of the `signal_detect` parameter.

Table 166–7—Signal detect value definitions

Receive conditions	Signal detect value
Average optical power at TP3 < –30 dBm	FAIL
Average optical power at TP3 > –24 dBm	OK
–30 dBm < Average optical power at TP3 < –24 dBm	Unspecified (uncertainty range)

166.6.3 PMD to MDI optical specifications

166.6.3.1 Operating range

The operating range for the BASE-AU PMDs is defined in Table 166–8. A compliant PMD operates on 50/125 μm multimode fibers, type A1a.2 (OM3), according to the specifications defined in Table 166–19. A PMD that exceeds the operating range requirement while meeting all other optical specifications is considered compliant (e.g., a 10GBASE-AU PMD operating at 60 m meets the operating range requirement of 0.2 m to 40 m).

Table 166–8—BASE-AU operating range

PMD type	Required operating range
2.5GBASE-AU	0.2 m to 40 m
5GBASE-AU	
10GBASE-AU	
25GBASE-AU	
50GBASE-AU	

166.6.3.2 BASE-AU transmitter optical characteristics

The BASE-AU PHY transmitter shall meet the specifications in Table 166–9 measured at TP2, per the parameter definitions in 166.6.4.

Table 166–9—BASE-AU PMD transmitter optical characteristics

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
Signaling rate (range)	2.65625 ±100 ppm	5.3125 ±100 ppm	10.625 ±100 ppm	26.5625 ±100 ppm		Gbd
Modulation format	NRZ				PAM4	—
Center wavelength (range)	970 to 990					nm
RMS spectral width ^a (max)	0.7					nm
Average launch power (max)	3.9				5	dBm
Average launch power (min)	−4.3	−3.3	−2.4	−1.4	−0.3	dBm
Optical Modulation Amplitude (OMA _{outer}) (max)	3.2				4.3	dBm
Optical Modulation Amplitude (OMA _{outer}) (min)	max (A, TDFOM + B)					dBm
A	−5	−4	−3.1	−2.1	−1	dBm
B	−5	−3.9	−2.7	−1.5	−0.4	dBm
Transmitter distortion figure of merit (TDFOM) (max)	1	1	2	2.5	3	dB
Transmitter distortion figure of merit (TDFOM) (min)	−0.3	−0.4	−1.0	−1.5	−1.7	dB
Average launch power of OFF transmitter (max)	−30					dBm
Extinction ratio (min)	4				3.5	dB
RIN ₁₂ OMA (max)	−120			−124	−131	dB/Hz
Optical return loss tolerance (max)	12					dB
Uncorrelated random jitter (<i>t_J</i>) (max)	0.02				0.008	UI
Encircled flux ^b	≥ 86% at 19 μm ≤ 30% at 4.5 μm					

^a RMS spectral width is the standard deviation of the spectrum.

^b If measured into type A1a.2 50 μm fiber in accordance with IEC 61280-1-4:2009.

166.6.3.3 BASE-AU receiver optical characteristics

The BASE-AU PHY receiver shall meet the specifications in Table 166–10 measured at TP3 per the parameter definitions in 166.6.4.

Table 166–10—BASE-AU PMD receiver optical characteristics

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
Signaling rate (range)	2.65625 ±100 ppm	5.3125 ±100 ppm	10.625 ±100 ppm	26.5625 ±100 ppm		Gbd
Modulation format	NRZ				PAM4	—
Center wavelength (range)	970 to 990					nm
Damage threshold ^a (max)	4.9				6	dBm
Average receive power (max)	3.9				5	dBm
Average receive power ^b (min)	−18	−17	−16.1	−13.1	−8.3	dBm
Receive power (OMA _{outer}) (max)	3.2				4.3	dBm
Receiver reflectance (max)	−12					dB
Stressed receiver sensitivity ^c (OMA _{outer}), condition 1 (max)	−15.1	−14	−11.8	−8.1	−2.8	dBm
Stressed receiver sensitivity ^d (OMA _{outer}), condition 2 (max)	−16.1	−15.1	−14.2	−11.2	−6.4	dBm
Receiver sensitivity ^e (OMA _{outer}) (max)	max (C, TDFOM + D)					dBm
C	−16.1	−15.1	−14.2	−11.2	−6.4	dBm
D	−16.1	−15	−13.8	−10.6	−5.8	dBm
Conditions of stressed receiver sensitivity test ^f :						
Stressed TDFOM (STDFOM), condition 1	1	1	2	2.5	3	dB
Stressed TDFOM (STDFOM), condition 2	0	−0.1	−0.4	−0.6	−0.6	dB

^a The receiver shall be able to tolerate, without damage, continuous exposure to an optical input signal having this average power level. The receiver does not have to operate correctly at this input power.

^b Average receive power (min) is informative and not the principal indicator of signal strength. A received power below this value cannot be compliant; however, a value above this does not ensure compliance.

^c Measured with condition 1 conformance test signal at TP3 (see 166.6.4.13.2) for the BER specified in 166.1.

^d Measured with condition 2 conformance test signal at TP3 (see 166.6.4.13.2) for the BER specified in 166.1.

^e Receiver sensitivity (OMA_{outer})(max) is optional.

^f These test conditions are for measuring stressed receiver sensitivity for condition 1 and 2 respectively. They are not characteristics of the receiver.

166.6.3.4 BASE-AU illustrative link power budget

An illustrative power budget and penalties for BASE-AU channels are shown in Table 166–11.

Table 166–11— BASE-AU illustrative link power budget

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
Effective modal bandwidth at 980 nm ^a	950					MHz·km
Power budget	13	12	11.1	9.1	5.4	dB
Operating distance (max)	0.2 to 40					m
Channel insertion loss ^b (max)	10.5			8.5	4.5	dB
Channel insertion loss (min)	0					dB
Allocation for penalties ^c	0.6				0.9	dB

^a Per IEC 60793-2-10.

^b The channel insertion loss is calculated including aging using the maximum distance specified in Table 166–8, cabled optical fiber attenuation of 2 dB/km at 980 nm plus an allocation of 0.4 dB for cable attenuation penalty and connection insertion loss given in 166.6.6.2.1.

^c The allocation for penalties considers addition of two factors, the receiver sensitivity loss caused by modal noise and the macro-bending loss. Maximum macro-bending loss budgeted is 0.2 dB.

As an example, the values of transmitter $\text{OMA}_{\text{outer}}(\text{max})$ and transmitter $\text{OMA}_{\text{outer}}(\text{min})$ versus TDFOM for 50GBASE-AU, and the receiver sensitivity ($\text{OMA}_{\text{outer}}(\text{max})$) versus STDFOM for 50GBASE-AU are illustrated in Figure 166–37.

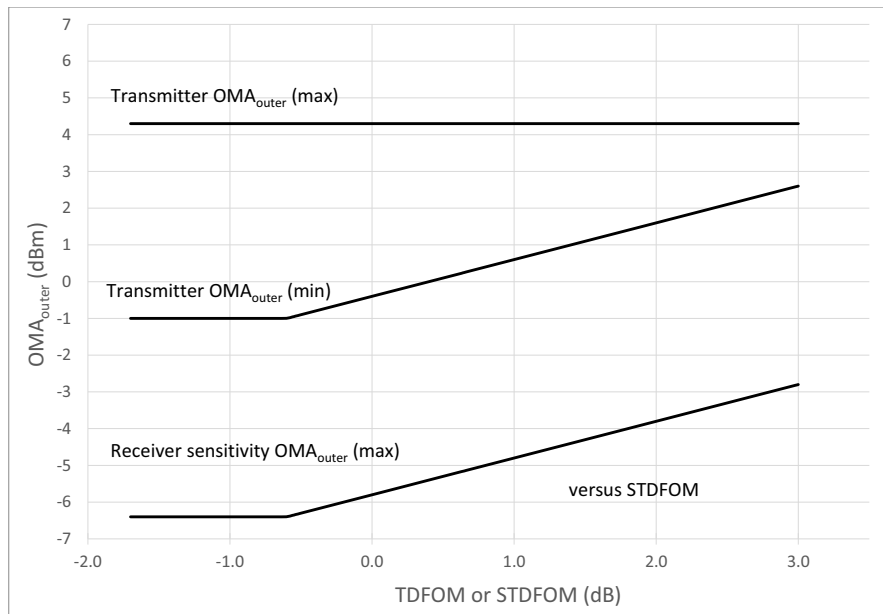


Figure 166–37—Transmitter $\text{OMA}_{\text{outer}}$ versus TDFOM, and receiver sensitivity ($\text{OMA}_{\text{outer}}$) versus STDFOM for 50GBASE-AU

166.6.4 Definition of optical parameters and measurement methods

166.6.4.1 Test patterns for optical parameters

While compliance is to be achieved in normal operation, specific test patterns are defined for measurement consistency and to enable measurement of some parameters. The test patterns used in this clause are shown in Table 166–12. Table 166–13 gives the test patterns to be used in each measurement, unless otherwise specified, and also lists references to the subclauses in which each parameter is defined. Any of the test patterns given for a particular test in Table 166–13 may be used to perform that test.

Table 166–12—Test patterns used for BASE-AU compliance

Pattern	Pattern description	Defined in
1	SSQWP square wave ($n_{sq} \{+1\}$, $n_{sq} \{-1\}$)	166.5.2
2	FSQWP square wave ($1 \{+1\}$, $1 \{-1\}$)	166.5.3
3	SSPR-NRZ	166.5.4
4	SSPR-PAM4	166.5.5
5	Pattern for stressed receiver sensitivity measurement	166.5.6

Table 166–13—Test pattern definitions and related subclauses

Parameter	Pattern (BASE-AU with $G = 1$)	Pattern (BASE-AU with $G = 2$)	Related subclause
Center wavelength and RMS spectral width	3, or valid BASE-AU signal	4, or valid BASE-AU signal	166.6.4.2
Average optical power	3, or valid BASE-AU signal	4, or valid BASE-AU signal	166.6.4.3
Outer Optical Modulation Amplitude (OMA_{outer})	1 or 3	1 or 4	166.6.4.4
Transmitter distortion figure of merit (TDFOM)	3	4	166.6.4.8
Extinction ratio	1 or 3	1 or 4	166.6.4.5
$RIN_{12}OMA$	1	1	166.6.4.6
Uncorrelated random jitter (t_J)	1 or 2	1 or 2	166.6.4.7
Stressed receiver conformance test signal calibration (STDFOM and ER)	3	4	166.6.4.13.2
Stressed receiver conformance test signal calibration (RIN and random jitter)	1	1	166.6.4.13.2
Stressed receiver sensitivity	5	5	166.6.4.13.3

166.6.4.2 Center wavelength and RMS spectral width

The center wavelength and RMS spectral width shall be within the range given in Table 166–9 if measured per IEC 61280-1-3. The center wavelength and RMS width is measured using the test pattern defined in Table 166–13.

166.6.4.3 Average optical power

The average optical power (AOP) shall be within the range given in Table 166–9 if measured per ANSI/TIA/EIA-455-95-2019 with a multimode fiber patch cord of 1 m to 3 m in length consistent with the PHY type under test (see 166.6.6.1). The average optical power is measured using the test pattern defined in Table 166–13, with the transmitter connected to an optical power meter.

Alternatively, the average optical power may be measured as the parameter α obtained in 166.6.4.8.

166.6.4.4 Outer Optical Modulation Amplitude (OMA_{outer})

The outer optical modulation shall be within the limits given in Table 166–9 if measured as defined in 166.6.4.4.1 using the test pattern defined in Table 166–13.

166.6.4.4.1 OMA_{outer} test setup

A block diagram for the OMA_{outer} measurement is shown in Figure 166–38. The patch cord is a multimode fiber of 1 m to 3 m in length consistent with the PHY type under test (see 166.6.6.1). Other equivalent measurement implementations may be used with suitable calibration.

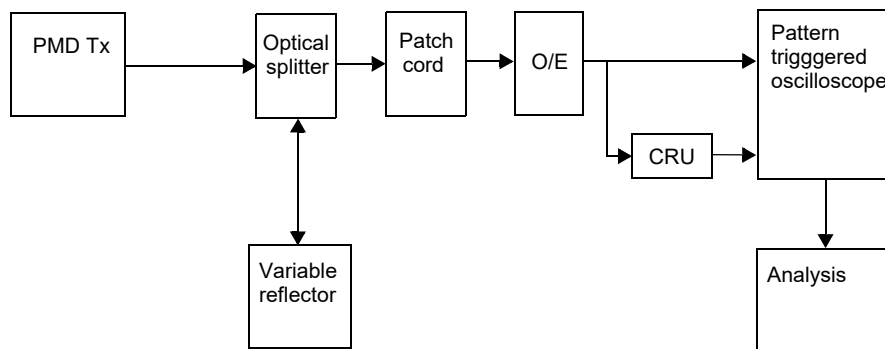


Figure 166–38—Conformance test block diagram for OMA, ER, RIN, and uncorrelated random jitter

The optical splitter and variable reflector are adjusted so that the transmitter is tested with the optical return loss specified in Table 166–9.

The combination of the O/E converter and the oscilloscope has a -3 dB bandwidth $BW_{-3\text{dB}}$ according to Table 166–14 with a fourth-order Bessel-Thomson low-pass filter response to at least $3 \times BW_{-3\text{dB}}$, and at frequencies above $3 \times BW_{-3\text{dB}}$, the response should not exceed -24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson response. BW_N is the equivalent noise bandwidth of fourth-order Bessel-Thomson filter response.

Table 166–14—Combined O/E and fourth-order Bessel BW_{-3dB} and BW_N

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
BW_{-3dB}	7.5			19.34		GHz
BW_N	7.8			20.11		GHz

The clock recovery unit (CRU) has a corner frequency of 0.1 MHz and a slope of 20 dB/decade for 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU. The CRU has a corner frequency of 1 MHz and a slope of 20 dB/decade for 25GBASE-AU and 50GBASE-AU. The CRU can be implemented in hardware or software depending on oscilloscope technology.

166.6.4.4.2 OMA_{outer} measurement

$P_{\max}$ is measured as the mean value of signal over the center 3% of the time interval where the signal is in the highest state.

$P_{\min}$ is measured as the mean value of signal over the center 3% of the time interval where the signal is in the lowest state.

Equation (166–6) specifies the OMA_{outer} of the PMD under test.

$$\text{OMA}_{\text{outer}} = P_{\max} - P_{\min} \quad (\text{W}) \quad (166-6)$$

Alternatively, the OMA_{outer} can be measured as defined in 166.6.4.9, Equation (166–18).

166.6.4.5 Extinction ratio

The extinction ratio shall be within the limits given in Table 166–9 if measured as described in 166.6.4.4.1 using the SSQWP test pattern (see Table 166–13).

Using $P_{\min}$ and $P_{\max}$ obtained in 166.6.4.4.2, the extinction ratio (ER) is calculated as defined in Equation (166–7).

$$\text{ER} = 10 \log_{10} \left(\frac{P_{\max}}{P_{\min}} \right) \quad (\text{dB}) \quad (166-7)$$

Alternatively, the ER can be measured as defined in 166.6.4.10, Equation (166–19).

166.6.4.6 Relative intensity noise (RIN_{12OMA})

The RIN_{12OMA} shall be within the limits given in Table 166–9 if measured as described in 166.6.4.4.1 and this subclause using the test pattern specified in Table 166–13.

Using the setup described in 166.6.4.4.1, RN_1 is measured as the standard deviation of signal over the center 3% of the time interval where the signal is in the highest state.

RN_0 is measured as the standard deviation of signal over the center 3% of the time interval where the signal is in the lowest state.

The RIN_{12OMA} is calculated using Equation (166–8), where OMA_{outer} is calculated as specified in 166.6.4.4.1, and BW_N is according to Table 166–14 and the BASE-AU PMD under test.

$$RIN_{12OMA} = 10 \log_{10} \left(\frac{(RN_1 + RN_0)^2}{(OMA_{outer})^2 BW_N} \right) \quad (\text{dB/Hz}) \quad (166-8)$$

166.6.4.7 Uncorrelated random jitter

The uncorrelated random jitter (t_j) shall be within the limits given in Table 166–9 if measured as described in 166.6.4.4.1 and this subclause using the test pattern specified in Table 166–13.

Using the setup described in 166.6.4.4.1, σ_{rise} is measured as the standard deviation of the time measurements where the rising edge of the pattern crosses the average level $(P_{max} + P_{min}) / 2$ in a horizontal measurement window of height 2% of $(P_{max} - P_{min})$.

σ_{fall} is measured as the standard deviation of the time measurements where the falling edge of the pattern crosses the average level $(P_{max} + P_{min}) / 2$ in a horizontal measurement window of height 2% of $(P_{max} - P_{min})$.

P_{max} and P_{min} are measured as specified in 166.6.4.4.2.

The uncorrelated random jitter (t_j) is calculated as specified in Equation (166–9).

$$t_j = \sqrt{\left(\frac{(\sigma_{rise})^2 + (\sigma_{fall})^2}{2} \right)} \quad (\text{UI}) \quad (166-9)$$

166.6.4.8 Transmitter distortion figure of merit (TDFOM)

TDFOM shall be within the limits given in Table 166–9 if measured using the method specified in 166.6.4.8.2 and the test setup specified in 166.6.4.8.1.

TDFOM measures the capability of a transmitter connected to a reference receiver (specified in 166.6.4.8.2) to comply with the required system BER (see 166.1) after a worst case optical channel (specified in 166.6.4.8.2), as measured through an optical to electrical converter (O/E) and oscilloscope with the combined frequency response given in 166.6.4.8.1. The reference receiver may be implemented in software or may be part of an oscilloscope.

Table 166–13 specifies the TDFOM test patterns to be used for BASE-AU.

166.6.4.8.1 TDFOM conformance test setup

A block diagram for the TDFOM conformance test is shown in Figure 166–39. The patch cord is a multimode fiber of 1 m to 3 m in length consistent with the PHY type under test (see 166.6.6.1). Other equivalent measurement implementations may be used with suitable calibration.

The optical splitter and variable reflector are adjusted so that the transmitter is tested with the optical return loss specified in Table 166–9.

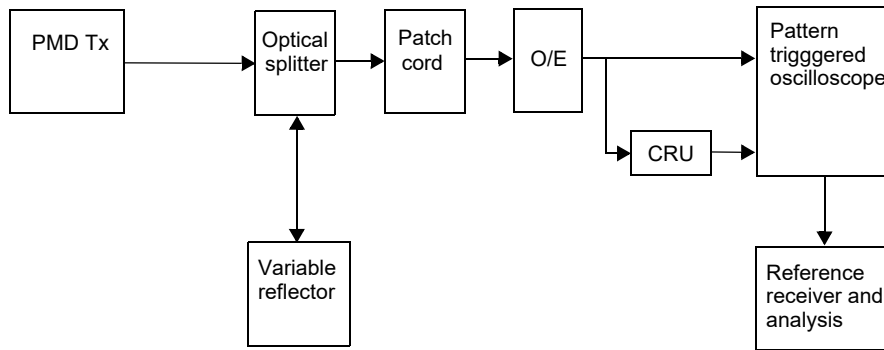


Figure 166–39—TDFOM conformance test block diagram

The combination of the O/E converter and the oscilloscope has a -3 dB bandwidth of 16.4 GHz with a fourth-order Bessel-Thomson low-pass response to at least 49.2 GHz, and at frequencies above 49.2 GHz, the response should not exceed -24 dB. Compensation may be made for any deviation from an ideal fourth-order Bessel-Thomson low-pass response.

The test pattern (specified in Table 166–13) is transmitted repetitively by the transmitter under test and the oscilloscope is set up to capture the complete pattern for TDFOM analysis as described in 166.6.4.8.2. The clock recovery unit (CRU) has a corner frequency of 0.1 MHz and a slope of 20 dB/decade for 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU. The CRU has a corner frequency of 1 MHz and a slope of 20 dB/decade for 25GBASE-AU and 50GBASE-AU. The CRU can be implemented in hardware or software depending on oscilloscope technology.

166.6.4.8.2 TDFOM measurement method

The oscilloscope is set up to capture samples from all symbols in the complete pattern with averaging with a number of samples per UI greater than 15.

A waveform with samples from all symbols of the complete pattern without Gaussian noise, DC unbiased, and gain normalized y is obtained by first averaging several captures, then subtracting the mean of the averaged waveform, denominated as α , and finally dividing the DC unbiased and averaged waveform by the maximum absolute value of its samples, denominated as β .

The number of captures necessary to perform the averaging is such that by adding a new capture to the averaging process, the difference of the new TDFOM calculated with respect to the previous one is less than 0.05 dB in absolute value.

Then, the reference receiver specified in Figure 166–40 is applied at each sample of y .

In the following, each of the blocks that compose the reference receiver are described.

- The sequence y is first processed by a second-order low-pass Butterworth filter with -3 dB bandwidth of $S \times 26.5625/5$ GHz for BASE-AU with $G = 1$ and $S \times 26.5625/3$ GHz for BASE-AU with $G = 2$.
- For BASE-AU with $G = 1$, the antialiasing filter is a single first-order low-pass filter with -3 dB bandwidth of $S \times 26.5625/2$ GHz.
- For BASE-AU with $G = 2$, the antialiasing filter is composed of the concatenation of two first-order low-pass filters with -3 dB bandwidth of $S \times 26.5625/2$ GHz for each one.

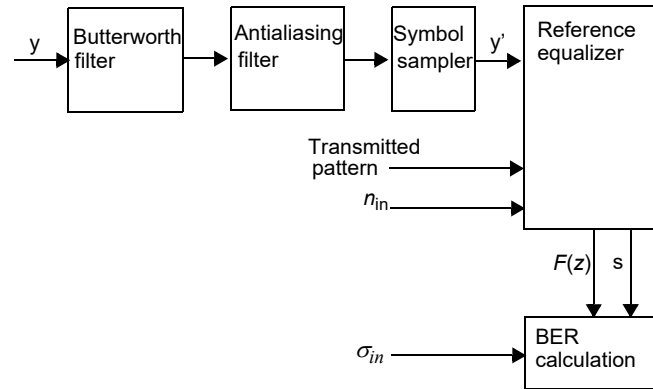


Figure 166-40—TDFOM reference receiver

- The symbol sampler selects a single sample per UI by choosing a sampler delay ϕ from the first sample in the UI.
- The white Gaussian random noise generator generates a sequence of random samples n_{in} spectrally white with standard deviation equal to σ_{in} .
- The reference equalizer is specified in 166.6.4.8.3, and generates the equalized sequences s and s_n .
- The BER calculation block calculates the BER from σ_{in} , the equalized sequence s , and the equalizer filter $F(z)$ as specified in 166.6.4.8.4.
- An iterative algorithm finds the maximum value of σ_{in} so that the calculated BER is equal to 1.757×10^{-4} .

166.6.4.8.3 TDFOM reference equalizer

The functional model of the reference equalizer is specified in Figure 166-41.

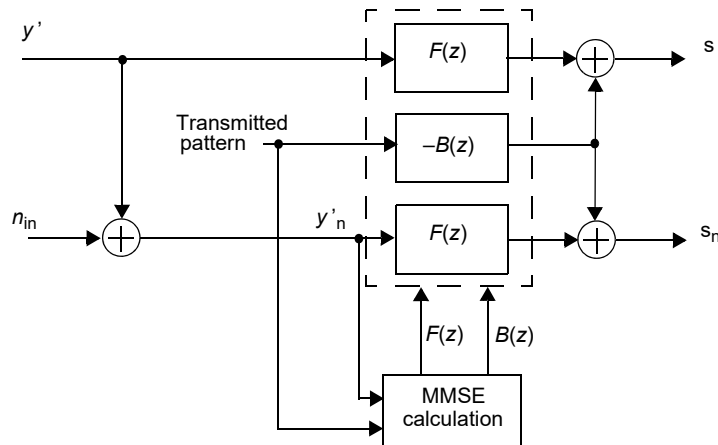


Figure 166-41—TDFOM reference equalizer

The noise sequence n_{in} is added to y' generating the noisy sequence y'_n .

The equalizer filters $F(z)$ and $B(z)$ are defined in Equation (166-10) and Equation (166-11), respectively, where z^{-i} represents a delay of i unit intervals.

$$F(z) = \sum_{i=0}^{N_F-1} f[i]z^{-i} \quad (166-10)$$

$$B(z) = \sum_{i=1}^{N_B} b[i]z^{-i} \quad (166-11)$$

Table 166–15 specifies the number of taps N_F and N_B for the filters $F(z)$ and $B(z)$ in the TDFOM reference equalizer for the BASE-AU type under test.

Table 166–15—BASE-AU TDFOM reference equalizer number of taps

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU
Number of taps of the $F(z)$ filter (N_F)	4			8	
Number of taps of the $B(z)$ filter (N_B)	2				1

Figure 166–42 and Figure 166–43 illustrate the $F(z)$ and $B(z)$ filters functional models, respectively.

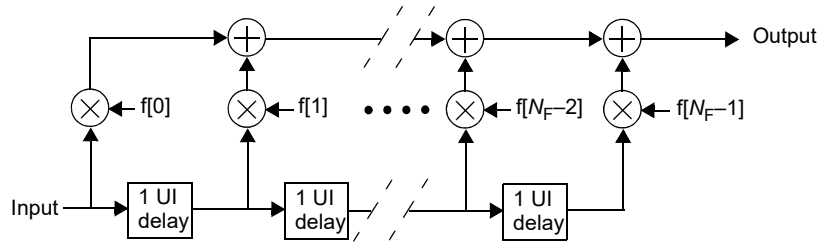


Figure 166–42—TDFOM $F(z)$ filter functional model

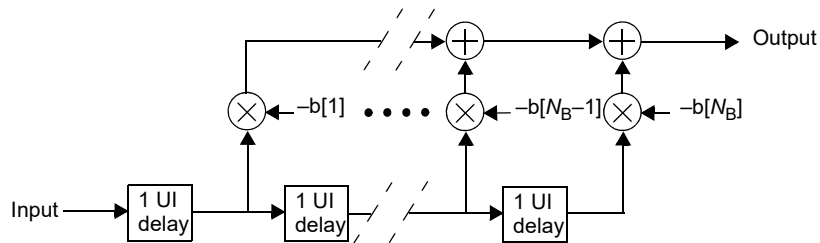


Figure 166–43—TDFOM $B(z)$ filter functional model

$F(z)$ and $B(z)$ are calculated to minimize the mean square error (MMSE) between the transmitted pattern (SSPR-NRZ for $G = 1$, or SSPR-PAM4 for $G = 2$) and the equalized sequence s_n .

Additionally, the selected sampler delay ϕ and $F(z)$ delay are optimized for minimum BER after equalization, calculated as specified in 166.6.4.8.4, at each value of σ_{in} .

Equation (166–12) defines the equalizer gain.

$$G_{eq} = \left| \frac{\sum_{i=0}^{N_F-1} f[i]}{N_B} \right| \left| 1 + \sum_{i=1} b[i] \right| \quad (166-12)$$

166.6.4.8.4 BER calculation

The BER calculation is obtained after the following steps:

- For BASE-AU with $G = 1$, define the threshold vector THv containing a single value equal to 0 and with length N_{th} equal to 1.
- For BASE-AU with $G = 2$, define the threshold vector THv containing three values equal to $-2/3$, 0 and $+2/3$, and with length N_{th} equal to 3.
- Calculate the histogram of sequence s , where the value of each bin $h(i)$ is normalized to relative probability, such that $h(i) = c(i)/N_h$, where $c(i)$ is the number of elements in the bin centered in $e(i)$ of width $\Delta e = (\max(s) - \min(s))/N_h$, and N_h is the number of bins of the histogram of sequence s , which are selected to be greater or equal to 500.
- For each element of the threshold vector THv , denoted as $THv(k)$, calculate ih_p as the bins that meet $e(i) > THv(k)$, ih_n as the bins that meet $e(i) \leq THv(k)$, and $BER_{th}(k)$ as specified in Equation (166–13).
- Calculate the total BER as the sum of the N_{th} values of $BER_{th}(k)$.

Equation (166–13) specifies how to calculate $BER_{th}(k)$:

$$BER_{th}(k) = \frac{1}{2G} \sum_{i=\min(ih_n)}^{\max(ih_n)} \left(h(i) \operatorname{erfc} \left(\frac{THv(k) - e(i)}{\sigma_n \sqrt{2}} \right) \right) + \frac{1}{2G} \sum_{i=\min(ih_p)}^{\max(ih_p)} \left(h(i) \operatorname{erfc} \left(\frac{e(i) - THv(k)}{\sigma_n \sqrt{2}} \right) \right) \quad (166-13)$$

where

- $\operatorname{erfc}(x)$ is the complementary error function defined in Equation (166–14)
- σ_n is calculated from σ_{in} using Equation (166–15)
- G depends on the PHY type as specified in Table 166–1

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_x^{\infty} e^{-t^2} dt \quad (166-14)$$

$$\sigma_n = \sigma_{in} \sqrt{\sum_{i=0}^{N_{(F)}-1} f^2[i]} \quad (166-15)$$

166.6.4.8.5 OMA calculation at equalization output

OMA_{eq} is defined as the OMA of sequence s at the equalization output, and is measured using continuous identical digits (CID).

The OMA_{eq} calculation is obtained after the following steps:

- For BASE-AU with $G = 1$, search for positive CID as continuous samples of s with value > 0 .
- For BASE-AU with $G = 2$, search for positive CID as continuous samples of s with value $> 2/3$.
- For BASE-AU with $G = 1$, search for negative CID as continuous samples of s with value < 0 .
- For BASE-AU with $G = 2$, search for negative CID as continuous samples of s with value $< -2/3$.
- Select CID sequences with length greater or equal to 14.
- Remove first six and last six samples from the selected CID sequences.
- For the remaining symbols of all the CID sequences, calculate the average value. For positive CID sequences, OMA_p is calculated. For negative CID sequences, OMA_n is calculated.
- Calculate OMA_{eq} as $OMA_p - OMA_n$.

166.6.4.8.6 TDFOM calculation

TDFOM is calculated as specified in Equation (166–16), where M , Q_0 , and $TDFOM_0$ depend on the BASE-AU under test as specified in Table 166–16.

$$TDFOM = 10\log_{10}\left(\frac{OMA_{in}}{2(M-1)\sigma_{in}Q_0}\right) - TDFOM_0 \quad (\text{dB}) \quad (166-16)$$

Table 166–16— BASE-AU TDFOM parameters

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
M	2				4	—
Q_0	3.5741 ^a				3.4981 ^a	—
$TDFOM_0$	3.59	3.61	3.63	3.92	2.83	dB

^a Consistent with $BER = 1.757 \times 10^{-4}$.

OMA_{in} is the OMA at the input of the reference equalizer, which is calculated as defined in Equation (166–17), where G_{eq} is given by Equation (166–12).

$$OMA_{in} = \frac{OMA_{eq}}{G_{eq}} \quad (\text{W}) \quad (166-17)$$

166.6.4.9 OMA_{outer} calculation

The OMA_{outer} can be calculated as defined in Equation (166–18), where OMA_{in} has been calculated as specified in 166.6.4.8.6, and β has been obtained as specified in 166.6.4.8.2.

$$OMA_{outer} = OMA_{in} \times \beta \quad (W) \quad (166-18)$$

166.6.4.10 Extinction ratio calculation

The extinction ratio can be calculated as defined in Equation (166–19), where OMA_p and OMA_n have been measured as specified in 166.6.4.8.5, and α and β have been obtained as specified in 166.6.4.8.2.

$$ER = 10 \log_{10} \left(\frac{OMA_p \times \beta + G_{eq} \times \alpha}{OMA_n \times \beta + G_{eq} \times \alpha} \right) \quad (dB) \quad (166-19)$$

166.6.4.11 Average optical power calculation

The average optical power (AOP) can be calculated as the α parameter obtained as specified in 166.6.4.8.2.

166.6.4.12 Receiver sensitivity

Receiver sensitivity is optional. The mandatory requirement for receivers is the stressed receiver sensitivity (see 166.6.4.13).

For 2.5GBASE-AU, receiver sensitivity constraints are defined by Equation (166–20) for any STDFOM value from –0.3 dB through 1 dB.

For 5GBASE-AU, receiver sensitivity constraints are defined by Equation (166–20) for any STDFOM value from –0.4 dB through 1 dB.

For 10GBASE-AU, receiver sensitivity constraints are defined by Equation (166–20) for any STDFOM value from –1 dB through 2 dB.

For 25GBASE-AU, receiver sensitivity constraints are defined by Equation (166–20) for any STDFOM value from –1.5 dB through 2.5 dB.

For 50GBASE-AU, receiver sensitivity constraints are defined by Equation (166–20) for any STDFOM value from –1.7 dB through 3 dB.

The receiver sensitivity limits given by Equation (166–20) for each PHY type are illustrated in Figure 166–44.

NOTE—The receiver sensitivity is defined over a range of STDFOM values, while the stressed receiver sensitivity is defined for specific values of STDFOM for each PHY type.

$$RS \leq \max(C, STDFOM + D) \quad (dBm) \quad (166-20)$$

where

RS	is the receiver sensitivity
C	is a parameter defined in Table 166–10 for each BASE-AU under test
$STDFOM$	is the STDFOM of the transmitter used to measure the receiver sensitivity
D	is a parameter defined in Table 166–10 for each BASE-AU under test

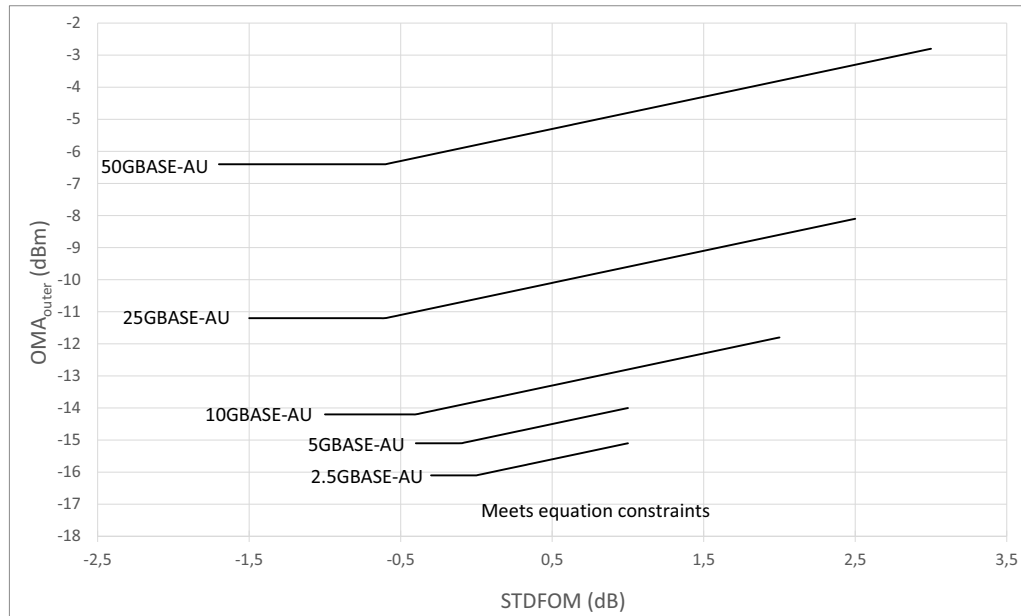


Figure 166-44—Illustration of receiver sensitivity

166.6.4.13 Stressed receiver sensitivity

Stressed receiver sensitivity shall be within the limits given in Table 166-10 if measured using the methodology defined in 166.6.4.13.1 and 166.6.4.13.3, with the conformance test signal at TP3 as described in 166.6.4.13.2, for conditions 1 and 2. The test patterns used for stressed receiver sensitivity are those specified in Table 166-13.

The pattern specified for stressed receiver sensitivity in Table 166-13 is sent from the calibrated stressed transmitter to the PHY receiver under test. The signal being transmitted by the PHY under test is asynchronous to the received signal.

166.6.4.13.1 Stressed receiver conformance test block diagram

A block diagram for the receiver conformance test is shown in Figure 166-45. The pattern used for the received conformance signal is specified in Table 166-13. The optical test signal is conditioned (stressed) using the stressed receiver methodology defined in 166.6.4.13.2 and has a sinusoidal jitter applied as specified in 166.6.4.13.4. A suitable test set is needed to characterize the signal used to test the receiver. Stressed receiver conformance test signal verification is described in 166.6.4.13.2.

The FFE filter $P(z)$ has 3 taps working at the nominal symbol rate of the test-pattern generator defined in Equation (166-21), where z^{-i} represents a delay of i unit intervals.

$$P(z) = -a + (1 + 2a)z^{-1} - az^{-2} \quad (166-21)$$

The Gaussian random jitter generator, the sinusoidal jitter, the Gaussian random noise generator, bias configuration, FFE filter, and the low-pass filter are adjusted so that the STDFOM, ER, RIN, and uncorrelated random jitter specified in Table 166-9 and Table 166-10 is met for the condition under test, according to the methods specified in 166.6.4.13.2. The E/O converter is connected to the optical attenuator by means of a 40 m long multimode patch cord, consistent with the PHY type under test (see 166.6.6.1).

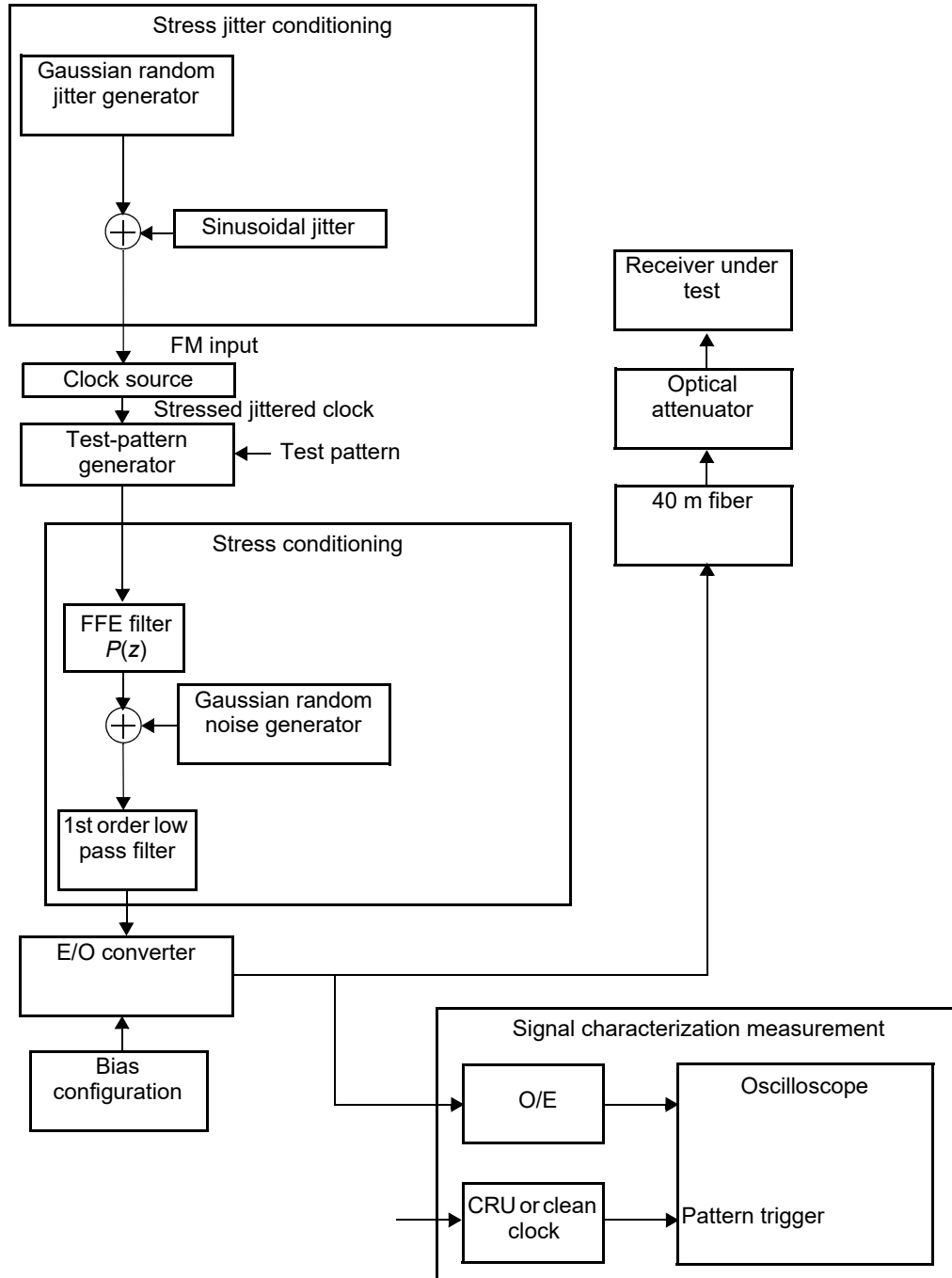


Figure 166-45—Stressed receiver conformance test block diagram

The stressed receiver sensitivity $\text{OMA}_{\text{outer}}$ at TP3 ($\text{SRS OMA}_{\text{outer}}$) is obtained after the following steps:

- The test pattern generator is configured to generate a specified pattern for stressed receiver sensitivity in Table 166-13.

- b) The optical attenuation is increased until the receiver under test reports one of the following conditions:
 - The RS-FEC codeword error ratio is over the limit specified in 166.3.4.6.1.
 - Local link margin reported in register 3.2350 (see 45.2.3.92) is lower than 0.
 - Local receiver status, link status, or local PHD reception status reported in register 3.2349 (see 45.2.3.91) is equal to 0.
- c) Optical attenuation is decreased in small steps until none of the above conditions are met.
- d) Measure average optical power at TP3 (AOP_{TP3}).

The stressed receiver sensitivity OMA_{outer} at TP3 (SRS OMA_{outer}) is calculated using Equation (166–22), where Γ_{tx} is obtained as specified in 166.6.4.13.2.

$$SRS\ OMA_{outer} = AOP_{TP3} \times \Gamma_{tx} \quad (W) \quad (166-22)$$

Alternatively, SRS OMA_{outer} can be measured using the method described in 166.6.4.4.

166.6.4.13.2 Stressed receiver conformance test signal characteristics and calibration

The conformance test signal is used to validate that the PMD receiver meets BER requirements with near worst-case waveforms at TP3 in two different conditions, named condition 1 and condition 2.

The primary parameters of the stressed receiver conformance test signals are their stressed TDFOM (STDFOM), ER, and RIN.

The following steps describe a possible method for setting up and calibrating a stressed receiver conformance test signal when using a stressed receiver conformance test setup as shown in Figure 166–45:

- 1) Measure and calibrate the STDFOM of the stressed receiver conformance test signal according to 166.6.4.8. An example stressed receiver conformance test setup is shown in Figure 166–45; however, alternative test setups that generate equivalent stress conditions may be used.
 - a) Set the signaling rate of the test pattern generator to meet the requirements in Table 166–10 for the PMD receiver under test.
 - b) Set the test pattern specified for stressed receiver conformance test signal calibration (STDFOM and ER) in Table 166–13.
 - c) Configure the oscilloscope for TDFOM analysis (see 166.6.4.8).
 - d) With the stress jitter conditioning and the Gaussian noise generator turned off, set the bias configuration, 1st order low pass filter, and the parameter a of the $P(z)$ FFE filter to get the required STDFOM of the condition under test as specified in Table 166–10. Modulation amplitude of the test pattern generator is also adjusted to obtain the minimum ER specified in Table 166–9.
 - e) Measure OMA_{outer} and AOP as specified in 166.6.4.9 and 166.6.4.11 to calculate $\Gamma_{tx} = OMA_{outer}/AOP$.
- 2) Measure and calibrate the RIN and uncorrelated random jitter of the stressed receiver conformance test signal according to 166.6.4.6.
 - a) Set the signaling rate of the test pattern generator to meet the requirements in Table 166–10 for the PMD receiver under test.
 - b) Set the test pattern specified for stressed receiver conformance test signal calibration (RIN and random jitter) in Table 166–13.
 - c) Configure the oscilloscope for RIN and random jitter analysis (see 166.6.4.6).

- d) Turn on the sinusoidal jitter and adjust its amplitude according to 166.6.4.13.4, using a clean clock instead of the CRU to trigger the oscilloscope.
 - e) The Gaussian random jitter generator is adjusted in amplitude to obtain the maximum uncorrelated random jitter specified in Table 166–9.
 - f) Turn on the Gaussian noise generator and adjust it in amplitude to obtain the maximum RIN_{12OMA} specified in Table 166–9.
- 3) Measure and calibrate again the STDFOM of the stressed receiver conformance test signal according to 166.6.4.8, adjusting the $P(z)$ FFE filter parameter a and the low pass filter, if necessary, to get the STDFOM specified in Table 166–10 for the condition under test.

The optical attenuator is adjusted to supply a signal at the input to the receiver under test and measure the stressed receiver sensitivity (OMA_{outer}) for the condition under test (1 or 2).

166.6.4.13.3 Stressed receiver conformance test signal verification

The STDFOM and ER of the stressed receiver conformance test signal are measured according to 166.6.4.8, and RIN and uncorrelated random jitter are measured according to 166.6.4.6 and 166.6.4.7, respectively. The clock output from the clock source in Figure 166–45 is modulated with the sinusoidal and random jitter. To use an oscilloscope to calibrate the final stressed signal that includes the sinusoidal jitter component, a separate clock source (clean clock of Figure 166–45) is required that is synchronized to the source clock, but not modified with the jitter source.

Care should be taken when characterizing the test signal because excessive noise/jitter in the measurement system would result in an input signal that does not fully stress the receiver under test. Running the receiver sensitivity test with a signal that is under-stressed may result in the deployment of non-compliant receivers. The noise/jitter introduced by the O/E, filters, and oscilloscope should be negligible or the results should be corrected for its effects. While the details of test equipment are beyond the scope of this standard, it is recommended that the implementer fully characterize the test equipment and apply appropriate guard bands so that the stressed receiver conformance input signal meets the stress and sinusoidal jitter specified in 166.6.4.13.2 and 166.6.4.13.4.

166.6.4.13.4 Sinusoidal jitter for receiver conformance test

The sinusoidal jitter is used in the stress jitter conditioning of the stress receiver sensitivity test setup in Figure 166–45. The amplitude of the applied sinusoidal jitter is dependent on frequency and on the BASE-AU PMD under test as specified in Table 166–10.

The sinusoidal jitter is dependent on the frequency f as specified in Table 166–17 for 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU, and in Table 166–18 for 25GBASE-AU and 50GBASE-AU. The parameter f is given in kHz for the equations in Table 166–17 and Table 166–18.

Table 166–17—Applied sinusoidal jitter for 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU

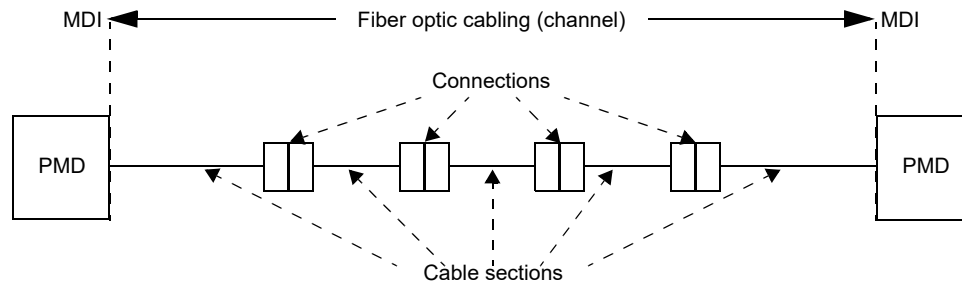
Frequency range	Sinusoidal jitter, peak-to-peak (UI)		
	2.5GBASE-AU	5GBASE-AU	10GBASE-AU
$f < 1$ kHz	Not specified		
$1 \text{ kHz} \leq f < 300 \text{ kHz}$	$15 \text{ kHz}/f$		
$300 \text{ kHz} \leq f \leq 1 \text{ MHz}$	0.05		

Table 166–18—Applied sinusoidal jitter for 25GBASE-AU and 50GBASE-AU

Frequency range	Sinusoidal jitter, peak-to-peak (UI)	
	25GBASE-AU	50GBASE-AU
$f < 1 \text{ kHz}$	Not specified	
$1 \text{ kHz} \leq f < 3 \text{ MHz}$	$150 \text{ kHz}/f$	$60 \text{ kHz}/f$
$3 \text{ MHz} \leq f \leq 10 \text{ MHz}$	0.05	0.02

166.6.5 Fiber optic cabling model

The fiber optic cabling model is shown in Figure 166–46.



NOTE—The actual number of connections supported depends on the total connection insertion loss as defined in Table 166–20.

Figure 166–46—BASE-AU fiber optic cabling model

The channel insertion loss is given in Table 166–11. A channel may contain additional connectors as long as the optical characteristics of the channel (such as attenuation, modal dispersion, reflections, and losses of all connectors and splices) meet the specifications. The fiber optic cabling model (channel) defined here is the same as a simplex fiber optic link segment. The term *channel* is used here for consistency with generic cabling standards.

166.6.6 Characteristics of the fiber optic cabling (channel)

The BASE-AU fiber optic cabling shall meet the specifications defined in Table 166–11. The fiber optic cabling consists of one or more sections of fiber optic cable and any intermediate connections required to connect sections together.

166.6.6.1 Optical fiber cable

The fiber contained within the BASE-AU fiber optic cabling shall comply with the requirements of IEC 60793-2-10 for optical fiber type A1a.2 (OM3) and the requirements of Table 166–19. For parameters where they differ, Table 166–19 prevails. A variety of multimode cable types may satisfy these requirements, provided the resulting channel also meets the specifications of Table 166–11.

Table 166–19— BASE-AU optical fiber and cable characteristics

Parameter	OM3 ^a	Units
Nominal wavelength	980	nm
Cabled optical attenuation (max)	2	dB/km
Effective modal bandwidth ^b (min)	950	MHz·km
Zero dispersion wavelength λ_0	1328	nm
Chromatic dispersion slope (max) S_0	0.093477	ps/(nm ² ·km)

^a IEC 60793-2-10 type A1a.2.

^b When measured with the launch conditions specified in Table 166–9.

166.6.6.2 Optical fiber connection

An optical fiber connection, as shown in Figure 166–46, consists of a mated pair of optical connectors.

166.6.6.2.1 Optical fiber connection insertion loss

The connection insertion loss is specified for a connection, which consists of a mated pair of optical connectors.

The maximum number of connections is calculated based on the allocation of total connection insertion loss shown in Table 166–20.

Table 166–20— BASE-AU total connection insertion loss

Parameter	2.5GBASE-AU	5GBASE-AU	10GBASE-AU	25GBASE-AU	50GBASE-AU	Units
Total connection insertion loss (max)	10			8	4	dB

For example, the allocation of total insertion loss for 25GBASE-AU supports four connections with a maximum insertion loss equal to 2 dB per connection.

Connections with lower loss characteristics may be used provided the requirements of Table 166–11 are met. However, the loss of a single connection shall not exceed 2 dB for 25GBASE-AU and 50GBASE-AU, and 2.5 dB for 2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU.

166.6.6.2.2 Maximum discrete reflectance

The maximum discrete reflectance shall be less than –20 dB.

166.6.7 Medium Dependent Interface (MDI)

The BASE-AU PMD is coupled to the fiber optic cabling at the MDI. The MDI is the interface between the PMD and the fiber optic cabling (as shown in Figure 166–46).

Specification of the MDI mechanical interface is beyond the scope of this standard.

Compliance testing is performed at TP2 and TP3 as defined in Figure 166–36 and 166.6.3.

166.7 BASE-U operations, administration, and maintenance (BASE-U OAM) channel

The optional OAM channel capability is announced by setting the field PHD.CAP.OAM (see Table 166–2) on the transmitted PHD to one. OAM channel functionality shall be enabled when both the transmitted and received fields PHD.CAP.OAM are equal to one, and disabled otherwise.

If supported and enabled, the OAM channel provides a mechanism to reliably exchange messages between station management entity (STA) peers attached to link partners, with the specifications in this subclause. If OAM is not supported or not enabled, all OAM fields shall be set to zero in the transmitted PHD, and ignored in the received PHD.

The BASE-U OAM message exchange occurs in the PCS, as part of the PHD, and does not impact the normal xMII to xMII data transmission. Moreover, the BASE-U OAM message exchange is not affected by EEE operation.

The BASE-U OAM channel utilizes BASE-U OAM transmit and receive registers accessible via the MDIO specified in 45.2.3.88 and 45.2.3.89.

The BASE-U OAM channel uses several control bits (TXO_REQ, TXO_MSGT, TXO_MERT, and TXO_PHYT) for requesting message transmission, message identification, message delivery notification, and for flow control.

The BASE-U OAM message transmission and reception protocol shall follow the same steps specified in 115.9.1 and 115.9.3, respectively for 1000BASE-H OAM, but using the transmit and receive BASE-U OAM registers specified in 45.2.3.88 and 45.2.3.89 instead of transmit and receive 1000BASE-H OAM registers.

The BASE-U OAM control bits TXO_REQ, TXO_MSGT, TXO_PHYT, and TXO_MERT specified in 45.2.3.88 shall reflect the BASE-U OAM transmitted message status as specified in 115.9.2, but using the transmit BASE-U OAM registers specified in 45.2.3.88 instead of transmit 1000BASE-H OAM registers.

The BASE-U OAM message transmission and reception protocol shall also conform to the transmit and receive state diagrams specified in 115.9.4, but using the transmit and receive BASE-U OAM registers specified in 45.2.3.88 and 45.2.3.89 instead of transmit and receive 1000BASE-H OAM registers.

166.8 Loopback modes

BASE-U loopback modes shall be selected using bits 3.2348.12:10 (see 45.2.3.90.2). Loopback modes support a MAC transmit-to-self that includes selected portions of the local PHY or the local and the remote PHYs.

The default value and value upon reset (bit 1.0.15 = 1) shall be no loopback (3.2348.12:10 = 000). When xMII level loopback mode is selected (3.2348.12:10 = 001), the BASE-U PCS shall accept data on the transmit data path from the xMII, looping the data back to the receive path of the xMII. In this mode, the PCS transmitter and receiver are not part of the communication path, and the loopback functionality is provided independent of a link having been established with the link partner. Operations of PCS, PMA, and PMD sublayers are not specified in this loopback mode.

When PMD interface level loopback mode is selected (3.2348.12:10 = 010), the loopback shall be implemented near the PMD service interface, completely exercising the PCS and PMA as in normal operation, although the PMA receiver ignores signals from the PMD receive function. PMD operation is not specified in this loopback mode.

When line loopback mode is selected (3.2348.12:10 = 011), received data shall be processed and looped back near the xMII interface toward the link partner with data going back through the PCS and PMA transmit path. The received signal is processed and decoded by the PMA and PCS sublayers. Then the received data stream is forwarded to the xMII receive interface as well as to the PCS transmitter. The xMII transmit interface signals are ignored during line loopback operation. The line loopback is only operative when a bidirectional link has been established. The PCS, PMA, and PMD functions work as in normal operation. The data loopback is implemented between the output of the PCS 64B/65B decoder and the input of the PCS 64B/65B encoder.

Figure 166–3 shows the location of each of the loopback modes in the BASE-AU PHY functional block.

166.9 Management interface

PHYs in the BASE-AU set shall provide the management capabilities described in this clause and the functionality provided by the referenced Clause 45 registers and bits.

The optional MDIO capability of Clause 45 describes several variables that provide control and status for and about the PHY. If the MDIO is not implemented, an implementation shall include the functionality provided by the specified MDIO registers.

PHYs in the BASE-AU set use some generic control bits common with other IEEE 802.3 PHY types. PHY variables shall be mapped as shown in Table 166–21. PHYs in the BASE-AU set also use specific registers (1.72, 1.901, and 3.2330 through 3.2353).

Table 166–21—BASE-AU variable mapping

MDIO variable/value	PMA/PMD register	Register bit number	PMA/PMD variable/value
Reset = 1	PCS control 1	3.0.15	pcs_reset = TRUE
Reset = 0	PCS control 1	3.0.15	pcs_reset = FALSE
Reset = 1	PMA/PMD control 1	1.0.15	pma_reset = ON
Reset = 0	PMA/PMD control 1	1.0.15	pma_reset = OFF
Low Power = 1	PMA/PMD control 1	1.0.11	link_control = DISABLE
Low Power = 0	PMA/PMD control 1	1.0.11	link_control = ENABLE

Table 166–21—BASE-AU variable mapping (*continued*)

MDIO variable/value	PMA/PMD register	Register bit number	PMA/PMD variable/value
Global PMD receive signal detect = 1	PMD receive signal detect	1.10.0	signal_detect = OK
Global PMD receive signal detect = 0	PMD receive signal detect	1.10.0	signal_detect = FAIL
Global PMD transmit disable = 1	PMD transmit disable	1.9.0	link_control = DISABLE
Global PMD transmit disable = 0	PMD transmit disable	1.9.0	link_control = ENABLE

In addition to the normal operation capabilities specified elsewhere in this clause, test modes and loopback modes use these registers and bits to facilitate testing.

166.10 Environmental specifications

166.10.1 Temperature classes

The optical link is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling appropriate to the intended environment (e.g., automotive). A compliant BASE-AU PHY shall clearly indicate the temperature class of Table 166–22 over which its compliance is claimed.

Table 166–22—BASE-AU temperature classes^a

Class	Low temperature (°C)	High temperature (°C)
Class 1	–40	+125
Class 2	–40	+105
Class 3	–40	+85

^a The temperatures in this table are the ambient operating temperatures, defined as the ambient temperature inside the electronic control unit (ECU) or the equipment in which a BASE-AU PHY is integrated.

166.10.2 General safety

Equipment subject to this clause shall conform to the general safety requirements in [J.2](#).

166.10.3 Environmental safety

All equipment subject to this clause, when used in an automotive environment, is expected to conform to the potential environmental stresses with respect to their mounting location, as defined in the following specifications:

- a) General loads: ISO 16750-1
- b) Electrical loads: ISO 16750-2, ISO 7637-2:2008, and ISO 8820-1
- c) Mechanical loads: ISO 16750-3, ASTM D4728, and ISO 12103-1
- d) Climatic loads: ISO 16750-4, IEC 60068-2-1, IEC 60068-2-27, IEC 60068-2-30, IEC 60068-2-38, IEC 60068-2-52, IEC 60068-2-64, and IEC 60068-2-78
- e) Chemical loads: ISO 16750-5 and ISO 20653

Automotive environmental conditions are generally more severe than those found in many commercial and industrial environments. The targeted automotive, industrial, or commercial environment(s) requires careful analysis prior to implementation.

166.10.4 Electromagnetic compatibility

A system integrating the BASE-AU PHY intended for automotive applications is expected to comply with all applicable local and national codes. In addition, the system may need to comply with more stringent requirements for the limitation of electromagnetic interference. When used in an automotive environment, a BASE-AU PHY is expected to meet the following motor vehicle EMC requirements:

- a) Radiated/Conducted Emissions: CISPR 25, IEC 61967-1, IEC 61967-4, and IEC 61000-4-21
- b) Radiated/Conducted Immunity: ISO 11452, IEC 62132-1, IEC 62132-4, and IEC 61000-4-21
- c) Electrostatic Discharge: ISO 10605, IEC 61000-4-2, and IEC 61000-4-3
- d) Electrical Disturbances: IEC 62215-3, ISO 7637-2, and ISO 7637-3

Exact test setup and test limit values may be adapted to each specific application.

166.10.5 Optical safety

BASE-AU optical transceivers shall conform to Hazard Level 1 laser requirements as defined in IEC 60825-1 and IEC 60825-2, under any condition of operation. This includes single fault conditions whether coupled into a fiber or out of an open bore.

Conformance to additional laser safety standards may be required for operation within specific geographic regions.

Laser safety standards and regulations require that the manufacturer of a laser product provide information about the product, which explicitly defines requirements and usage restrictions on the host system necessary to meet these safety certifications.

166.11 Delay constraints

In full duplex mode, predictable operation of the MAC Control PAUSE operation ([Clause 31](#), [Annex 31B](#)) also demands an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementers conform to certain maximum delay, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices.

The sum of the transmit and receive data delays for an implementation of the PHY shall not exceed the limits shown in Table 166–23. Transmit data delay is defined from the input of a given unit of data at the xMII to the presentation of the same unit of data by the PHY to the MDI. Receive data delay is defined from the input of a given unit of data at the MDI to the presentation of the same unit of data by the PHY to the xMII.

NOTE—The physical medium and in-line connectors between two PHYs introduce additional delay.

Table 166–23—Delay constraints

PHY	Bit times	Pause Quanta	Delay (ns)
2.5GBASE-AU	11 264	22	4505.6
5GBASE-AU	11 264	22	2252.8
10GBASE-AU	11 264	22	1126.4
25GBASE-AU	11 264	22	450.56
50GBASE-AU	14 848	29	296.96

166.12 Protocol implementation conformance statement (PICS) proforma for Clause 166, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU⁹

166.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 166, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

166.12.2 Identification

166.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

166.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3cz-2023, Clause 166, Physical Coding Sublayer (PCS), Physical Medium Attachment (PMA) sublayer, and Physical Medium Dependent (PMD) sublayer, types 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, 25GBASE-AU, and 50GBASE-AU
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3cz-2023.)	

Date of Statement	
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⁹*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

166.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*XGMII	PHY with exposed XGMII	46	Interface is supported.	O	Yes [] No []
*25GMII	PHY with exposed 25GMII	106	Interface is supported.	O	Yes [] No []
*50GMII	PHY with exposed 50GMII	132	Interface is supported.	O	Yes [] No []
*2.5G-AU	2.5GBASE-AU PHY	166.1		O.1	Yes [] No []
*5G-AU	5GBASE-AU PHY	166.1		O.1	Yes [] No []
*10G-AU	10GBASE-AU PHY	166.1		O.1	Yes [] No []
*25G-AU	25GBASE-AU PHY	166.1		O.1	Yes [] No []
*50G-AU	50GBASE-AU PHY	166.1		O.1	Yes [] No []
*BUOAM	BASE-U OAM channel	166.7	BASE-U OAM channel implementation.	O	Yes [] No []
*EEE	EEE capability	166.4	EEE implementation.	O	Yes [] No []
TG1	Temperature grade 1	166.10	Implementation of BASE-AU temperature grade 1.	O/2	Yes [] No []
TG2	Temperature grade 2	166.10	Implementation of BASE-AU temperature grade 2.	O/2	Yes [] No []
TG3	Temperature grade 3	166.10	Implementation of BASE-AU temperature grade 3.	O/2	Yes [] No []
MDIO	MDIO implemented	166.9		O	Yes [] No []

166.12.4 General

Item	Feature	Subclause	Value/Comment	Status	Support
*G1	Parameter <i>G</i> value for 2.5GBASE-AU, 5GBASE-AU, 10GBASE-AU, and 25GBASE-AU PHY equal to 1	166.1.1	$G = 1$ per Table 166–1.	2.5G-AU: M 5G-AU: M 10G-AU: M 25G-AU: M	Yes [] N/A []
*G2	Parameter <i>G</i> value for 50GBASE-AU PHY equal to 2	166.1.1	$G = 2$ per Table 166–1.	50G-AU: M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
G3	BASE-AU PHY transmitter composition	166.1.4	A BASE-AU PHY transmitter is composed of PCS, PMA, and PMD sublayers specified for the same data rate.	M	Yes []
G4	BASE-AU PHY receiver composition	166.1.4	A BASE-AU PHY receiver is composed of PCS, PMA, and PMD sublayers specified for the same data rate.	M	Yes []

166.12.5 Physical Coding Sublayer (PCS)

Item	Feature	Subclause	Value/Comment	Status	Support
PCS1	Physical header data (PHD) structure	166.2.2.1.1	The PHD consists of the fields detailed in Table 166–2.	M	Yes []
PCS2	PHD transmission	166.2.2.1.1	The PHD transmission conforms to the physical header data transmit bit ordering in Figure 166–10.	M	Yes []
PCS3	Transmit PHD commit point	166.2.2.1.1	LOC PHD fields assigned by the state diagrams are sampled at the beginning of a Transmit Block by the PHD Builder to create the PHD included in that current Transmit Block.	M	Yes []
PCS4	Receive PHD commit point	166.2.2.1.1	The new values of REMPHD fields are available to the state diagrams and registers immediately after reception, decoding, and validation of the entire PHD and before decoding the first RS-FEC codeword of the next received Transmit Block.	M	Yes []
PCS5	PHD building	166.2.2.1.2	The 224-bit PHD is built from the physical layer control information as specified in 166.2.2.1.1.	M	Yes []
PCS6	PHD 16-bit cyclic redundancy check (CRC16) encoding	166.2.2.1.3	A 16-bit cyclic redundancy check (CRC16) is calculated from the PHD as specified in 166.2.2.1.3.	M	Yes []
PCS7	PHD splitting function	166.2.2.1.4	The splitting function specified in 166.2.2.1.4 is used to split the concatenation of the PHD and the CRC16 redundancy bits into 20-bit sub-blocks.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
PCS8	PHD three repetition code (TRC) encoding	166.2.2.1.5	The TRC specified in 166.2.2.1.5 is used to code each 20-bit sub-block into three 20-bit PHD encoded sub-blocks by repetition.	M	Yes []
PCS9	PHD CRC16 input	166.2.2.1.3	CRC16 is computed from PHD bits.	M	Yes []
PCS10	PHD CRC16 operation	166.2.2.1.3	CRC16 compliant with Figure 166–6.	M	Yes []
PCS11	PHD CRC16 initialization	166.2.2.1.3	CRC16 shift register initialized to 0x0000 for each PHD.	M	Yes []
PCS12	Transmission payload data path	166.2.2.2	The incoming data from the xMII is encapsulated and encoded into 65-bit blocks for transmission as specified by the PCS 64B/65B transmit state diagram per 166.2.2.8 and Figure 166–16.	M	Yes []
PCS13	Transmit ordering	166.2.2.3	The PCS transmit ordering follows each sequence of 80×65 -bit blocks with a 20-bit encoded PHD sub-block per Figure 166–11.	M	Yes []
PCS14	RS-FEC encoder	166.2.2.4	The RS-FEC encoder generates the parity symbols per 166.2.2.4.	M	Yes []
PCS15	Scrambler operation	166.2.2.5	Scrambler per Figure 166–9.	M	Yes []
PCS16	Scrambler initialization for $G = 1$	166.2.2.5	Shift register of the scrambler is initialized with 0x0FB9659 at the beginning of each Transmit Block.	G1:M	Yes [] N/A []
PCS17	Scrambler initialization for $G = 2$	166.2.2.5	Shift register of the scrambler is initialized with 0x020492C at the beginning of each Transmit Block.	G2:M	Yes [] N/A []
PCS18	PHD transmit order	166.2.2.6	The PCS transmit function conforms to the PCS physical header data transmit bit order per Figure 166–10.	M	Yes []
PCS19	PCS transmit order	166.2.2.6	The PCS transmit function conforms to the PCS transmit bit order per Figure 166–11.	M	Yes []
PCS20	Mapping from the XGMII or 25GMII to a 65-bit block	166.2.2.6	Mapping per Figure 166–12.	XGMII:M 25GMII:M	Yes [] N/A []
PCS21	Mapping from the 50GMII to a 65-bit block	166.2.2.6	Mapping per Figure 166–13.	50GMII:M	Yes [] N/A []
PCS22	PCS transmit function	166.2.2.7.2	PCS transmit function per 166.2.2.9 and Figure 166–16.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
PCS23	XGMII or 25GMII transmitted control codes	166.2.2.7.4	Control codes not in Table 166–4 are not to be transmitted.	XGMII:M 25GMII:M	Yes [] N/A []
PCS24	50GMII transmitted control codes	166.2.2.7.4	Control codes not in Table 166–5 are not to be transmitted.	50GMII:M	Yes [] N/A []
PCS25	XGMII or 25GMII received control codes	166.2.2.7.4	Control codes not in Table 166–4 are treated as an error if received.	XGMII:M 25GMII:M	Yes [] N/A []
PCS26	50GMII received control codes	166.2.2.7.4	Control codes not in Table 166–5 are treated as an error if received.	50GMII:M	Yes [] N/A []
PCS27	LPI transmission	166.2.2.7.4	IfEEE capability is not enabled, LPI is not transmitted.	M	Yes []
PCS28	LPI reception	166.2.2.7.4	IfEEE capability is not enabled, LPI is treated as an error if received.	M	Yes []
PCS29	XGMII or 25GMII idle control characters insert and deletion	166.2.2.7.5	To occur in groups of four.	XGMII:M 25GMII:M	Yes [] N/A []
PCS30	50GMII idle control characters insert and deletion	166.2.2.7.5	To occur in groups of eight.	50GMII:M	Yes [] N/A []
PCS31	Addition of idle control characters	166.2.2.7.5	Not to be added while data is being received.	M	Yes []
PCS32	XGMII or 25GMII idle control characters deletion after /T/	166.2.2.7.5	When deleting /I/s, the first four characters after a /T/ are not deleted.	XGMII:M 25GMII:M	Yes [] N/A []
PCS33	XGMII or 25GMII LPI control characters insert and deletion	166.2.2.7.6	To occur in groups of four.	XGMII:M 25GMII:M	Yes [] N/A []
PCS34	50GMII LPI control characters insert and deletion	166.2.2.7.6	To occur in groups of eight.	50GMII:M	Yes [] N/A []
PCS35	Addition of LPI control characters	166.2.2.7.6	Not to be added while data is being received.	M	Yes []
PCS36	Deletion of sequence ordered sets	166.2.2.7.9	Occurs only when two consecutive sequence ordered sets have been received and delete only one of the two.	M	Yes []
PCS37	PCS ENCODE function	166.2.2.8.3	The ENCODE function encodes the block as specified in 166.2.2.7.	M	Yes []
PCS38	PCS receive function state diagram	166.2.3	PCS receive function per state diagram in Figure 166–21 and Figure 166–22.	M	Yes []
PCS39	PHD receive bit ordering	166.2.3.4	PHD receive bit ordering per Figure 166–17.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
PCS40	PCS receive bit ordering	166.2.3.6	PCS receive bit ordering per Figure 166–18.	M	Yes []
PCS41	PCS receive function blocks	166.2.3	PCS receive function carries out descrambling, RS-FEC decoding, PHD decoding, and 64B/65B decoding per 166.2.3.	M	Yes []
PCS42	PCS descrambler	166.2.3.1	Reverses the effect of the scrambler per 166.2.3.1.	M	Yes []
PCS43	PCS RS-FEC decoder functionality	166.2.3.2	RS-FEC decoding of received codewords per 166.2.3.2.	M	Yes []
PCS44	PCS RS-FEC decoder error correction and detection capability	166.2.3.2	RS-FEC decoder capable of correcting any combination of up to $t = 11$ symbol errors in a codeword and detecting any combination of up to $2t = 22$ symbol errors in a codeword.	M	Yes []
PCS45	PCS RS-FEC decoder uncorrectable codeword marking	166.2.3.2	The resulting erroneous bits are marked as corrupt by setting the R_BLOCK_TYPE of the affected 65-bit blocks equal to E.	M	Yes []
PCS46	PCS RS-FEC decoder uncorrectable codeword count	166.2.3.2	The uncorrectable codeword is counted as an RS-FEC codeword error and reflected in the RS-FEC codeword error counter (see 45.2.3.95).	M	Yes []
PCS47	PCS receiver ordering	166.2.3.3	Per 166.2.3.3 and Figure 166–18.	M	Yes []
PCS48	Received PHD field contents	166.2.3.4	Only available if the CRC16 computation is correct.	M	Yes []
PCS49	PHD receive bit order	166.2.3.4	Per Figure 166–17.	M	Yes []
PCS50	PCS receive bit order	166.2.3.6	Per Figure 166–18.	M	Yes []
PCS51	Mapping from 65-bit block to the XGMII or 25GMII	166.2.3.6	Mapping per Figure 166–19.	XGMII:M 25GMII:M	Yes [] N/A []
PCS52	Mapping from 65-bit block to the 50GMII	166.2.3.6	Mapping per Figure 166–20.	50GMII:M	Yes [] N/A []
PCS53	PCS DECODE function	166.2.3.7.3	The DECODE function decodes the rx_block as specified in 166.2.2.7.4.	M	Yes []

166.12.6 Physical Medium Attachment (PMA)

Item	Feature	Subclause	Value/Comment	Status	Support
PMA1	PMA transmit function mapping	166.3.1	Per 166.3.1 and Table 166–6, depending on the value of parameter <i>G</i> .	M	Yes []
PMA2	PHY Tx control state diagram	166.3.4.2	Per Figure 166–23.	M	Yes []
PMA3	PHY Rx control state diagram	166.3.4.3	Per Figure 166–24.	M	Yes []
PMA4	Coarse timing recovery	166.3.4.3	Transmit Block synchronization is performed during coarse timing recovery step.	M	Yes []
PMA5	Fine timing recovery	166.3.4.3	Stable sampling clock for reliable reception is obtained after fine timing recovery step.	M	Yes []
PMA6	Equalization training	166.3.4.3	In the case that equalizers are implemented, the PHY receiver trains them when the clock is stable.	M	Yes []
PMA7	Link monitor state diagram	166.3.4.4	Per Figure 166–25.	M	Yes []
PMA8	Maximum time from deassertion of <code>pma_reset</code> or <code>pcs_reset</code> until the system is ready to transmit and receive Ethernet frames, transmission side	166.3.4.4	The time from the fulfillment of the following conditions together: <code>pma_reset</code> is equal to OFF, and <code>pcs_reset</code> is equal to FALSE until a compliant optical signal is presented at TP2 is less than 5 ms.	M	Yes []
PMA9	Maximum time from deassertion of <code>pma_reset</code> or <code>pcs_reset</code> until the system is ready to transmit and receive Ethernet frames, reception side	166.3.4.4	The time from the fulfillment of the following conditions together: <code>pma_reset</code> is equal to OFF, <code>pcs_reset</code> is equal to FALSE, and a compliant optical signal is presented at TP3 until the assertion of the <code>link_status</code> variable to OK is less than 20 ms.	M	Yes []
PMA10	PHD monitor state diagrams	166.3.4.5	Per Figure 166–26, Figure 166–27, and Figure 166–28.	M	Yes []
PMA11	PHY quality monitor state diagrams	166.3.4.6.4	Per Figure 166–29.	M	Yes []

166.12.7 Energy-Efficient Ethernet (EEE)

Item	Feature	Subclause	Value/Comment	Status	Support
EEE1	EEE capability enable	166.4.1	EEE capability is enabled when the field PHD.CAP.LPI of both the transmitted and received PHD are equal to one, and disabled otherwise.	EEE:M	Yes [] N/A []
EEE2	Detection of the condition to enter LPI operation mode in transmission	166.4.3	Monitor 65-bit blocks generated by the PCS transmit function to detect the condition to enter LPI operation mode.	EEE:M	Yes [] N/A []
EEE3	Detection of the condition to resume to normal operation mode in transmission	166.4.3	Monitor 65-bit blocks generated by the PCS transmit function to detect the condition to resume to normal operation mode.	EEE:M	Yes [] N/A []
EEE4	LPI codewords transmission when in LPI operation mode	166.4.3	PCS transmit function in LPI operation mode transmits LPI codewords per Figure 166–30 instead of RS-FEC codewords.	EEE:M	Yes [] N/A []
EEE5	PCS PHD transmit function when in LPI operation mode	166.4.3	PCS PHD transmit function in LPI operation mode is the same as in normal operation mode.	EEE:M	Yes [] N/A []
EEE6	LPI refresh codewords	166.4.3.1	Per Figure 166–30 aggregating 80 LPI refresh 65-bit blocks.	EEE:M	Yes [] N/A []
EEE7	LPI wake codewords	166.4.3.2	Per Figure 166–30 aggregating 80 LPI wake 65-bit blocks.	EEE:M	Yes [] N/A []
EEE8	PCS transmit function condition to enter LPI operation mode	166.4.3.3	Enter LPI mode when last 65-bit block of an RS-FEC codeword encodes eight control characters with value /LI/.	EEE:M	Yes [] N/A []
EEE9	Alignment of PCS transmit function LPI operation mode	166.4.3.3	The PCS transmit function enters LPI operation mode aligned with the next RS-FEC codeword boundary after transmission of the RS-FEC codeword with last 65-bit block encoding eight control characters with value /LI/.	EEE:M	Yes [] N/A []
EEE10	PCS transmit function in LPI mode	166.4.3.3	PCS transmit function in LPI mode transmits LPI refresh codewords.	EEE:M	Yes [] N/A []
EEE11	LPI wake codeword transmission	166.4.3.4	PCS transmit function in LPI mode transmits a LPI wake codeword when last 65-bit block of an RS-FEC codeword encodes eight control characters with value /I/.	EEE:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
EEE12	Alignment of PCS transmit function normal operation mode	166.4.3.4	After transmission of an LPI wake codeword, next codeword is an RS-FEC codeword resulting from normal operation of the PCS transmit function, that resumes its normal operation.	EEE:M	Yes [] N/A []
EEE13	PCS receive function in LPI operation mode	166.4.4	When the PCS 64B/65B receive function is in any of the states described in Figure 166–22.	EEE:M	Yes [] N/A []
EEE14	PCS receive function in LPI operation mode process	166.4.4	PCS receive function in LPI operation mode processes LPI codewords per Figure 166–33.	EEE:M	Yes [] N/A []
EEE15	PCS PHD receive function when in LPI operation mode	166.4.4	PCS PHD receive function in LPI operation mode is the same as in normal operation mode.	EEE:M	Yes [] N/A []
EEE16	LPI wake codeword reception	166.4.4	PCS receive function in LPI mode receives a LPI wake codeword, the corresponding 80×65 -bit blocks are set as LPI wake 65-bit blocks.	EEE:M	Yes [] N/A []
EEE17	PHD sub-block decoding in LPI operation mode	166.4.4	From each LPI codeword received, the 20-bit encoded PHD sub-block is decoded by majority voting using a number of repetitions equal to or less than 11.	EEE:M	Yes [] N/A []

166.12.8 Test modes

Item	Feature	Subclause	Value/Comment	Status	Support
TM1	BER test mode	166.5	BER test mode implemented per 166.5.1.	M	Yes []
TM2	SSQWP test pattern	166.5	SSQWP test pattern implemented per 166.5.2.	M	Yes []
TM3	FSQWP test pattern	166.5	FSQWP test pattern implemented per 166.5.3.	M	Yes []
TM4	SSPR-NRZ test pattern	166.5	SSPR-NRZ test pattern implemented per 166.5.4.	G1:M	Yes [] N/A []
TM5	SSPR-PAM4 test pattern	166.5	SSPR-PAM4 test pattern implemented per 166.5.5.	G2:M	Yes [] N/A []
TM6	Test modes and patterns configuration	166.5	The test modes and patterns are configured by setting the BASE-U PCS control register, operation mode bits defined in 45.2.3.90.	G2:M	Yes [] N/A []

166.12.9 Physical Medium Dependent (PMD)

Item	Feature	Subclause	Value/Comment	Status	Support
PMD1	PMD transmit function conversion	166.6.2.2	The PMD transmit function converts the communication signal amplitude given by the tx_signal parameter into optical signal p at TP2 according to Equation (166–7).	M	Yes []
PMD2	Transmitter optical specifications	166.6.2.2	Optical signal p at TP2 complies with the transmitter optical specifications in 166.6.3.2.	M	Yes []
PMD3	PMD receive function conversion	166.6.2.3	The PMD receive function converts the optical signal received at the MDI into the communication signal amplitude given by the rx_signal parameter.	M	Yes []
PMD4	Receiver optical specifications	166.6.2.3	Optical signal at TP3 complies with the receive optical specifications in 166.6.3.3.	M	Yes []
PMD5	Signal detect function	166.6.2.4	The value of the signal_detect parameter is generated in response to the average optical power (AOP) present at TP3 according to the conditions defined in Table 166–7.	M	Yes []

166.12.10 Optical measurement methods

Item	Feature	Subclause	Value/Comment	Status	Support
OM1	Center wavelength and RMS spectral width	166.6.4.2	Measured per IEC 61280-1-3.	M	Yes []
OM2	Average optical power	166.6.4.3	Measured per ANSI/TIA/EIA-455-95-2019, or α obtained per 166.6.4.8.	M	Yes []
OM3	Outer Optical Modulation Amplitude	166.6.4.4	Per 166.6.4.4.	M	Yes []
OM4	Extinction ratio	166.6.4.5	Per 166.6.4.5, or as defined in 166.6.4.8.5, Equation (166–19).	M	Yes []
OM5	Relative intensity noise	166.6.4.6	Per 166.6.4.6.	M	Yes []
OM6	Uncorrelated random jitter	166.6.4.7	Per 166.6.4.7.	M	Yes []
OM7	Transmitter distortion figure of merit (TDFOM)	166.6.4.8	Per 166.6.4.8.	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
OM8	2.5GBASE-AU receiver sensitivity	166.6.4.12	Meets Equation (166–20) for any value from –0.3 dB through 1 dB.	2.5G-AU:M	Yes [] N/A []
OM9	5GBASE-AU receiver sensitivity	166.6.4.12	Meets Equation (166–20) for any value from –0.4 dB through 1 dB.	5G-AU:M	Yes [] N/A []
OM10	10GBASE-AU receiver sensitivity	166.6.4.12	Meets Equation (166–20) for any value from –1 dB through 2 dB.	10G-AU:M	Yes [] N/A []
OM11	25GBASE-AU receiver sensitivity	166.6.4.12	Meets Equation (166–20) for any value from –1.5 dB through 2.5 dB.	25G-AU:M	Yes [] N/A []
OM12	50GBASE-AU receiver sensitivity	166.6.4.12	Meets Equation (166–20) for any value from –1.7 dB through 3 dB.	50G-AU:M	Yes [] N/A []
OM13	Stressed receiver sensitivity	166.6.4.13	Per 166.6.4.13.	M	Yes []

166.12.11 Characteristics of the fiber optic cabling (channel)

Item	Feature	Subclause	Value/Comment	Status	Support
CH1	Fiber optic cabling specifications	166.6.6	Meets the specifications defined in Table 166–11.	M	Yes []
CH2	Optical fiber cable specifications	166.6.6	Per IEC 60793-2-10 for optical fiber Type A1a.2 (OM3) or Table 166–19 where they differ.	M	Yes []
CH3	2.5GBASE-AU, 5GBASE-AU, and 10GBASE-AU single connection loss	166.6.6.2.1	Not exceed 2.5 dB.	2.5G-AU:M 5G-AU:M 10G-AU:M	Yes [] N/A []
CH4	25GBASE-AU and 50GBASE-AU single connection loss	166.6.6.2.1	Not exceed 2 dB.	50G-AU:M 25G-AU:M	Yes [] N/A []
CH5	Maximum discrete reflectance	166.6.6.2.2	Less than –20 dB.	M	Yes []

166.12.12 BASE-U operations, administration, and maintenance (BASE-U OAM) channel

Item	Feature	Subclause	Value/Comment	Status	Support
OC1	OAM channel capability enable	166.7	OAM channel capability is enabled when the field PHD.CAP.OAM of both the transmitted and received PHD are equal to one, and disabled otherwise.	BUOAM:M	Yes [] N/A []
OC2	Message transmission and reception protocol	166.7	Per 115.9.1 and 115.9.3 respectively.	BUOAM:M	Yes [] N/A []
OC3	OAM control bits	166.7	Per 115.9.2, but using BASE-U OAM registers specified in 45.2.3.88.	BUOAM:M	Yes [] N/A []
OC4	OAM transmit and receive state diagrams	166.7	Per 115.9.4, but using BASE-U OAM registers specified in 45.2.3.88 and 45.2.3.89.	BUOAM:M	Yes [] N/A []

166.12.13 Loopback modes

Item	Feature	Subclause	Value/Comment	Status	Support
LB1	Loopback mode selection	166.8	Selected using bits 3.2348.12:10.	M	Yes []
LB2	Loopback mode reset	166.8	Default and reset value is no loopback.	M	Yes []
LB3	xMII level loopback	166.8	When xMII level loopback mode is selected, xMII Tx is looped back to xMII Rx.	M	Yes []
LB4	PMD interface level loopback	166.8	When PMD interface level loopback is selected, transmit path is looped back to receive path near PMD service interface, with PCS and PMA operating as in normal mode, and the PMA receive ignoring PMD service signals.	M	Yes []
LB5	Line loopback	166.8	When line loopback is selected, receive data is looped back to the transmit path (toward link partner) near xMII interface.	M	Yes []

166.12.14 Management Interface

Item	Feature	Subclause	Value/Comment	Status	Support
MI1	BASE-AU management	166.9	PHY provides Clause 166 management capabilities and functionality provided by referenced Clause 45 registers and bits.	M	Yes []
MI2	MDIO registers functionality	166.9	PHY provides control and status functionalities of MDIO registers.	M	Yes []
MI3	MDIO interface mapping	166.9	PHY implements variable to register mapping as specified in Table 166–21.	M	Yes []

166.12.15 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
E1	Temperature grade indication	166.10.1	A compliant BASE-AU PHY clearly indicates the temperature grade of Table 166–21 over which its compliance is claimed.	M	Yes []
E2	General safety	166.10.2	Equipment subject to this clause conforms to the general safety requirements in J.2 .	M	Yes []
E3	Optical safety	166.10.5	BASE-AU optical transceiver conforms to Hazard Level 1 laser requirements per IEC 60825-1 and IEC 60825-2, under any condition of operation.	M	Yes []

166.12.16 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
DLY1	Delay constraint	166.11	The sum of the transmit and receive data delays for an implementation of the PHY do not exceed the limits shown in Table 166–23.	M	Yes []

Insert new annexes Annex 166A and Annex 166B as follows:

Annex 166A

(informative)

BASE-U scrambler sequence

166A.1 Introduction

Clause 166 specifies the BASE-U scrambler, where the output of the RS-FEC encoder is mod-2 added with a binary sequence that can be generated by a linear feedback register (see Figure 166–9). Table 166A–1 and Table 166A–2 contain hexadecimal presentations of the binary sequence tx_scrambler<0:195839> produced by the BASE-U scrambler shift register. The tables are read in the same order tx_scrambler<0:195839> is generated by the scrambler during a Transmit Block transmission, from left to right within each row starting from the top row and ending at the bottom row. The most significant bit of each hexadecimal symbol is generated first by the scrambler. Therefore, the most significant bit of the first hexadecimal symbol is tx_scrambler<0>, and the least significant bit of the last hexadecimal symbol is tx_scrambler<195839>. An underscore (“_”) symbol has been inserted between 64-bit groups to improve readability of the tables.

166A.2 2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U scrambler sequence

Table 166A–1 shows the hexadecimal representation of the first and last 2048 bits of tx_scrambler<0:195839> for 2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U.

Table 166A–1—2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U scrambler sequence

Index<i:j>	tx_scrambler<i:j>
<0:255>	249393000785801D_D4C079B961DA3937_85F845D51D35BEFA_4C249416801A5140
<256:511>	646DA1A380565FC1_7250E5706BE7799A_6E5AA48E47C07E3C_E1C7EDF7FC41240F
<512:767>	24023B0209F18923_3EE01D64F07B21B9_D1163BACD3F36526_34260BCA8B2ACF69
<768:1023>	3E5A10664549AA37_827DC5C8C9F5C743_2DFEAD2105E21495_79581F59CC72CBD7
<1024:1279>	F24AB4300F8AD839_EE0CFB4F35909B4C_8A20946D921B8081_63C2453EE83664D4
<1280:1535>	C2A1396BD4791ABB_D8E7B28FED137842_C45D2A735A3CF4C5_EDA97540716FA1FD
<1536:1791>	09570AA1F7AFD725_68B70716879FD15D_B0ADD81AE18C66F6_F7A8A0A576D2C760
<1792:2047>	227E3098C7DA2F7C_85AA4C547C177BCE_5652DE726216F0B9_50BAB9EAB7BB5F85
...	...
<193792:194047>	11CD9A4FD0A418AA_C26EFE688C869674_5912EB48C2D8876A_0E5E1D3E577A6676
<194048:194303>	54AAE278FEF8DF84_9F11D033CFA8D6D9_7730096758212ECC_942C541AA77A67CE

**Table 166A–1—2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U scrambler sequence
(continued)**

Index<i:j>	tx_scrambler<i:j>
<194304:194559>	54ACDE78E516DFE6_D01068284994B502_988E8B2E7D692ECB_102C49C8A703C6CF
<194560:194815>	8EF859FC9D4B0C3B_89B6F3E200B67902_82D88B4A0E688D3E_96726512F0A6C2BA
<194816:195071>	C86BB6459B0234A1_89C8D6E3C730FEFF_5B8482C3D04A6EA9_0C8DF0B4713A8BFC
<195072:195327>	776A0FE61D386B7A_7D9854C8AD7946E3_59B8FCCA3F8D4DE1_F39177378D6745F3
<195328:195583>	2EB5352D8E4E20FE_1F93877187DE7EDC_96C41C10727E49F0_C6033B7B0D7051B3
<195584:195839>	796E14590F5B48BA_C086B6625982B84A_CBB0E4B2ABE093F_A720614F11B59BCE

NOTE—A CSV file, Annex_166A_G1_binary_scrambler_sequence.csv, containing the entire 2.5GBASE-U, 5GBASE-U, 10GBASE-U, and 25GBASE-U scrambler sequence is available at <https://standards.ieee.org/downloads/802.3/>.

166A.3 50GBASE-U scrambler sequence

Table 166A–2 shows the hexadecimal representation of the first and last 2048 bits of tx_scrambler<0:195839> for 50GBASE-U.

Table 166A–2—50GBASE-U scrambler sequence

Index<i:j>	tx_scrambler<i:j>
<0:255>	4801130404C59211_74814D6845931534_85DE485596057C93_174C05D69615B113
<256:511>	5C1CC4DE7D7116CB_7CD0484D29051230_96C1DA10678549AD_D78261B5C8B60DC6
<512:767>	8331F94D5F1993D3_EA86A65F59CA52CB_CC624AD7B80E353C_3FCE6EE0DE8CF315
<768:1023>	75B5DF6C0D920630_811BDA44E28431FB_52DF10E213CBF946_CA19B84D6A3D131D
<1024:1279>	EAC5F95E7519D6EE_EBB0CCDB1B5501E0_FE8773855E77D7D6_E4B4B0E0889BF266
<1280:1535>	2230AB99DAFBAB86_937BD90452889363_66043C2B12EEB9C2_CDBBEA50325C68D0
<1536:1791>	5F9729519731EE97_5F4D16D292D02302_289D89B638E203FF_F90E0018BF006EA3
<1792:2047>	818DDFC6F190F8BE_8B9EA56BB5C71B0D_FFE1B100761C81E3_7C477C4F3E4F1B66
...	...
<193792:194047>	651ECF26F45B08AB_41A6F8A6489ECA06_344D1BCB12E2C9C2_FA43EA942E5F1AAE
<194048:194303>	53E7EE666C4EAA87_1DFF5FF902D0188A_286E6DB58E800CFD_40358BA0CCEB5355
<194304:194559>	D8E4FD8FE188F876_E79DE0EDB973C039_76E0F960F39933B7_A857057577976F65
<194560:194815>	960A24932D800520_C016136053443164_B2DD20921A120165_410527A4960D4013
<194816:195071>	33A045575137F6EC_4420C732937F5304_42E5932AE4853EE0_5664F172A1BD73D6

Table 166A-2—50GBASE-U scrambler sequence (*continued*)

Index<i:j>	tx_scrambler<i:j>
<195072:195327>	2B76B3B861973DB6_976801161404D35A_1124C54C01779605_6593172485D70055
<195328:195583>	B7817C05C54E15F7_9F5D25B2DA041205_124116C024D06081_29B244321032D148
<195584:195839>	D22D8721A0DF1653_13D265C6A0A5F9D2_C51BA276E358E0FC_CFF38D5837F3CCC4

NOTE—A CSV file, Annex_166A_G2_binary_scrambler_sequence.csv, containing the entire 50GBASE-U scrambler sequence is available at <https://standards.ieee.org/downloads/802.3/>.

Annex 166B

(informative)

RS-FEC(544,522) codeword example

166B.1 Introduction

This annex provides an example RS-FEC(544,522) codeword produced by the encoder defined in Clause 166 (see 166.2.2.4), presented in tabular form.

166B.2 Output of the RS-FEC(544,522) encoder

Table 166B–1 contains a 5440-bit RS-FEC(544,522) codeword in hexadecimal representation. Each row contains 256 bits except the last one that contains 64 bits. Underscore (“_”) symbols separate 64-bit groups to improve readability.

The transmission order is from left to right within each row, starting from the top row and ending at the bottom row, where the most significant bit of each hexadecimal symbol is transmitted first. Therefore, the most significant bit of the first hexadecimal symbol is CW<0>, and the least significant bit of the last hexadecimal symbol is CW<5439>.

The first 5220 transmitted bits of the codeword, CW<5219:0>, constitute the message portion of the codeword. The parity is computed using the encoder defined in 166.2.2.4 and it is appended to the message to complete the 5440-bit codeword.

Table 166B–1—RS-FEC(544,522) codeword example

Index<i:j>	CW<i:j>
<0:255>	D5000040000000040_6A80002000000020_3540001000000010_1AA0000800000008
<256:511>	0D500004000000004_06A8000200000002_0354000100000001_01AA000080000000
<512:767>	80D5000040000000_406A800020000000_2035400010000000_101AA00008000000
<768:1023>	080D500004000000_0406A80002000000_0203540001000000_0101AA0000800000
<1024:1279>	0080D50000400000_00406A8000200000_0020354000100000_00101AA000080000
<1280:1535>	00080D5000040000_000406A800020000_0002035400010000_000101AA00008000
<1536:1791>	000080D500004000_0000406A80002000_0000203540001000_0000101AA0000800
<1792:2047>	0000080D50000400_00000406A8000200_0000020354000100_00000101AA000080
<2048:2303>	00000080D5000040_000000406A800020_0000002035400010_000000101AA00008
<2304:2559>	000000080D500004_0000000406A80002_0000000203540001_0000000101AA0000
<2560:2815>	8000000080D50000_40000000406A8000_2000000020354000_10000000101AA000
<2816:3071>	08000000080D5000_040000000406A800_0200000002035400_010000000101AA00

Table 166B–1—RS-FEC(544,522) codeword example (*continued*)

Index<i:j>	CW<i:j>
<3072:3327>	008000000080D500_0040000000406A80_0020000000203540_0010000000101AA0
<3328:3583>	000800000080D50_00040000000406A8_0002000000020354_00010000000101AA
<3584:3839>	00008000000080D5_000040000000406A_8000200000002035_400010000000101A
<3840:4095>	A0000800000080D_5000040000000406_A800020000000203_5400010000000101
<4096:4351>	AA00008000000080_D500004000000040_6A80002000000020_3540001000000010
<4352:4607>	1AA0000800000008_0D50000400000004_06A8000200000002_0354000100000001
<4608:4863>	01AA000080000000_80D5000040000000_406A800020000000_2035400010000000
<4864:5119>	101AA00008000000_080D500004000000_0406A80002000000_0203540001000000
<5120:5375>	0101AA0000800000_00809FF00871EDA1_72A3B2F6682F3A95_C89C57D10DC551E1
<5376:5439>	8A16FC41E197A922

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